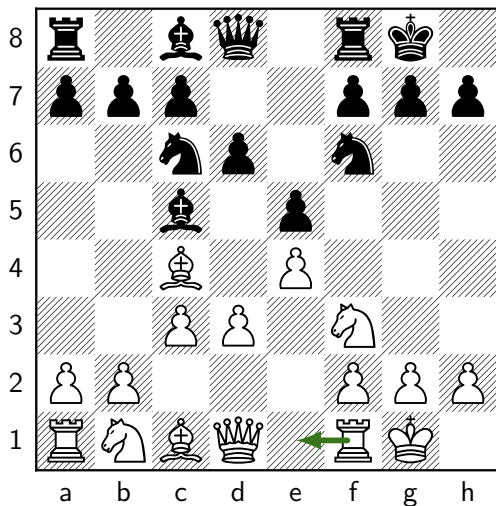


# Melanie-ANALOGY — Italian, Giuoco Pianissimo

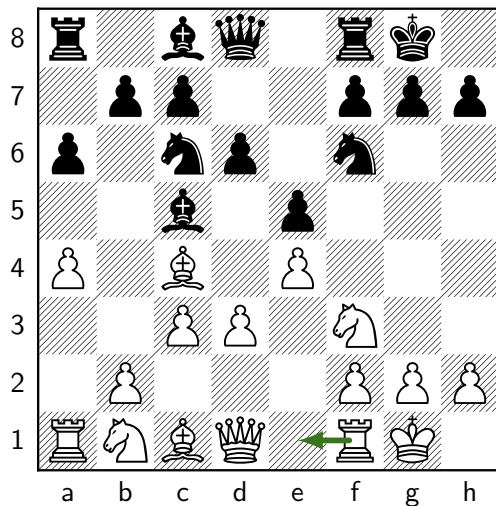
Melanie (Deep\_Vision)

## Abstract

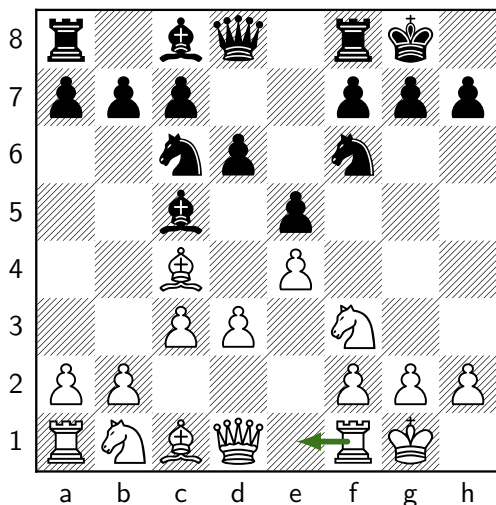
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under  $\mu$  – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



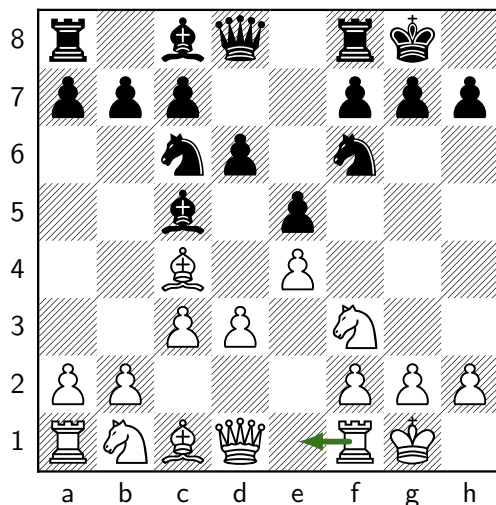
(a)



(b)

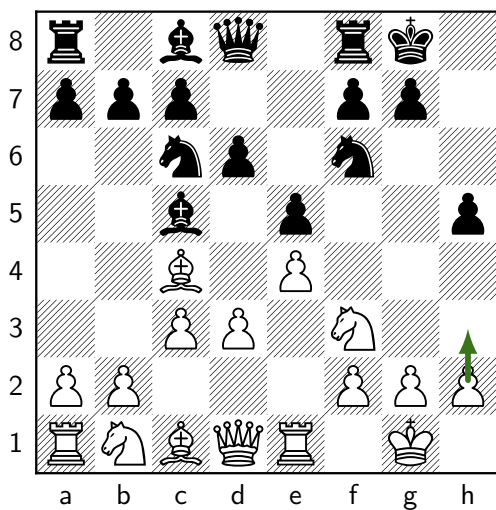


(c)

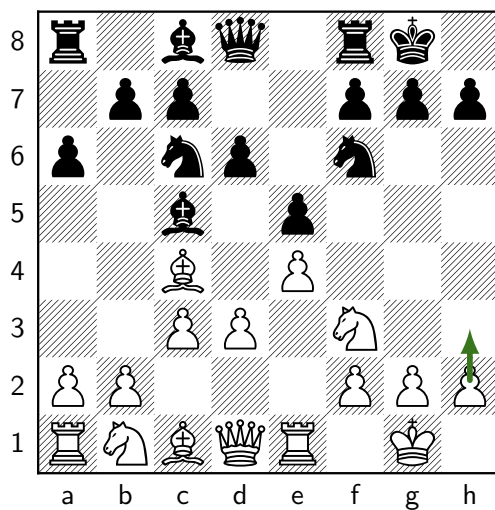


(d)

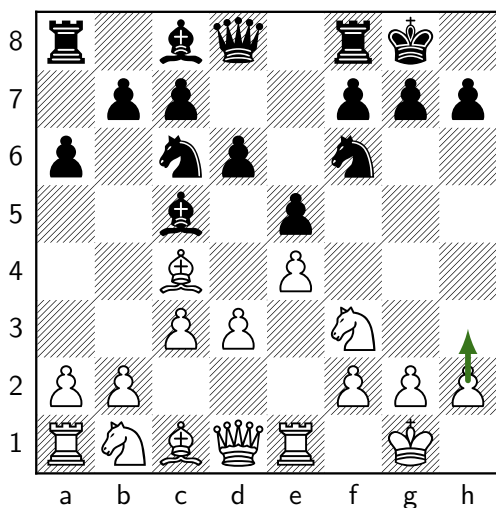
Figure 1: **Move 7, White: Re1**. (a) Our position; Melanie plays Re1 (f1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#547` (ply 14) [structural distance  $d=20$ , QAP obj=178]: the master played Re1; under  $\mu$  our *Rf1* plays the role of that game’s *Rf1*; transferred move f1e1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#138` (ply 12) [structural distance  $d=0$ , QAP obj=176]: the master played Re1; under  $\mu$  our *Rf1* plays the role of that game’s *Rf1*; transferred move f1e1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#916` (ply 12) [structural distance  $d=0$ , QAP obj=176]: the master played Re1; under  $\mu$  our *Rf1* plays the role of that game’s *Rf1*; transferred move f1e1 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Re1 (summed QAP objective 1672, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



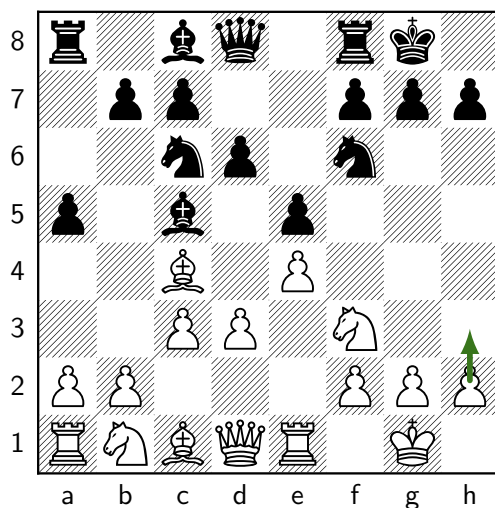
(a)



(b)

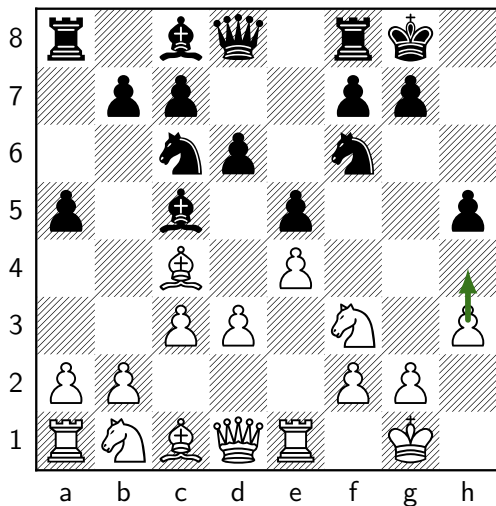


(c)

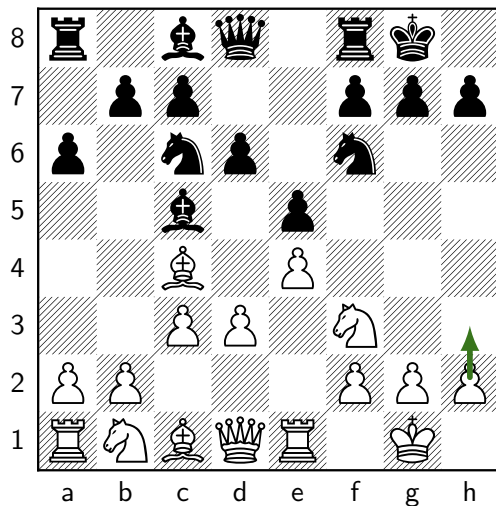


(d)

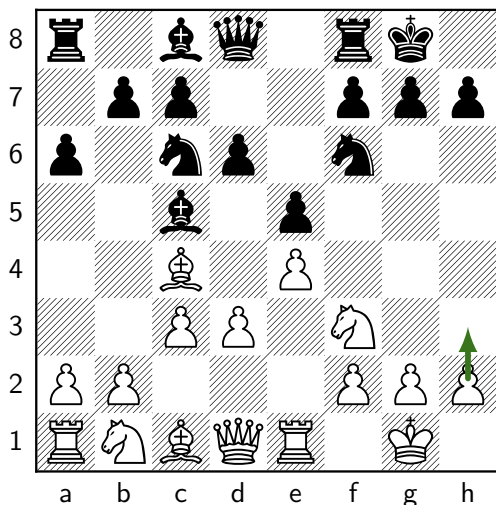
Figure 2: **Move 8, White: h3.** (a) Our position; Melanie plays h3 (h2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0003#923` (ply 14) [structural distance  $d=12$ , QAP obj=178]: the master played h3; under  $\mu$  our *Ph2* plays the role of that game’s *Ph2*; transferred move h2h3 (**endorses this move**). (c) Retrieved from `cctl_top_p0003#924` (ply 14) [structural distance  $d=12$ , QAP obj=178]: the master played h3; under  $\mu$  our *Ph2* plays the role of that game’s *Ph2*; transferred move h2h3 (**endorses this move**). (d) Retrieved from `cctl_top_p0000#916` (ply 14) [structural distance  $d=5$ , QAP obj=165]: the master played h3; under  $\mu$  our *Ph2* plays the role of that game’s *Ph2*; cross-type roles: our bc5 as their pa5, our bc8 as their pb7; transferred move h2h3 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to h3 (summed QAP objective 2597, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



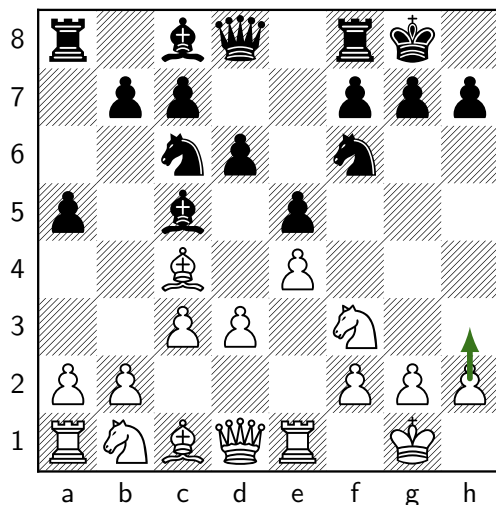
(a)



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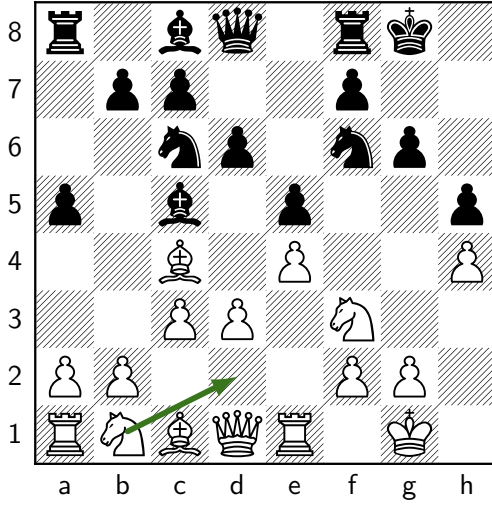


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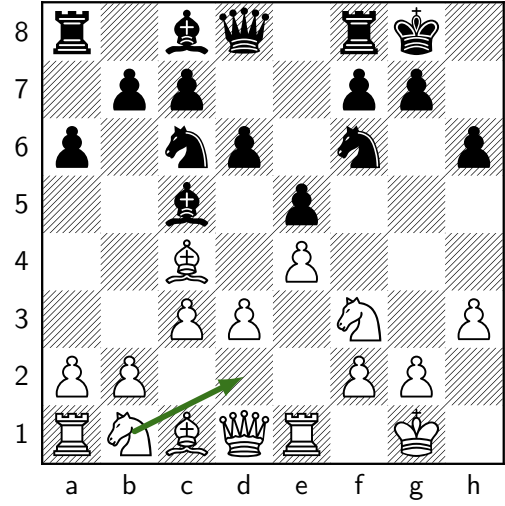


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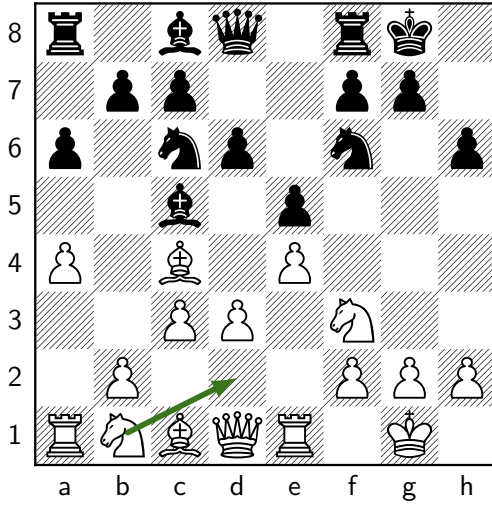
Figure 3: **Move 9, White: h4.** (a) Our position; Melanie plays **h4** (**h3h4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#923` (ply 14) [structural distance  $d=27$ , QAP obj=177]: the master played **h3**; under  $\mu$  our *Ph3* plays the role of that game’s *Ph2*; transferred move **h3h4** (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#924` (ply 14) [structural distance  $d=27$ , QAP obj=177]: the master played **h3**; under  $\mu$  our *Ph3* plays the role of that game’s *Ph2*; transferred move **h3h4** (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#916` (ply 14) [structural distance  $d=21$ , QAP obj=168]: the master played **h3**; under  $\mu$  our *Ph3* plays the role of that game’s *Ph2*; transferred move **h3h4** (**endorses this move**). Synthesis: 15 retrieved analogs transferred to **h4** (summed QAP objective 2438, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



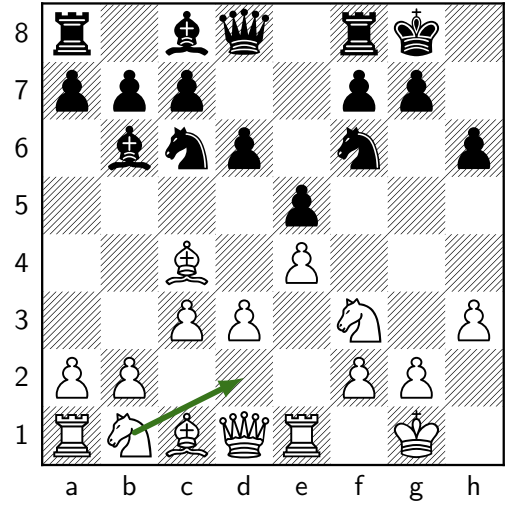
(a)



(b)

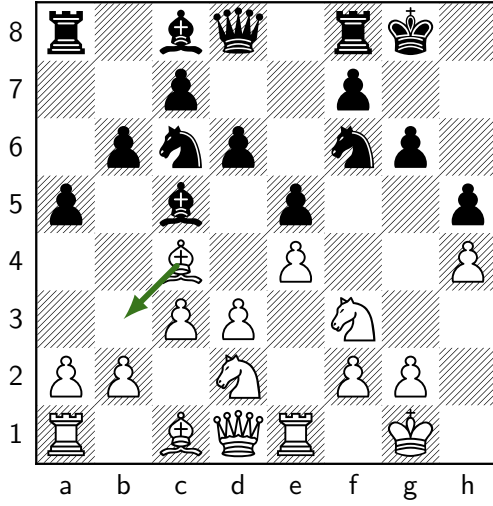


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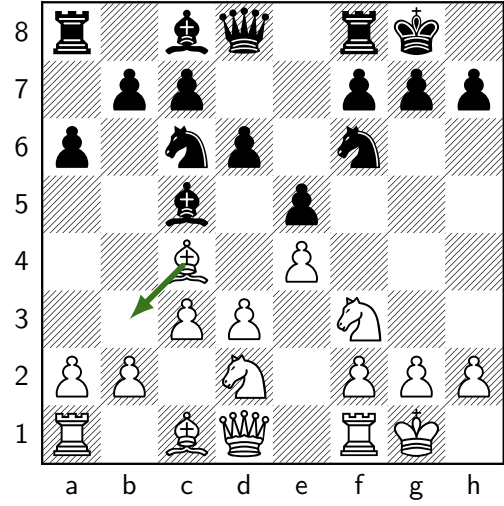


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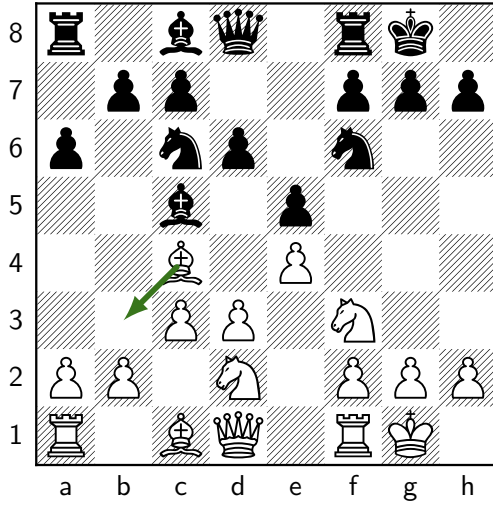
Figure 4: **Move 10, White: Nbd2.** (a) Our position; Melanie plays Nbd2 (b1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3310` (ply 16) [structural distance  $d=44$ , QAP obj=173]: the master played Nbd2; under  $\mu$  our *Nb1* plays the role of that game’s *Nb1*; cross-type roles: our *Nf3* as their *Bc1*, our *qd8* as their *pg7*; transferred move b1d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2841` (ply 16) [structural distance  $d=39$ , QAP obj=171]: the master played Nbd2; under  $\mu$  our *Nb1* plays the role of that game’s *Nb1*; transferred move b1d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#138` (ply 16) [structural distance  $d=34$ , QAP obj=147]: the master played Nbd2; under  $\mu$  our *Nb1* plays the role of that game’s *Nb1*; cross-type roles: our *nf6* as their *Bc1*, our *bc8* as their *Ph3*; transferred move b1d2 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Nbd2 (summed QAP objective 1396, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



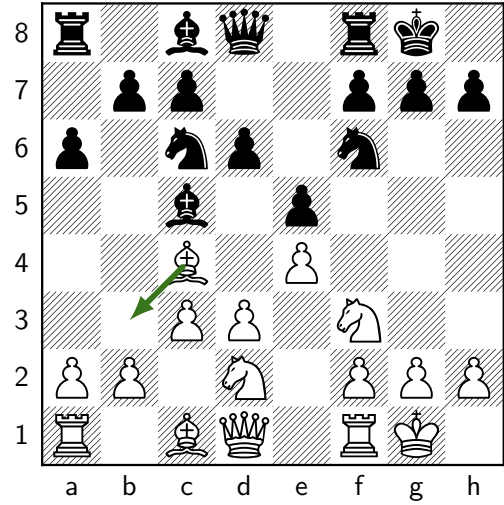
(a)



(b)

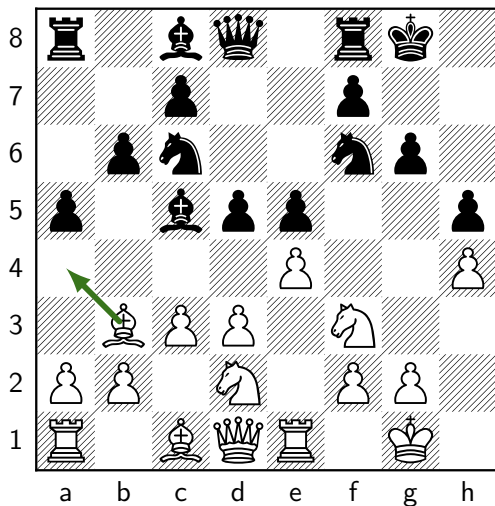


(c)

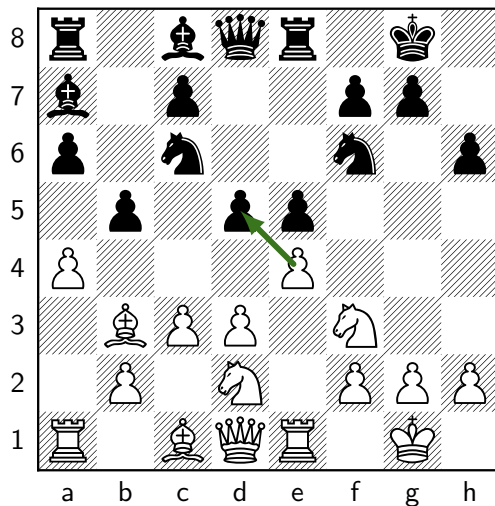


(d)

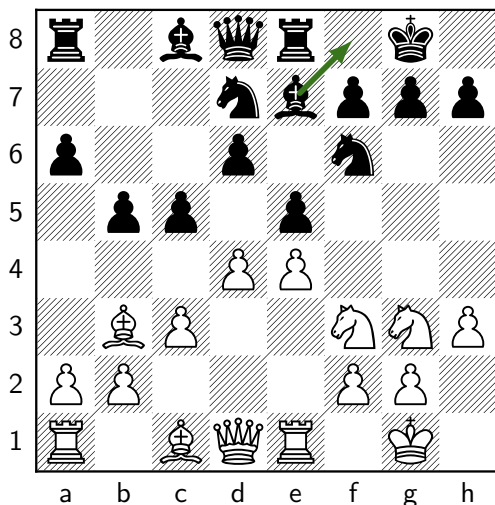
Figure 5: **Move 11, White: Bb3.** (a) Our position; Melanie plays Bb3 (c4b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3460` (ply 14) [structural distance  $d=62$ , QAP obj=179]: the master played Bb3; under  $\mu$  our  $Bc4$  plays the role of that game’s  $Bc4$ ; cross-type roles: our  $Bc1$  as their  $Ra1$ , our  $pb6$  as their  $Bc1$ ; transferred move c4b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3461` (ply 14) [structural distance  $d=62$ , QAP obj=179]: the master played Bb3; under  $\mu$  our  $Bc4$  plays the role of that game’s  $Bc4$ ; cross-type roles: our  $Bc1$  as their  $Ra1$ , our  $pb6$  as their  $Bc1$ ; transferred move c4b3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1299` (ply 14) [structural distance  $d=62$ , QAP obj=179]: the master played Bb3; under  $\mu$  our  $Bc4$  plays the role of that game’s  $Bc4$ ; cross-type roles: our  $Bc1$  as their  $Ra1$ , our  $pb6$  as their  $Bc1$ ; transferred move c4b3 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Bb3 (summed QAP objective 1605, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



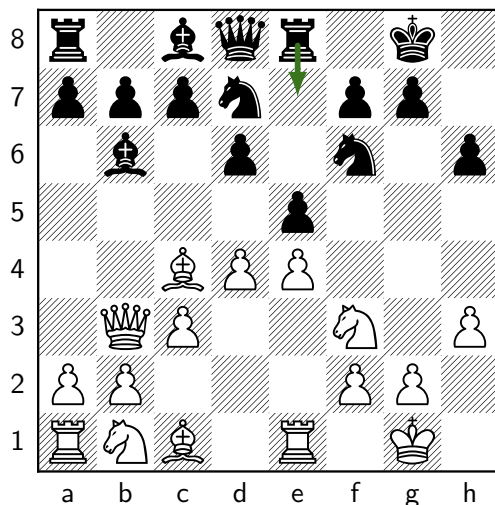
(a)



(b)

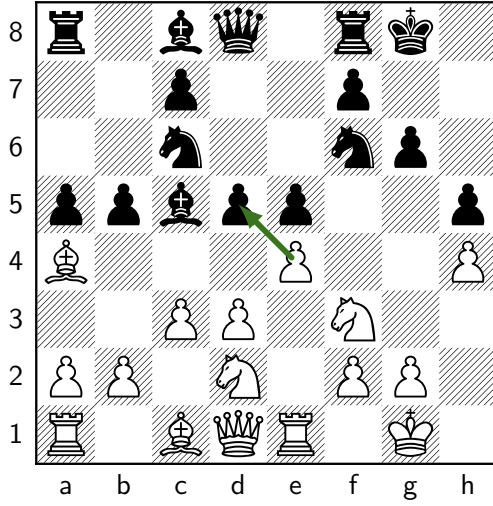


(c)

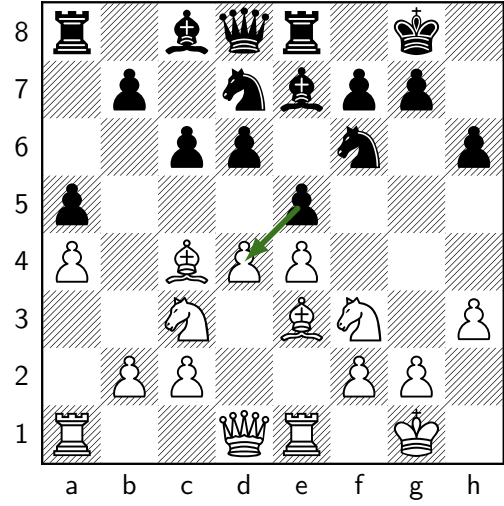


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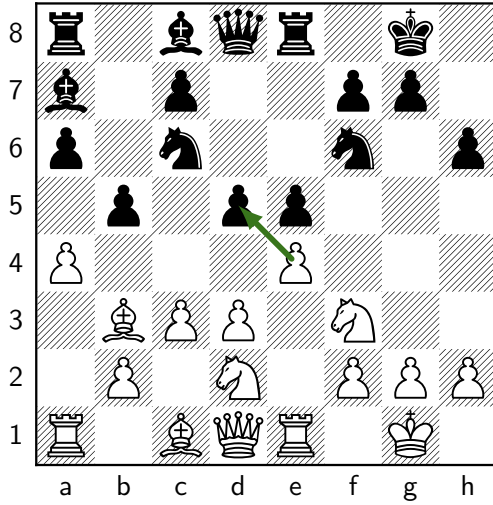
Figure 6: **Move 12, White: Ba4.** (a) Our position; Melanie plays Ba4 (b3a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#14` (ply 22) [structural distance  $d=52$ , QAP obj=170]: the master played `exd5`; under  $\mu$  our *Pe4* plays the role of that game’s *Pe4*; cross-type roles: our *pf7* as their *Ra1*, our *qd8* as their *pf7*; transferred move `e4d5`. (c) Retrieved from `ccr1_top_p0003#548` (ply 25) [structural distance  $d=58$ , QAP obj=166]: the master played `Bf8`; under  $\mu$  our *Bb3* plays the role of that game’s *Be2*; transferred move `b3c4`. (d) Retrieved from `ccr1_top_p0003#839` (ply 19) [structural distance  $d=65$ , QAP obj=135]: the master played `Re7`; under  $\mu$  our *Re1* plays the role of that game’s *Re1*; cross-type roles: our *Pa2* as their *Bc8*, our *Pe4* as their *Bb6*; transferred move `e1e2`. Synthesis: no retrieved analog transferred to Ba4 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



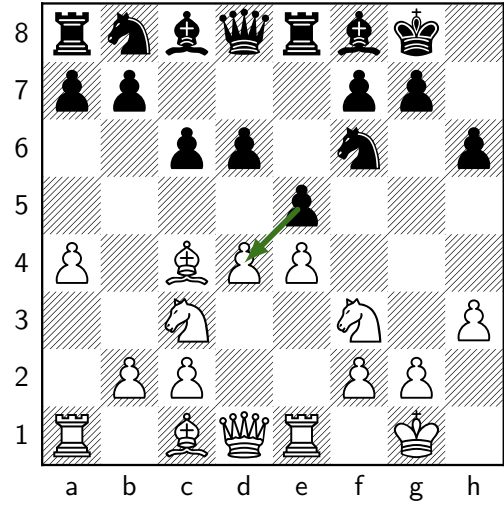
(a)



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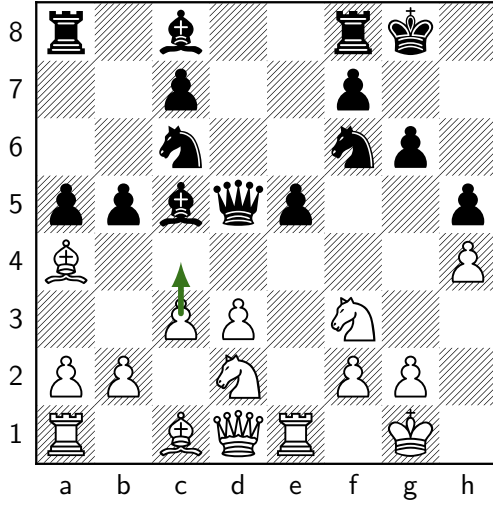
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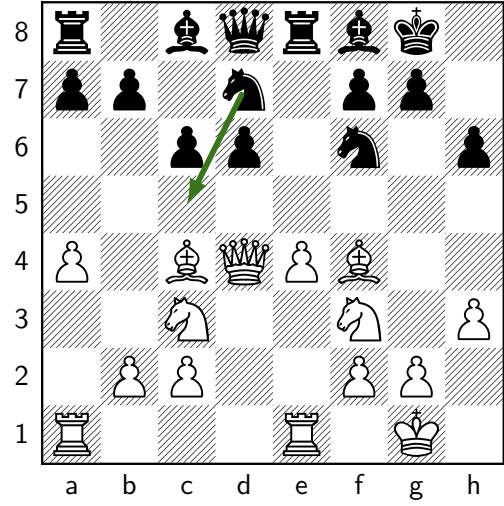
(d)

Figure 7: **Move 13, White: exd5.** (a) Our position; Melanie plays **exd5 (e4d5)**. Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2679` (ply 21) [structural distance  $d=49$ , QAP obj=63]: the master played **exd4**; under  $\mu$  our  $Pe_4$  plays the role of that game’s  $Pe_4$ ; cross-type roles: our  $Pc_3$  as their  $Ra_8$ , our  $Pa_2$  as their  $Bc_8$ ; transferred move **e4d5 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#14` (ply 22) [structural distance  $d=43$ , QAP obj=55]: the master played **exd5**; under  $\mu$  our  $Pe_4$  plays the role of that game’s  $Pe_4$ ; cross-type roles: our  $Qd1$  as their  $Ra_1$ , our  $ra_8$  as their  $Qd1$ ; transferred move **e4d5 (endorses this move)**. (d) Retrieved from `ccr1_top_p0001#2678` (ply 19) [structural distance  $d=96$ , QAP obj=52]: the master played **exd4**; under  $\mu$  our  $Pe_4$  plays the role of that game’s  $Pe_4$ ; cross-type roles: our  $pb_5$  as their  $Nb_8$ , our  $Pc_3$  as their  $Bc_8$ ; transferred move **e4d5 (endorses this move)**. Synthesis: 8 retrieved analogs transferred to **exd5** (summed QAP objective 366, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.

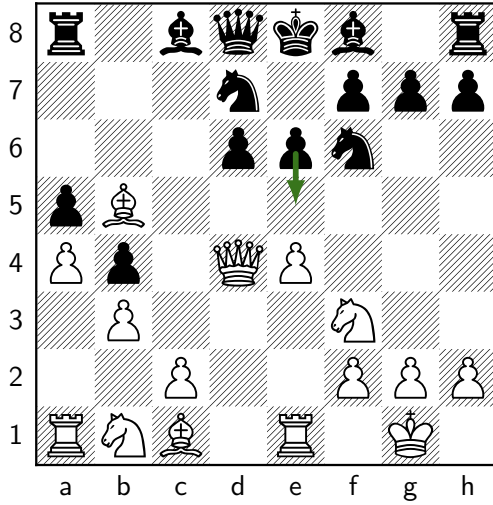




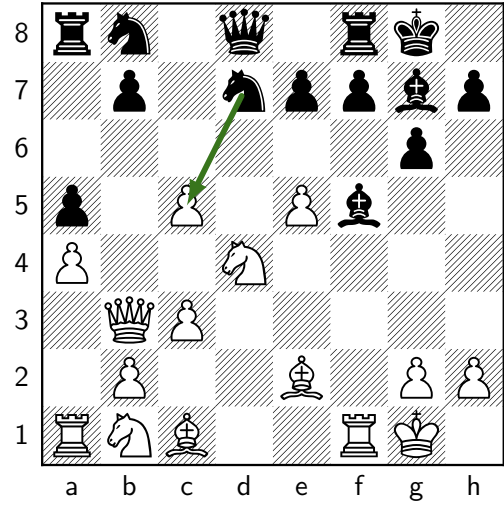
(a)



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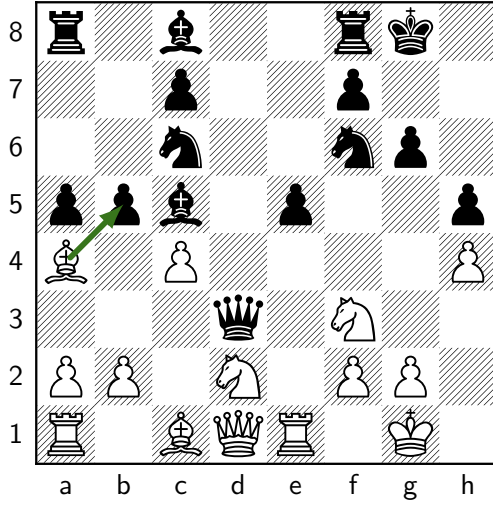


(c)

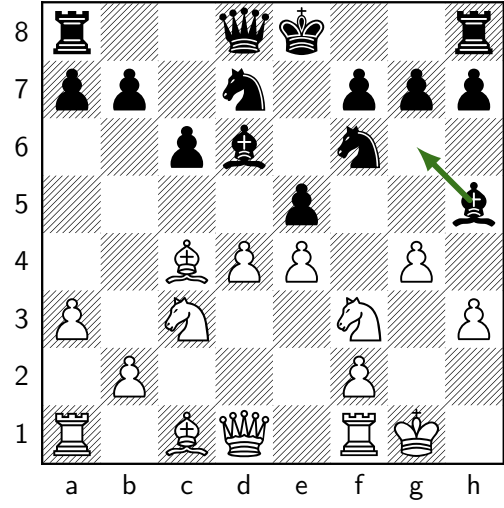


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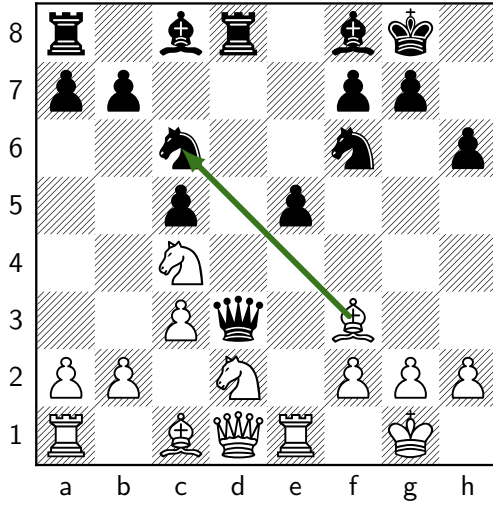
Figure 8: **Move 14, White: c4 (c3c4).** (a) Our position; Melanie plays c4 (c3c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2678` (ply 23) [structural distance  $d=69$ , QAP obj=91]: the master played Nc5; under  $\mu$  our  $Pc3$  plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Nd2 as their Qd8, our Pg2 as their Bf8; transferred move c3c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3571` (ply 21) [structural distance  $d=95$ , QAP obj=69]: the master played e5; under  $\mu$  our  $Pc3$  plays the role of that game’s Pe3; cross-type roles: our Ph4 as their Ra8, our pa5 as their Bc8; transferred move c3c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1304` (ply 25) [structural distance  $d=113$ , QAP obj=51]: the master played Nxc5; under  $\mu$  our  $Pc3$  plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Qd1 as their Nb8, our Pa2 as their Qd8; transferred move c3c4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to c4 (summed QAP objective 332, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



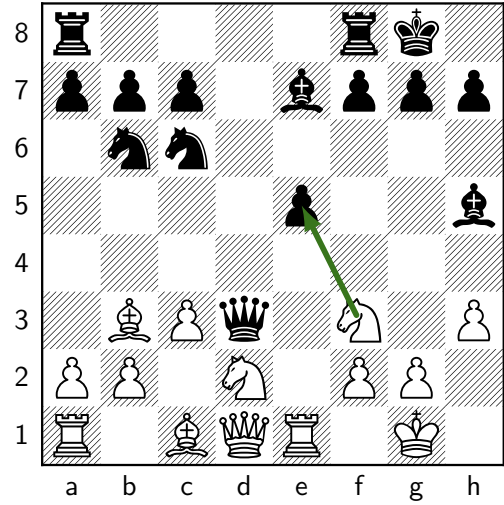
(a)



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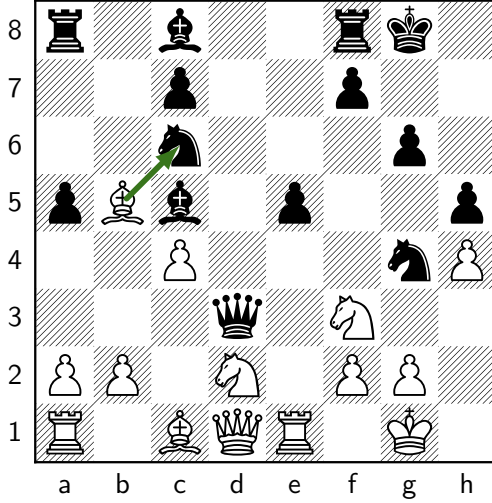


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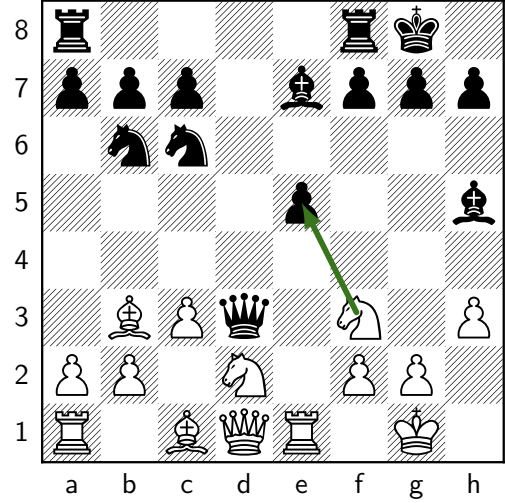


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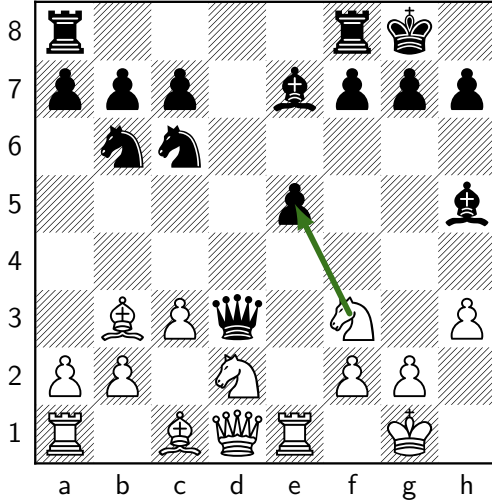
Figure 9: **Move 15, White: Bxb5.** (a) Our position; Melanie plays Bxb5 (a4b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#743` (ply 21) [structural distance  $d=133$ , QAP obj=49]: the master played Bg6; under  $\mu$  our Ba4 plays the role of that game’s Bh4; cross-type roles: our Qd1 as their Ra8, our Nf3 as their Qd8; transferred move a4b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2224` (ply 24) [structural distance  $d=114$ , QAP obj=122]: the master played Bxc6; under  $\mu$  our Nf3 plays the role of that game’s Bf3 (a **cross-type** correspondence); cross-type roles: our bc5 as their Ph2, our Nf3 as their Pc3; transferred move f3e5. (d) Retrieved from `ccr1_top_p0001#826` (ply 22) [structural distance  $d=106$ , QAP obj=116]: the master played Nxe5; under  $\mu$  our Nf3 plays the role of that game’s Nf3; cross-type roles: our Pc4 as their Bc1, our Bc1 as their Pb2; transferred move f3e5. Synthesis: 1 retrieved analog transferred to Bxb5 (summed QAP objective 49, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



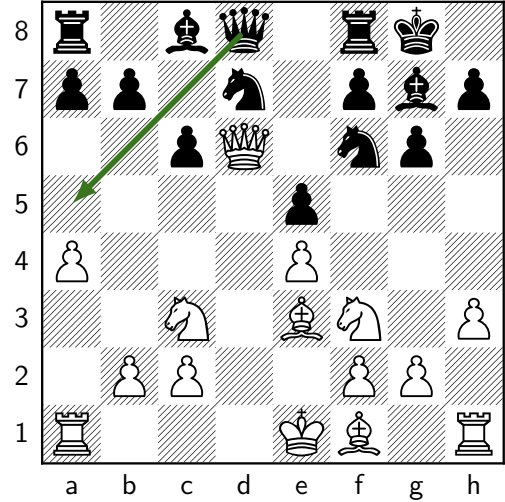
(a)



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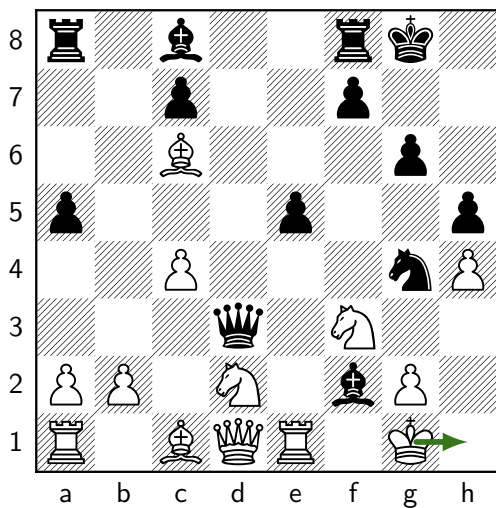


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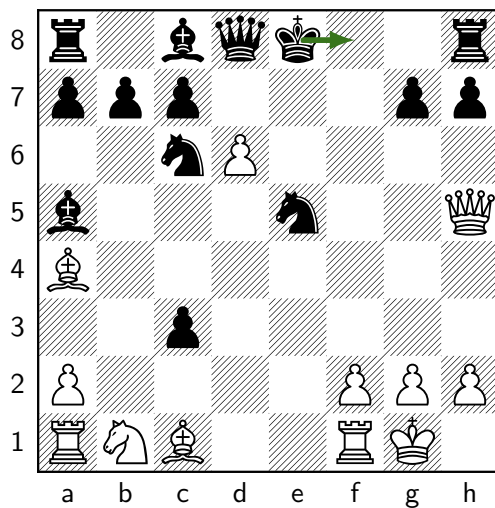


(d)

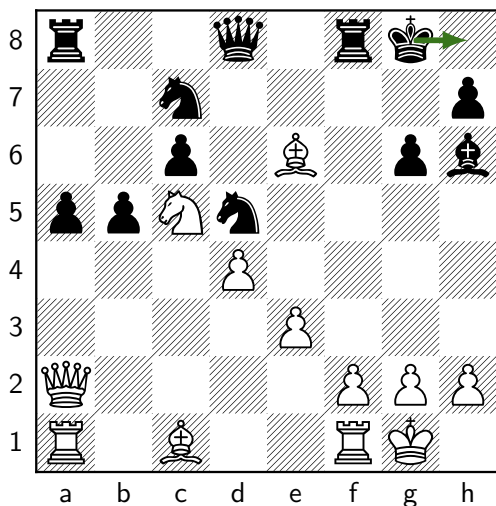
Figure 10: **Move 16, White: Bxc6.** (a) Our position; Melanie plays Bxc6 (b5c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#826` (ply 22) [structural distance  $d=95$ , QAP obj=156]: the master played Nxe5; under  $\mu$  our Nf3 plays the role of that game’s Nf3; cross-type roles: our Ph4 as their Ra1, our pc7 as their Bc1; transferred move f3g5. (c) Retrieved from `ccr1_top_p0001#827` (ply 22) [structural distance  $d=95$ , QAP obj=156]: the master played Nxe5; under  $\mu$  our Nf3 plays the role of that game’s Nf3; cross-type roles: our Ph4 as their Ra1, our pc7 as their Bc1; transferred move f3g5. (d) Retrieved from `ccr1_top_p0000#539` (ply 19) [structural distance  $d=103$ , QAP obj=142]: the master played Qa5; under  $\mu$  our Qd1 plays the role of that game’s Qd1; cross-type roles: our pc7 as their Bc8, our Nf3 as their Rf8; transferred move d1a4. Synthesis: no retrieved analog transferred to Bxc6 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



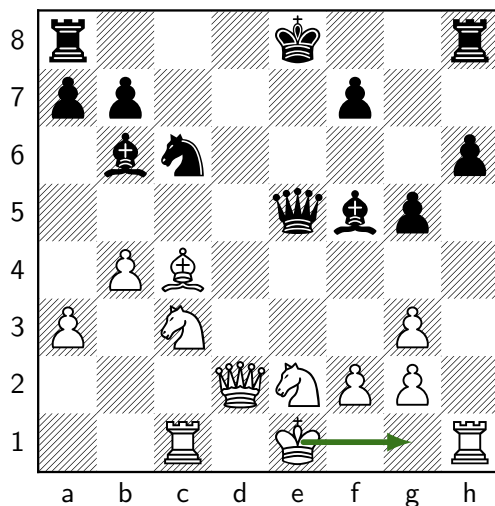
(a)



(b)

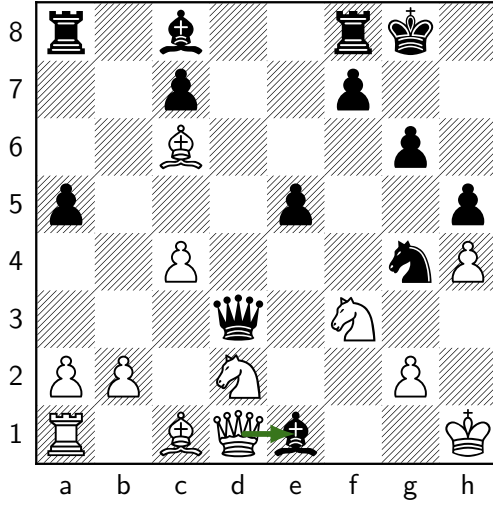


(c)

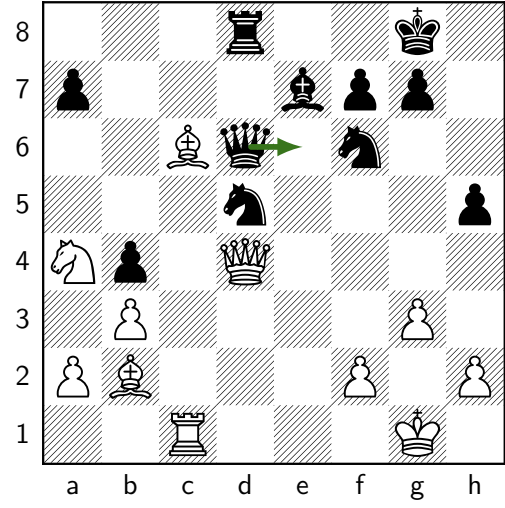


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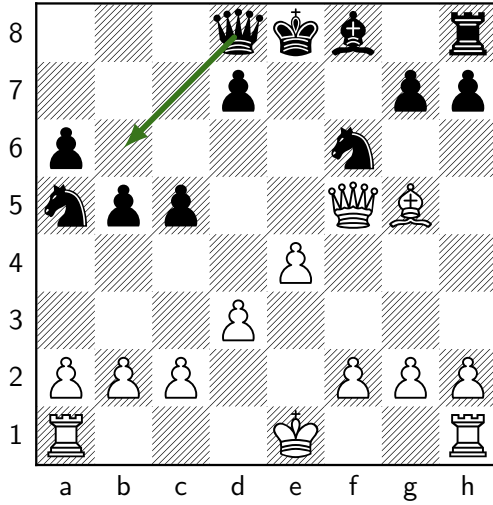
Figure 11: **Move 17, White: Kh1 (g1h1).** (a) Our position; Melanie plays Kh1 (g1h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1.top_p0003#1824` (ply 27) [structural distance  $d=131$ , QAP obj=41]: the master played Kf8; under  $\mu$  our Kg1 plays the role of that game's Ke1; cross-type roles: our ra8 as their Bc8, our Nf3 as their Qd8; transferred move g1h1 (**endorses this move**). (c) Retrieved from `ccr1.top_p0003#300` (ply 35) [structural distance  $d=126$ , QAP obj=40]: the master played Kh8; under  $\mu$  our Kg1 plays the role of that game's Kg1; cross-type roles: our Pg2 as their Ra8, our Pc4 as their Nc7; transferred move g1h1 (**endorses this move**). (d) Retrieved from `ccr1.top_p0000#516` (ply 38) [structural distance  $d=123$ , QAP obj=32]: the master played 0-0; under  $\mu$  our Kg1 plays the role of that game's Ke1; cross-type roles: our Bc6 as their Rh1, our pg6 as their bf5; transferred move g1h1 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kh1 (summed QAP objective 260, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



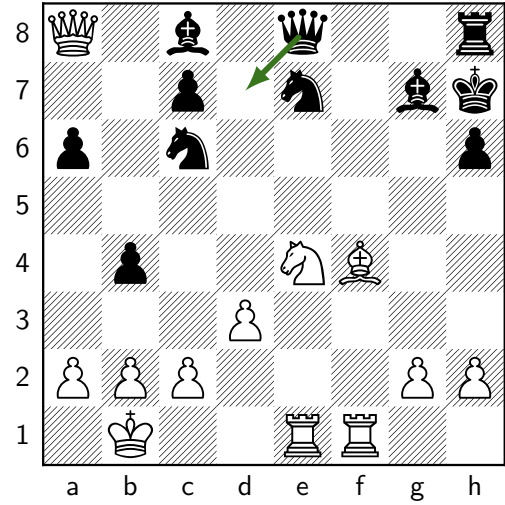
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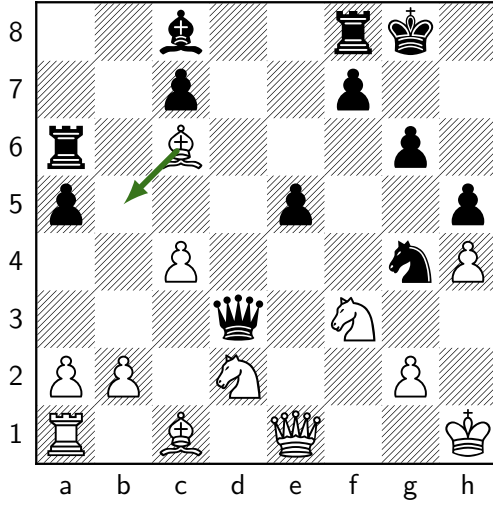


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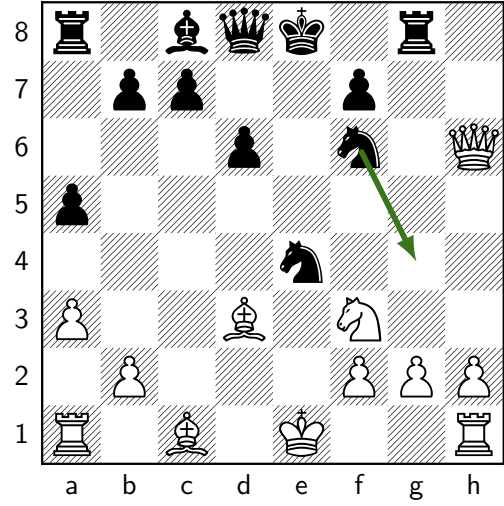


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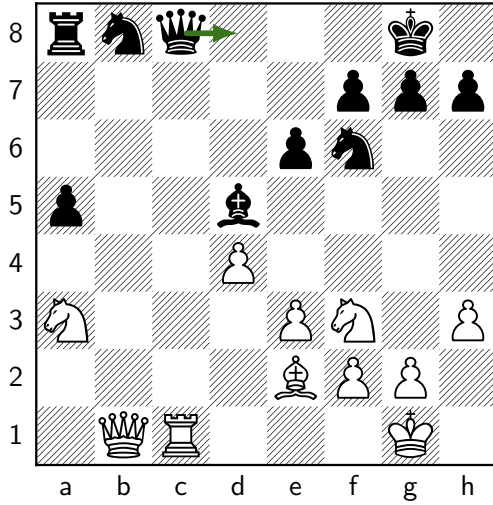
Figure 12: **Move 18, White: Qxe1.** (a) Our position; Melanie plays **Qxe1** (**d1e1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1717` (ply 47) [structural distance  $d=126$ , QAP obj=148]: the master played **Qe6**; under  $\mu$  our *Qd1* plays the role of that game’s *Qd3*; cross-type roles: our *Ph4* as their *Rd8*, our *ng4* as their *Pf7*; transferred move **d1e1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1908` (ply 35) [structural distance  $d=135$ , QAP obj=67]: the master played **Qb6**; under  $\mu$  our *Qd1* plays the role of that game’s *Qd1*; cross-type roles: our *Nf3* as their *Bf8*, our *Bc1* as their *Pg7*; transferred move **d1e1** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2149` (ply 47) [structural distance  $d=125$ , QAP obj=55]: the master played **Qd7**; under  $\mu$  our *Qd1* plays the role of that game’s *Qe1*; cross-type roles: our *Nd2* as their *Bc8*, our *Pg2* as their *Rh8*; transferred move **d1e1** (**endorses this move**). Synthesis: 8 retrieved analogs transferred to **Qxe1** (summed QAP objective 443, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



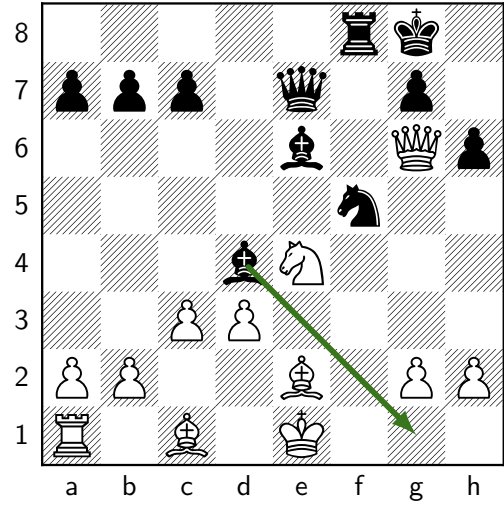
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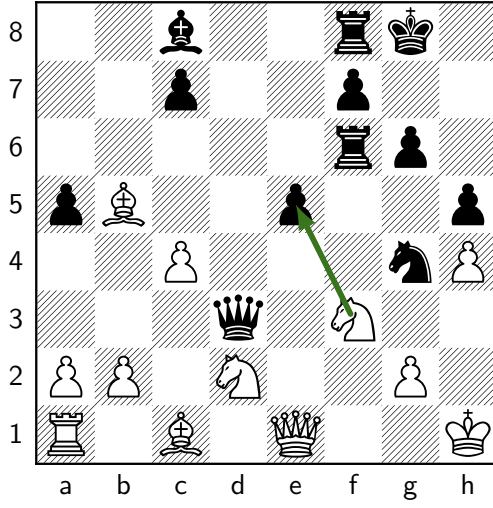


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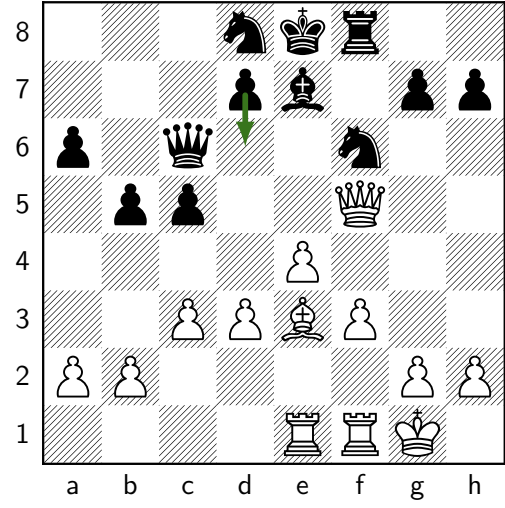


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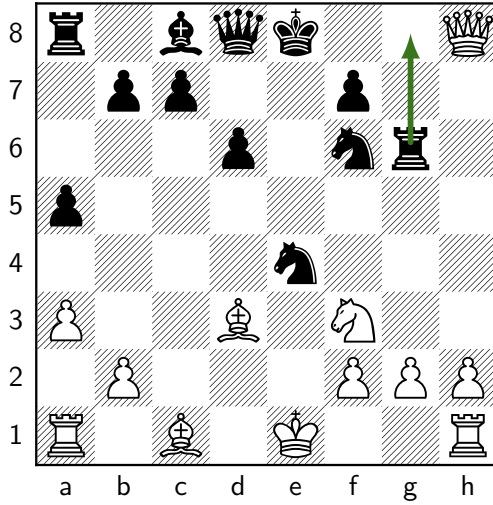
Figure 13: **Move 19, White: Bb5.** (a) Our position; Melanie plays Bb5 (c6b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#857` (ply 35) [structural distance  $d=105$ , QAP obj=47]: the master played Ng4; under  $\mu$  our Bc6 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Nf3 as their Ra8, our Ph4 as their Bc8; transferred move c6b5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3081` (ply 35) [structural distance  $d=115$ , QAP obj=21]: the master played Qd8; under  $\mu$  our Bc6 plays the role of that game’s Qc1 (a **cross-type** correspondence); cross-type roles: our pg6 as their Ra8, our Ra1 as their Nb8; transferred move c6b5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#261` (ply 35) [structural distance  $d=113$ , QAP obj=74]: the master played Bg1; under  $\mu$  our Bc6 plays the role of that game’s Bd5; cross-type roles: our Qe1 as their Rf8, our Ra1 as their Pa7; transferred move c6e8. Synthesis: 3 retrieved analogs transferred to Bb5 (summed QAP objective 114, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



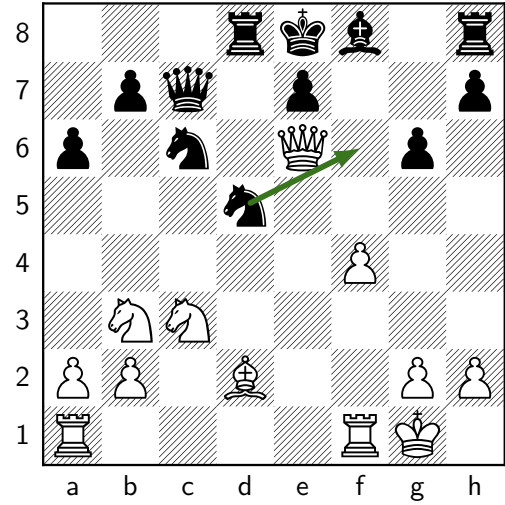
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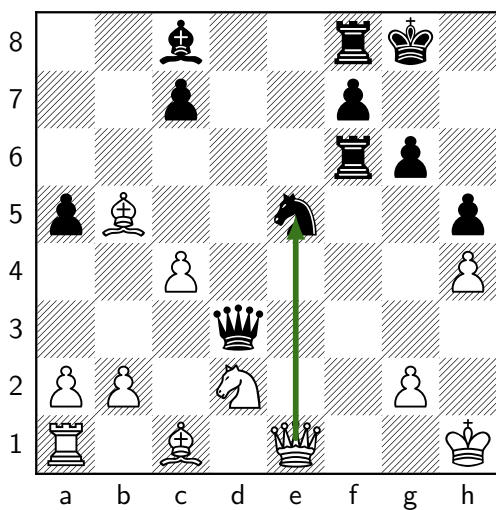
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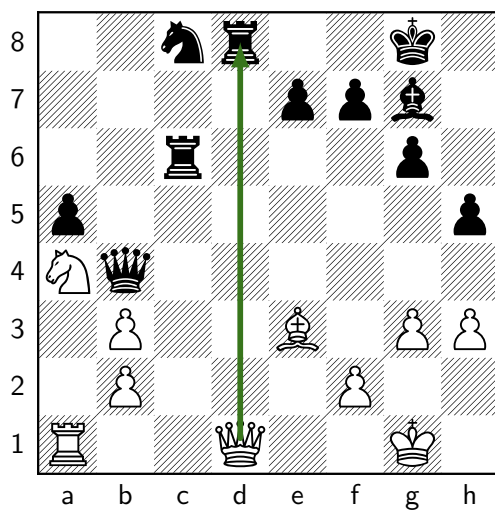
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Figure 14: **Move 20, White: Nxe5.** (a) Our position; Melanie plays Nxe5 (f3e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1908` (ply 47) [structural distance  $d=119$ , QAP obj=44]: the master played d6; under  $\mu$  our Nf3 plays the role of that game’s Pd2 (a **cross-type** correspondence); cross-type roles: our ph5 as their Nd8, our Qe1 as their Rf8; transferred move f3e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#857` (ply 29) [structural distance  $d=104$ , QAP obj=18]: the master played Rg8; under  $\mu$  our Nf3 plays the role of that game’s Rg3 (a **cross-type** correspondence); cross-type roles: our ph5 as their Ra8, our pa5 as their Bc8; transferred move f3e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1633` (ply 35) [structural distance  $d=114$ , QAP obj=114]: the master played Nf6; under  $\mu$  our Nd2 plays the role of that game’s Nd4; cross-type roles: our Bb5 as their Rd8, our Pa2 as their Rh8; transferred move d2e4. Synthesis: 2 retrieved analogs transferred to Nxe5 (summed QAP objective 62, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

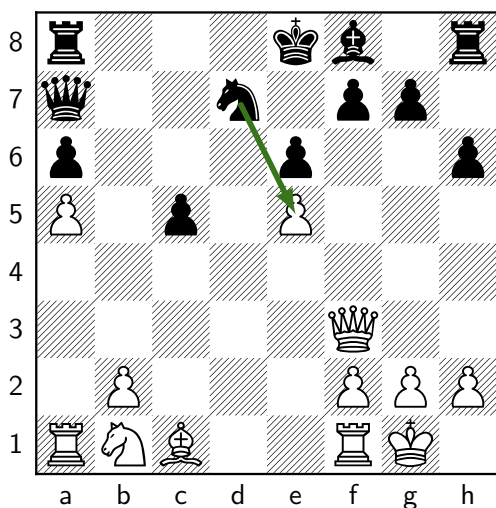




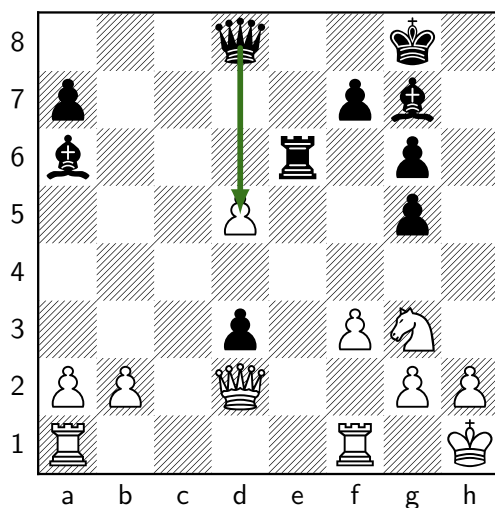
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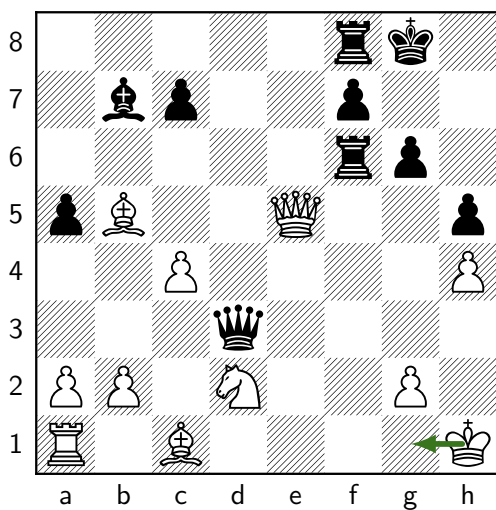
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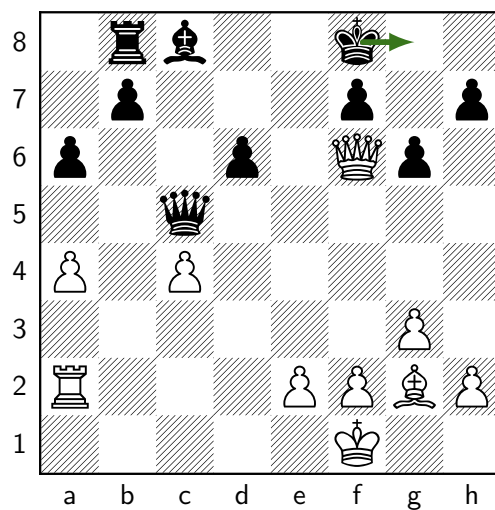
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Figure 15: **Move 21, White: Qxe5.** (a) Our position; Melanie plays Qxe5 (e1e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1770` (ply 52) [structural distance  $d=98$ , QAP obj=33]: the master played Qxd8+; under  $\mu$  our *Qe1* plays the role of that game’s *Qd1*; cross-type roles: our *Bc1* as their *Ra1*, our *bc8* as their *Pf2*; transferred move e1e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2897` (ply 33) [structural distance  $d=105$ , QAP obj=23]: the master played Nxe5; under  $\mu$  our *Qe1* plays the role of that game’s *Nd2* (a **cross-type** correspondence); cross-type roles: our *Nd2* as their *Ra8*, our *ne5* as their *Rh8*; transferred move e1e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#851` (ply 41) [structural distance  $d=107$ , QAP obj=14]: the master played Qxd5; under  $\mu$  our *Qe1* plays the role of that game’s *Qd1*; cross-type roles: our *ph5* as their *Bg7*, our *Nd2* as their *Re6*; transferred move e1e5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qxe5 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

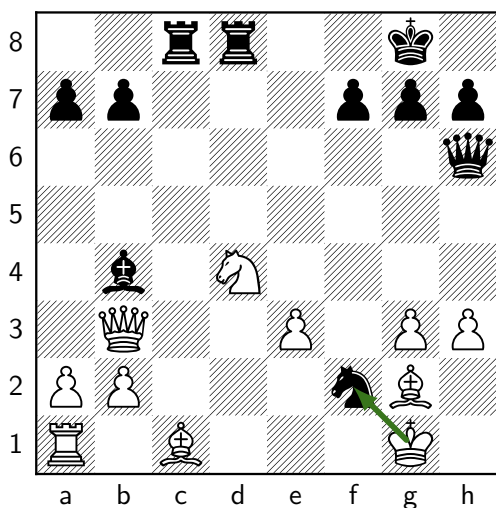




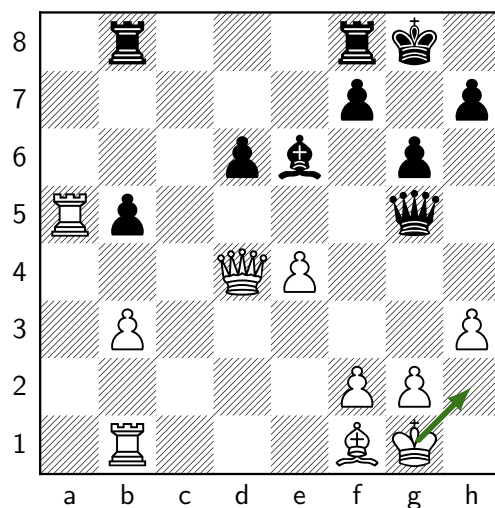
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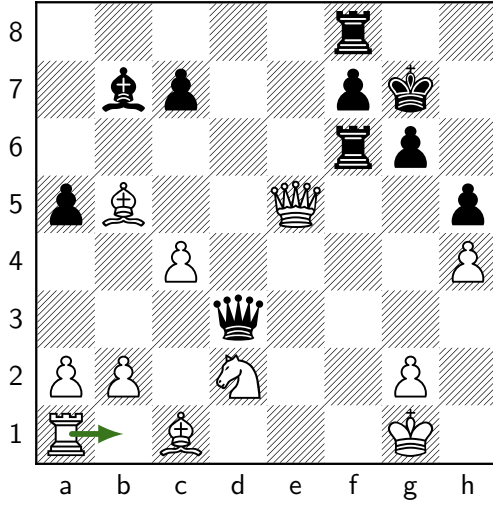


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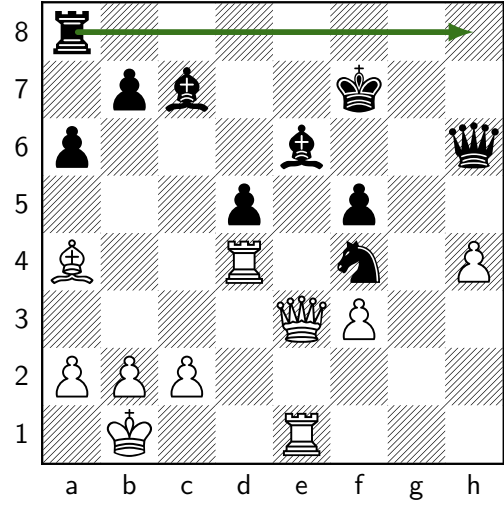


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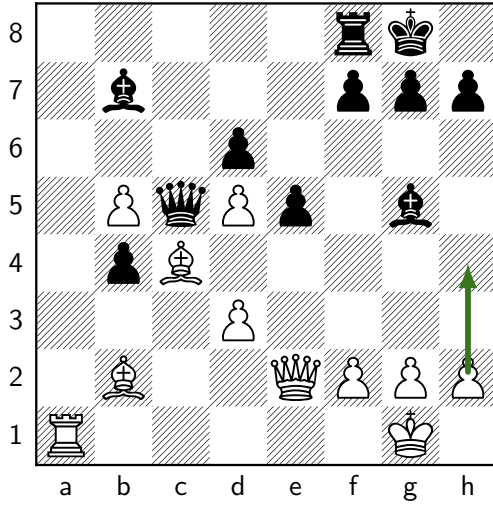
Figure 16: **Move 22, White: Kg1**. (a) Our position; Melanie plays Kg1 (**h1g1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2942` (ply 43) [structural distance  $d=84$ , QAP obj=40]: the master played Kg8; under  $\mu$  our Kh1 plays the role of that game's Kf1; cross-type roles: our rf6 as their Bc8, our Qe5 as their Pb7; transferred move h1g1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2852` (ply 46) [structural distance  $d=85$ , QAP obj=27]: the master played Kxf2; under  $\mu$  our Kh1 plays the role of that game's Kg1; cross-type roles: our Bc1 as their Ra1, our Ra1 as their Pa2; transferred move h1g1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1157` (ply 48) [structural distance  $d=85$ , QAP obj=27]: the master played Kh2; under  $\mu$  our Kh1 plays the role of that game's Kg1; cross-type roles: our Nd2 as their Pb3, our qd3 as their Ra5; transferred move h1g1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg1 (summed QAP objective 107, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



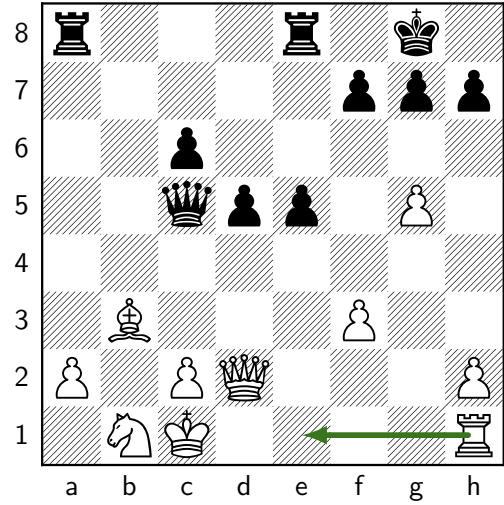
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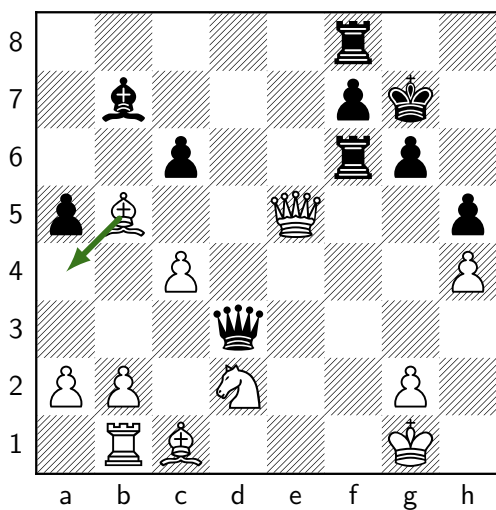


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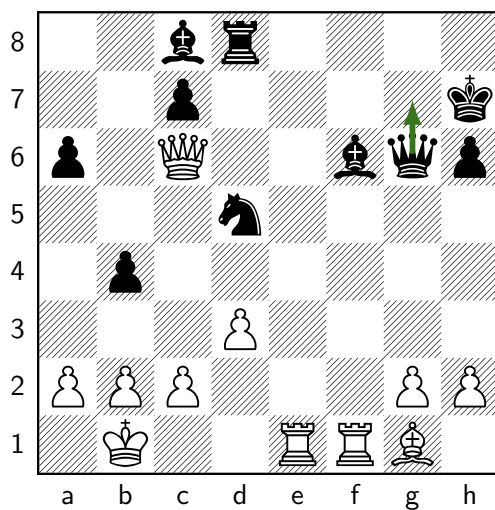


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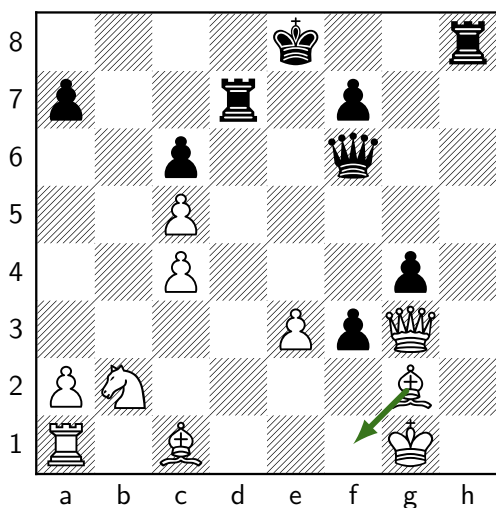
Figure 17: **Move 23, White: Rb1.** (a) Our position; Melanie plays Rb1 (a1b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1849` (ply 57) [structural distance  $d=91$ , QAP obj=37]: the master played Rh8; under  $\mu$  our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *Bc1* as their *Pb7*, our *qd3* as their *Pa6*; transferred move a1b1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1506` (ply 44) [structural distance  $d=96$ , QAP obj=22]: the master played h4; under  $\mu$  our *Ra1* plays the role of that game’s *Ph2* (a **cross-type** correspondence); cross-type roles: our *Bc1* as their *Ra1*, our *rf6* as their *Bc4*; transferred move a1b1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1726` (ply 58) [structural distance  $d=96$ , QAP obj=7]: the master played Re1; under  $\mu$  our *Ra1* plays the role of that game’s *Rh1*; cross-type roles: our *rf6* as their *pd5*, our *rf8* as their *pg7*; transferred move a1b1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rb1 (summed QAP objective 73, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



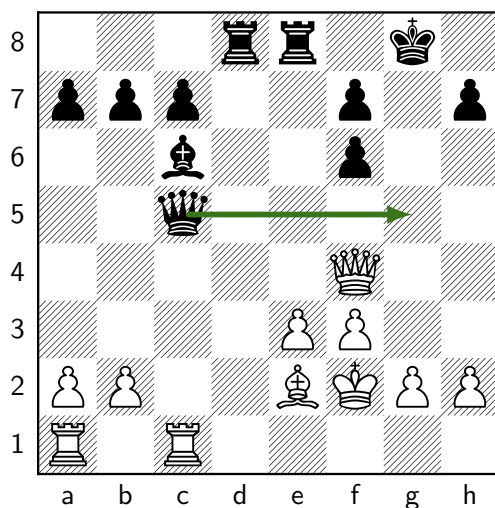
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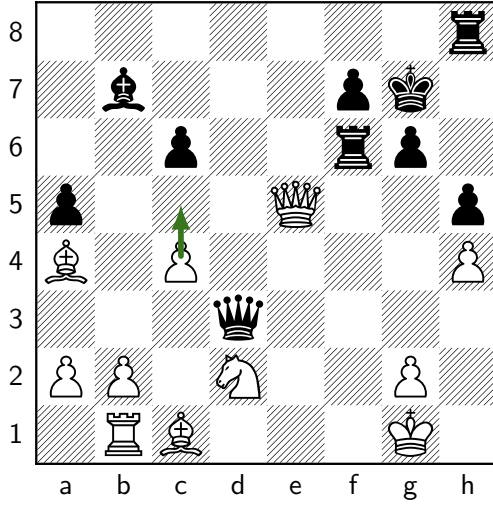


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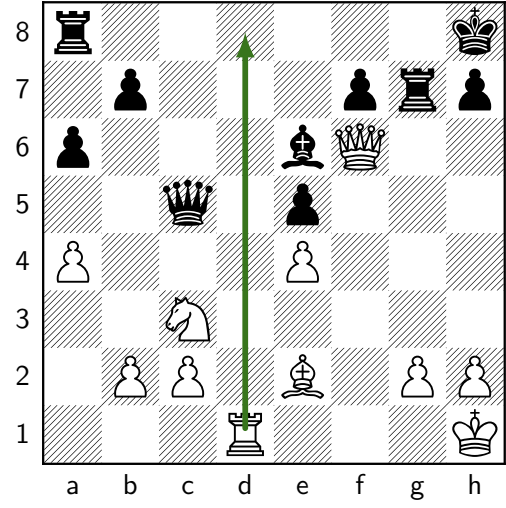


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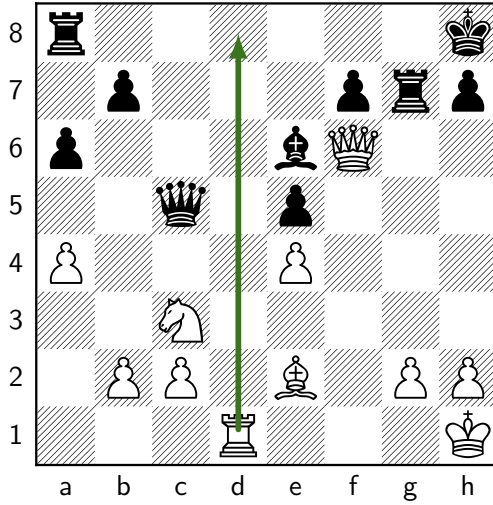
Figure 18: **Move 24, White: Ba4.** (a) Our position; Melanie plays Ba4 (b5a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2149` (ply 63) [structural distance  $d=89$ , QAP obj=29]: the master played Qg7; under  $\mu$  our Bb5 plays the role of that game’s Qg3 (a **cross-type** correspondence); cross-type roles: our ph5 as their Rd8, our Rb1 as their Bf6; transferred move b5a4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#445` (ply 48) [structural distance  $d=111$ , QAP obj=16]: the master played Bf1; under  $\mu$  our Bb5 plays the role of that game’s Bg2; cross-type roles: our Nd2 as their Pa2, our rf6 as their Nb2; transferred move b5a4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#19` (ply 33) [structural distance  $d=112$ , QAP obj=11]: the master played Qg5; under  $\mu$  our Bb5 plays the role of that game’s Qc4 (a **cross-type** correspondence); cross-type roles: our pf7 as their Rd8, our qd3 as their Re8; transferred move b5a4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ba4 (summed QAP objective 56, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



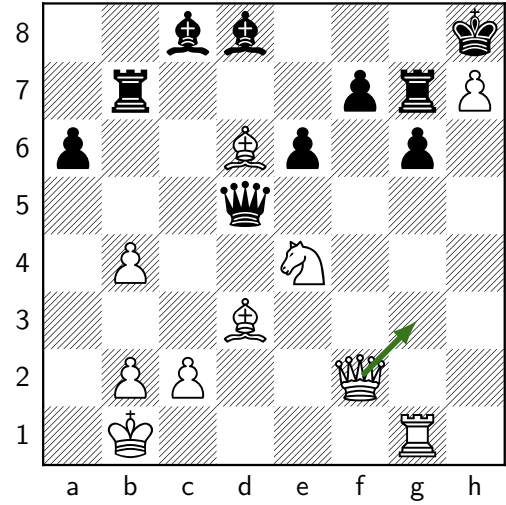
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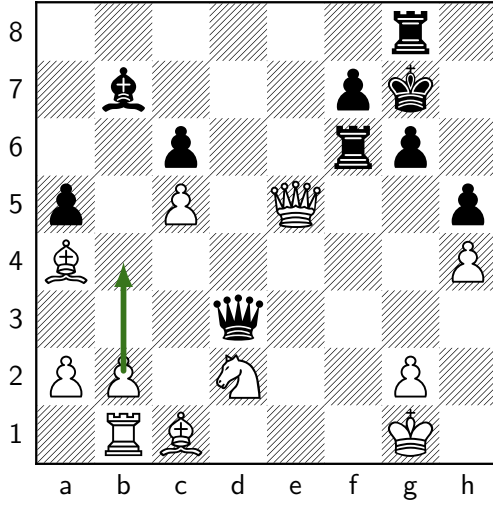


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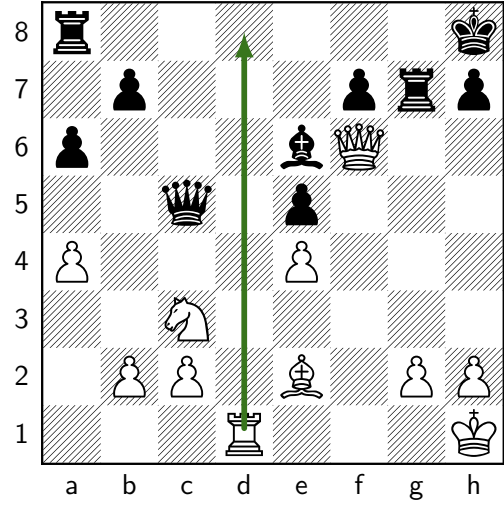


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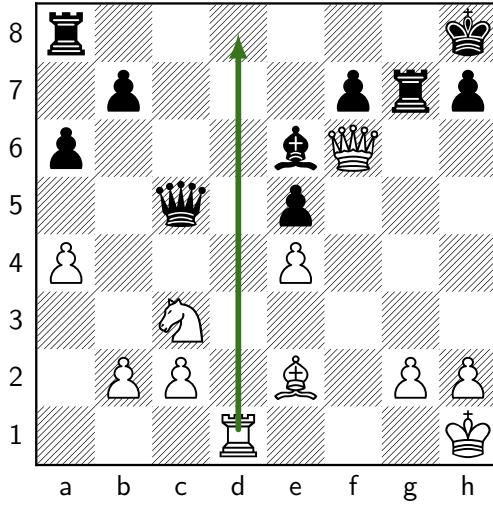
Figure 19: **Move 25, White: c5.** (a) Our position; Melanie plays c5 (c4c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#244` (ply 44) [structural distance  $d=87$ , QAP obj=80]: the master played Rd8+; under  $\mu$  our  $Pc4$  plays the role of that game’s  $Rd1$  (a **cross-type** correspondence); cross-type roles: our  $Nd2$  as their  $Pb2$ , our  $Rb1$  as their  $Nc3$ ; transferred move c4c5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2472` (ply 44) [structural distance  $d=87$ , QAP obj=80]: the master played Rd8+; under  $\mu$  our  $Pc4$  plays the role of that game’s  $Rd1$  (a **cross-type** correspondence); cross-type roles: our  $Nd2$  as their  $Pb2$ , our  $Rb1$  as their  $Nc3$ ; transferred move c4c5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1128` (ply 60) [structural distance  $d=111$ , QAP obj=40]: the master played Qg3; under  $\mu$  our  $Pc4$  plays the role of that game’s  $Qf2$  (a **cross-type** correspondence); cross-type roles: our  $Qe5$  as their  $Rg1$ , our  $Bc1$  as their  $Pb2$ ; transferred move c4c5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to c5 (summed QAP objective 229, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



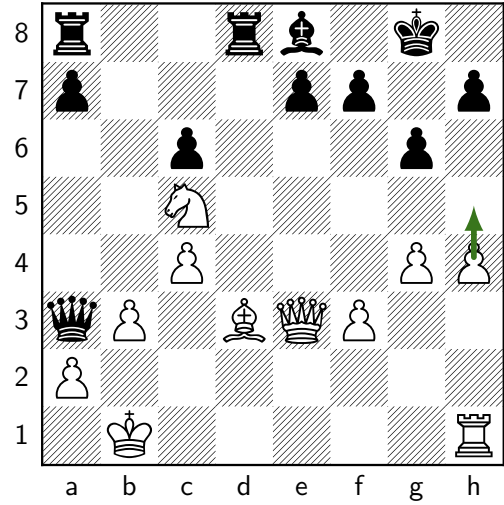
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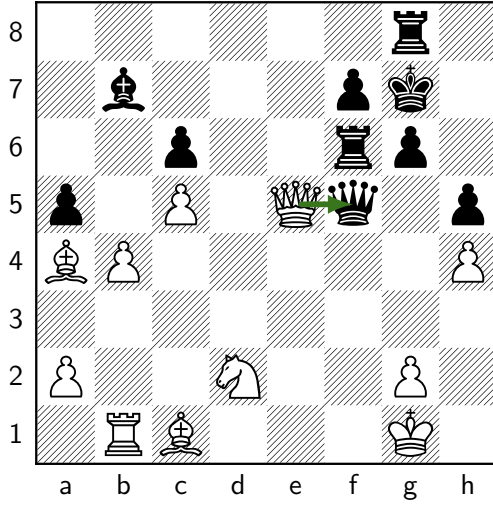


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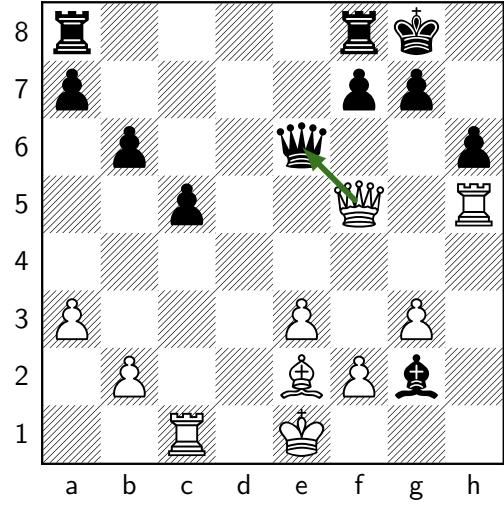


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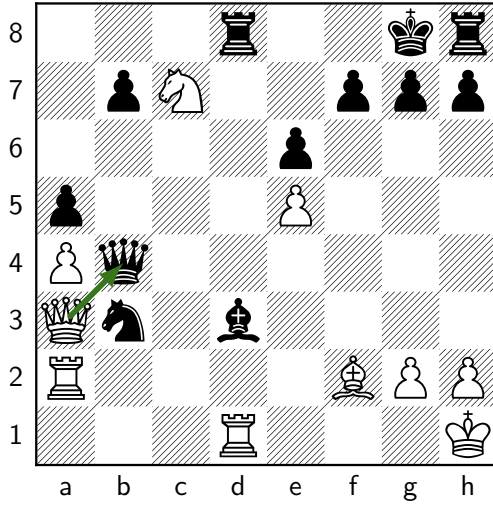
Figure 20: **Move 26, White: b4.** (a) Our position; Melanie plays b4 (b2b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#244` (ply 44) [structural distance  $d=97$ , QAP obj=79]: the master played  $Rd8+$ ; under  $\mu$  our  $Pb2$  plays the role of that game’s  $Rd1$  (a **cross-type** correspondence); cross-type roles: our  $Nd2$  as their  $Pb2$ , our  $pc6$  as their  $Be2$ ; transferred move b2b4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2472` (ply 44) [structural distance  $d=97$ , QAP obj=79]: the master played  $Rd8+$ ; under  $\mu$  our  $Pb2$  plays the role of that game’s  $Rd1$  (a **cross-type** correspondence); cross-type roles: our  $Nd2$  as their  $Pb2$ , our  $pc6$  as their  $Be2$ ; transferred move b2b4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#340` (ply 48) [structural distance  $d=97$ , QAP obj=12]: the master played  $h5$ ; under  $\mu$  our  $Pb2$  plays the role of that game’s  $Ph4$ ; cross-type roles: our  $qd3$  as their  $Bd3$ , our  $Ph4$  as their  $Qe3$ ; transferred move b2b4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to b4 (summed QAP objective 170, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



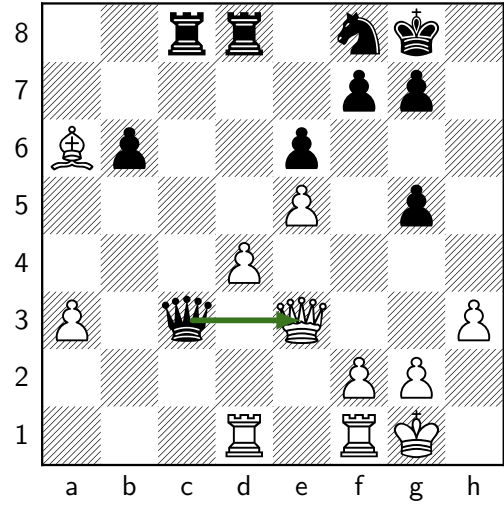
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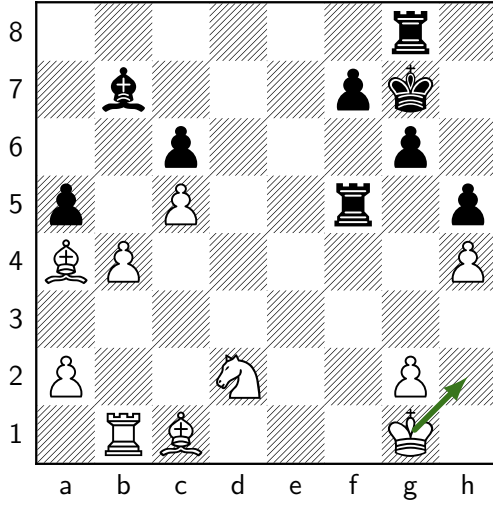


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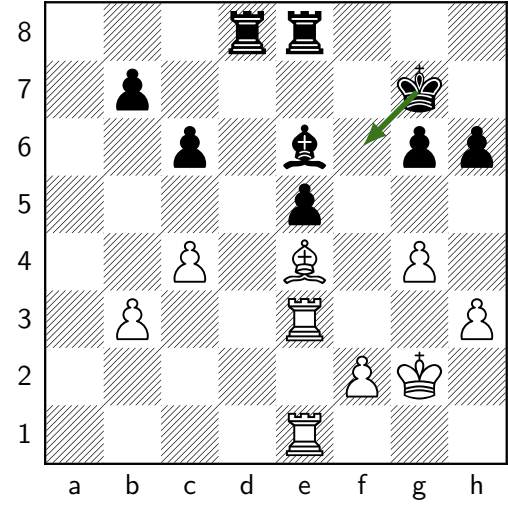


(d)

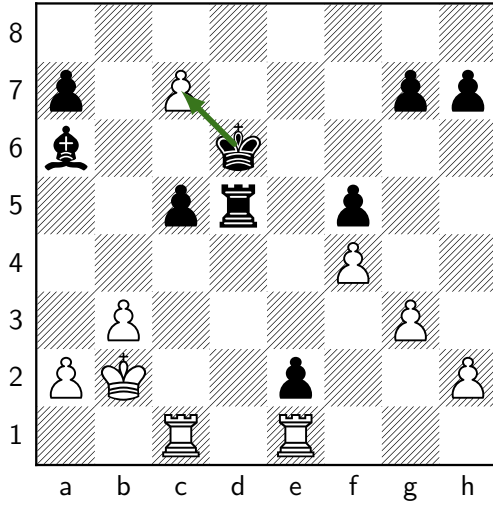
Figure 21: **Move 27, White: Qxf5**. (a) Our position; Melanie plays Qxf5 (e5f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1965` (ply 42) [structural distance  $d=86$ , QAP obj=67]: the master played Qxe6; under  $\mu$  our Qe5 plays the role of that game’s Qf5; cross-type roles: our Ba4 as their Rc1, our bb7 as their Pb2; transferred move e5f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#251` (ply 46) [structural distance  $d=84$ , QAP obj=66]: the master played Qxb4; under  $\mu$  our Qe5 plays the role of that game’s Qa3; cross-type roles: our Pb4 as their Rd1, our Pc5 as their Ra2; transferred move e5f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1771` (ply 49) [structural distance  $d=80$ , QAP obj=35]: the master played Qxe3; under  $\mu$  our Qe5 plays the role of that game’s Qc6; cross-type roles: our Pc5 as their Rc8, our ph5 as their Rd8; transferred move e5f5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qxf5 (summed QAP objective 168, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



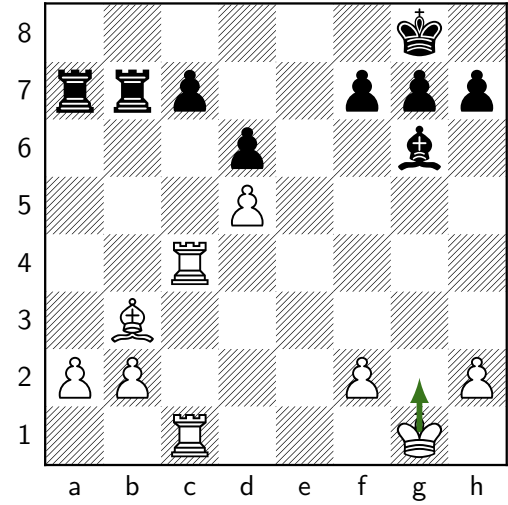
(a)



(b)



(c)



(d)

Figure 22: **Move 28, White: Kh2.** (a) Our position; Melanie plays Kh2 (**g1h2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3474` (ply 55) [structural distance  $d=50$ , QAP obj=32]: the master played Kf6; under  $\mu$  our *Kg1* plays the role of that game’s *Kg2*; cross-type roles: our *Pa2* as their *Re8*, our *Nd2* as their *Ph6*; transferred move **g1h2 (endorses this move)**. (c) Retrieved from `ccrl_top_p0000#3081` (ply 55) [structural distance  $d=54$ , QAP obj=26]: the master played Kxc7; under  $\mu$  our *Kg1* plays the role of that game’s *Kd3*; cross-type roles: our *Rb1* as their *Pa7*, our *Nd2* as their *pc7*; transferred move **g1h2 (endorses this move)**. (d) Retrieved from `ccrl_top_p0003#16` (ply 56) [structural distance  $d=54$ , QAP obj=22]: the master played Kg2; under  $\mu$  our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *pf7* as their *Rc1*, our *Rb1* as their *Pf2*; transferred move **g1h2 (endorses this move)**. Synthesis: 6 retrieved analogs transferred to Kh2 (summed QAP objective 138, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



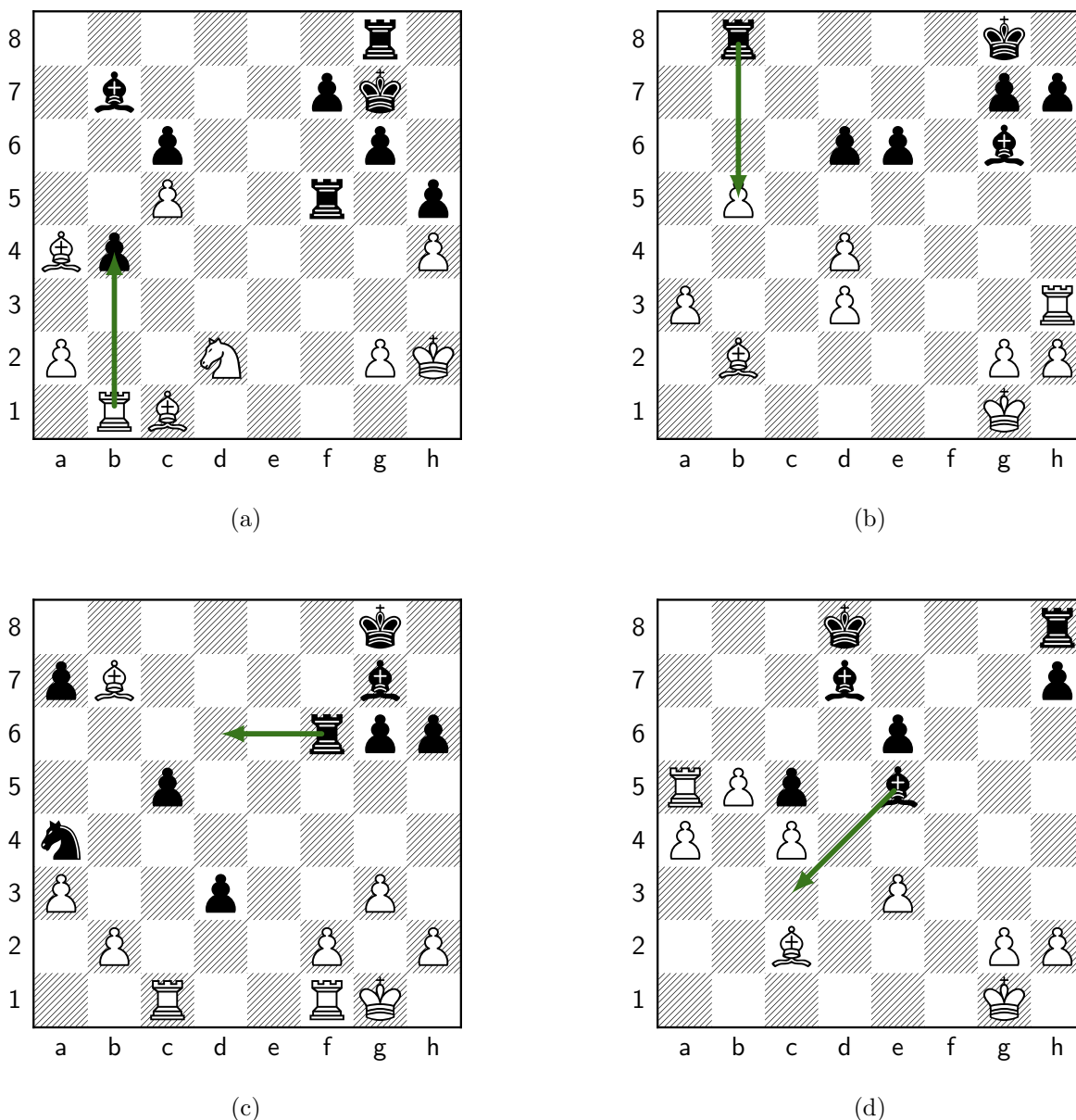
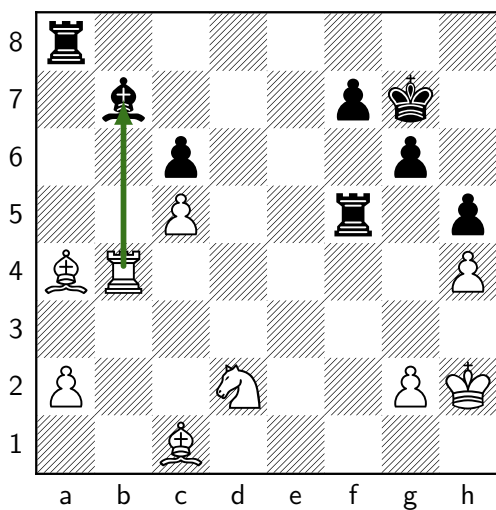
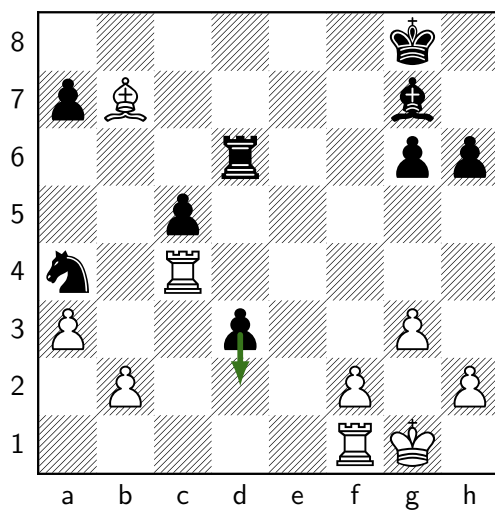


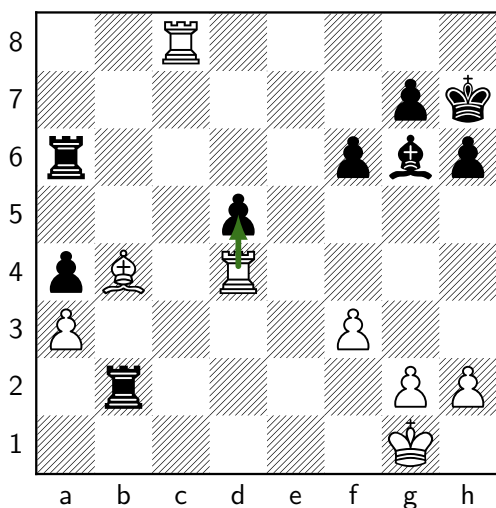
Figure 23: **Move 29, White: Rxb4**. (a) Our position; Melanie plays Rxb4 (b1b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1918` (ply 63) [structural distance  $d=50$ , QAP obj=55]: the master played Rxb5; under  $\mu$  our *Rb1* plays the role of that game’s *Rb1*; cross-type roles: our *Bc1* as their *pa3*, our *Nd2* as their *pg2*; transferred move b1b4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3191` (ply 47) [structural distance  $d=48$ , QAP obj=53]: the master played Rd6; under  $\mu$  our *Rb1* plays the role of that game’s *Rf3*; cross-type roles: our *ph5* as their *bb7*, our *Ph4* as their *Bg7*; transferred move b1b4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3142` (ply 57) [structural distance  $d=56$ , QAP obj=41]: the master played Bc3; under  $\mu$  our *Rb1* plays the role of that game’s *Be4* (a **cross-type** correspondence); cross-type roles: our *Bc1* as their *Rh8*, our *Nd2* as their *Pe6*; transferred move b1b4 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Rxb4 (summed QAP objective 327, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



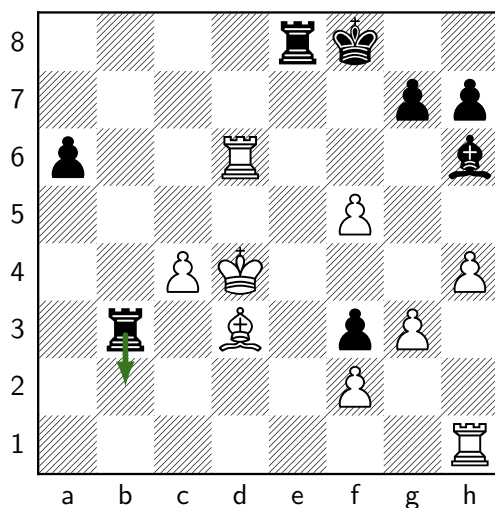
(a)



(b)



(c)



(d)

Figure 24: **Move 30, White: Rxb7**. (a) Our position; Melanie plays Rxb7 (b4b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3191` (ply 49) [structural distance  $d=50$ , QAP obj=37]: the master played d2; under  $\mu$  our *Rb4* plays the role of that game’s *Pd6* (a **cross-type** correspondence); cross-type roles: our *Bc1* as their *Pa7*, our *rf5* as their *bb7*; transferred move b4b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2478` (ply 64) [structural distance  $d=50$ , QAP obj=24]: the master played Rxd5; under  $\mu$  our *Rb4* plays the role of that game’s *Rd4*; cross-type roles: our *rf5* as their *Ph2*, our *bb7* as their *pd5*; transferred move b4b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2365` (ply 65) [structural distance  $d=46$ , QAP obj=23]: the master played Rb2; under  $\mu$  our *Rb4* plays the role of that game’s *Rb6*; cross-type roles: our *ph5* as their *Re8*, our *Bc1* as their *ph4*; transferred move b4b7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Rxb7 (summed QAP objective 135, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

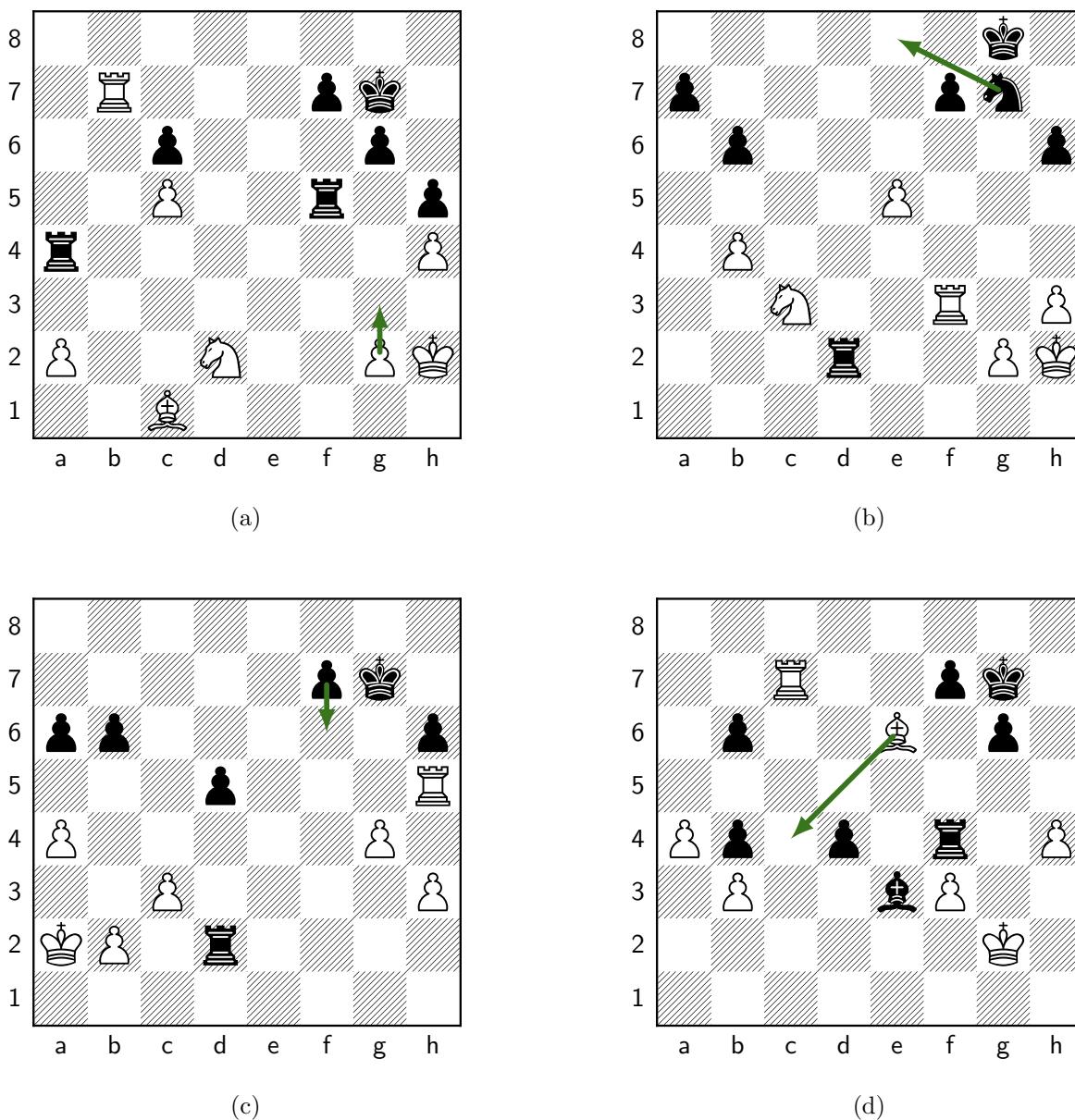
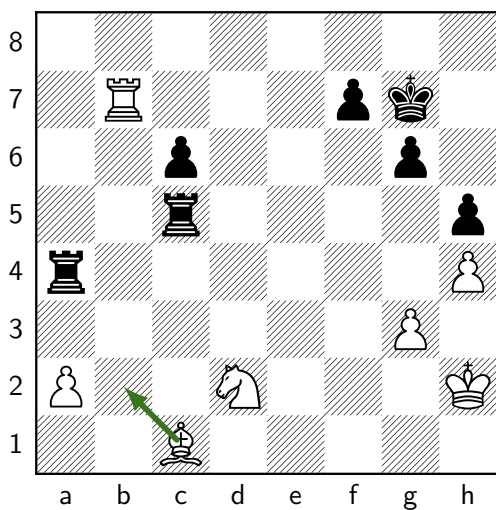
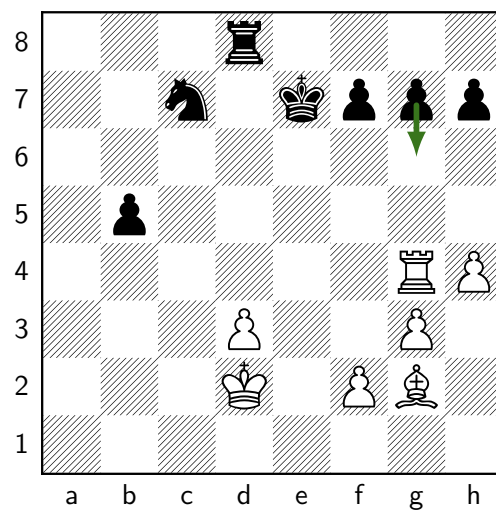


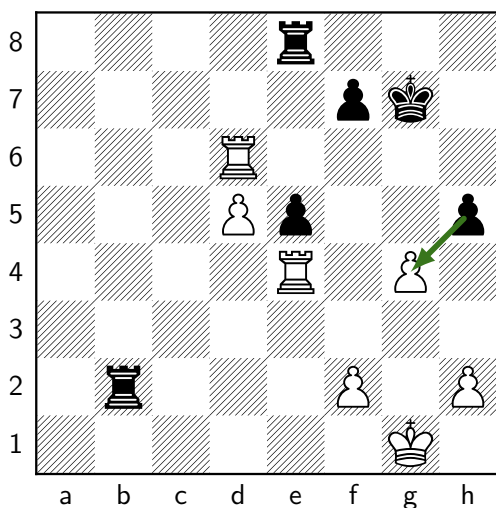
Figure 25: **Move 31, White: g3.** (a) Our position; Melanie plays g3 (g2g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1993` (ply 73) [structural distance  $d=41$ , QAP obj=39]: the master played Ne8; under  $\mu$  our Pg2 plays the role of that game’s Ng2 (a **cross-type** correspondence); cross-type roles: our Nd2 as their Pa7, our ra4 as their Pb6; transferred move g2g3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2617` (ply 77) [structural distance  $d=42$ , QAP obj=28]: the master played f6; under  $\mu$  our Pg2 plays the role of that game’s Pf2; cross-type roles: our rf5 as their Pb6, our Nd2 as their pg4; transferred move g2g3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1703` (ply 76) [structural distance  $d=42$ , QAP obj=22]: the master played Bc4; under  $\mu$  our Pg2 plays the role of that game’s Be6 (a **cross-type** correspondence); cross-type roles: our Nd2 as their Pb3, our pg6 as their be3; transferred move g2g3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to g3 (summed QAP objective 114, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



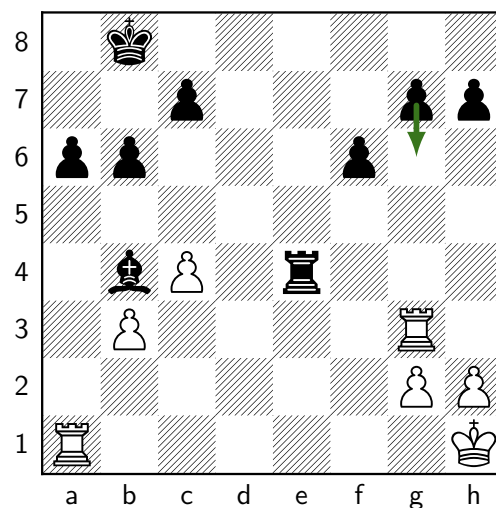
(a)



(b)

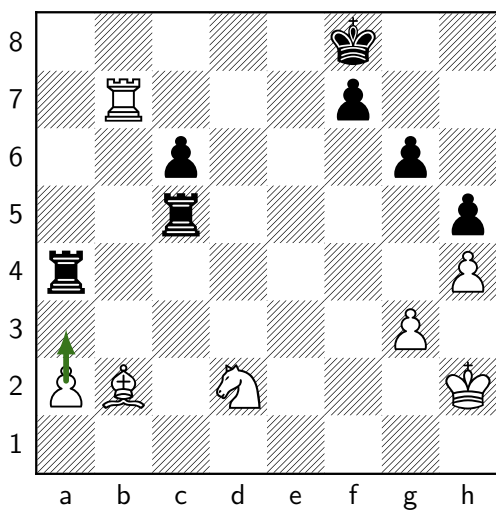


(c)

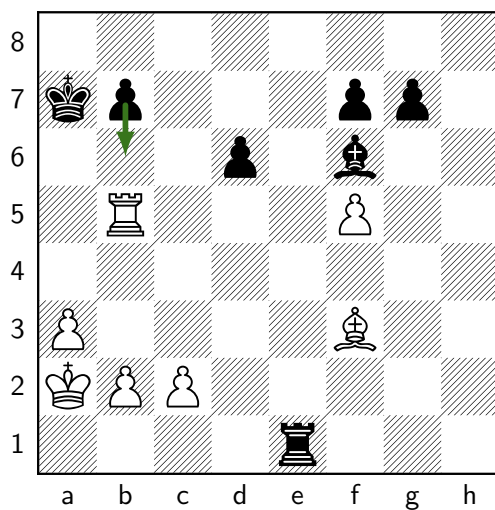


(d)

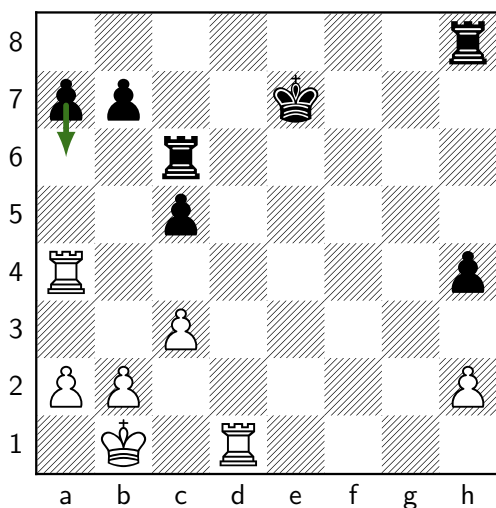
Figure 26: **Move 32, White: Bb2+.** (a) Our position; Melanie plays Bb2+ (c1b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3880` (ply 79) [structural distance  $d=40$ , QAP obj=36]: the master played g6; under  $\mu$  our *Bc1* plays the role of that game’s *Pg2* (a **cross-type** correspondence); cross-type roles: our *Pg3* as their *Nc7*, our *Nd2* as their *Ph7*; transferred move c1b2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1863` (ply 61) [structural distance  $d=38$ , QAP obj=30]: the master played hxg4; under  $\mu$  our *Bc1* plays the role of that game’s *Ph4* (a **cross-type** correspondence); cross-type roles: our *Pg3* as their *Re8*, our *pg6* as their *rd6*; transferred move c1b2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2780` (ply 65) [structural distance  $d=35$ , QAP obj=24]: the master played g6; under  $\mu$  our *Bc1* plays the role of that game’s *Pg2* (a **cross-type** correspondence); cross-type roles: our *Nd2* as their *pb3*; transferred move c1b2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bb2+ (summed QAP objective 90, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



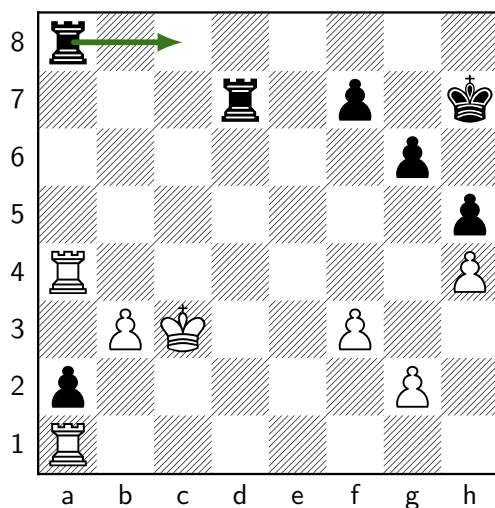
(a)



(b)

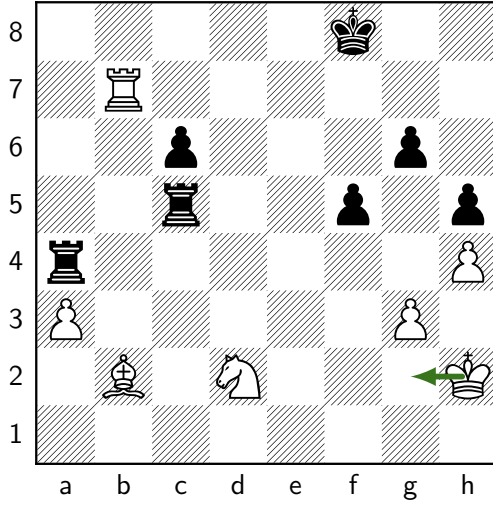


(c)

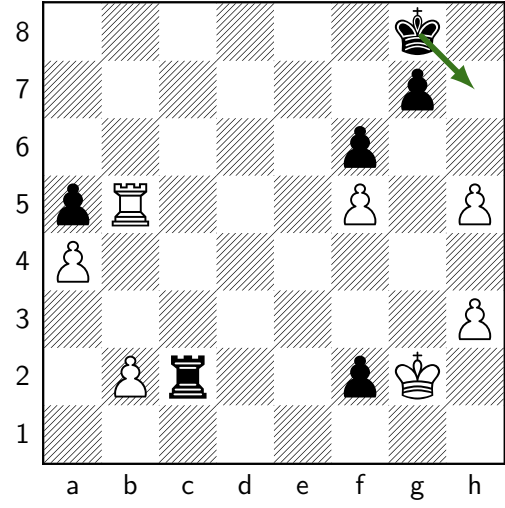


(d)

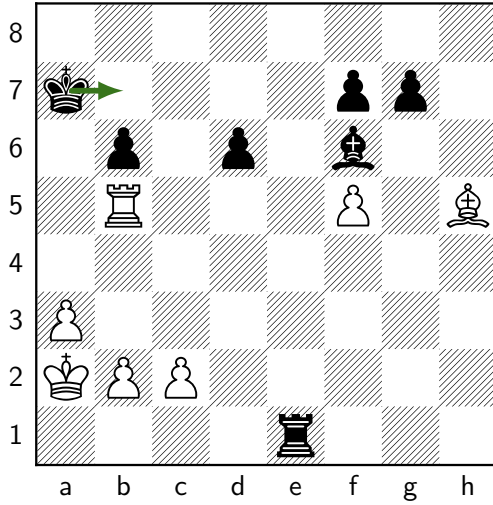
Figure 27: **Move 33, White: a3.** (a) Our position; Melanie plays a3 (a2a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3559` (ply 69) [structural distance  $d=29$ , QAP obj=36]: the master played b6; under  $\mu$  our *Pa2* plays the role of that game’s *Pb2*; cross-type roles: our *Bb2* as their *Pg7*, our *Rb7* as their *Bf6*; transferred move a2a3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1114` (ply 59) [structural distance  $d=31$ , QAP obj=19]: the master played a6; under  $\mu$  our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *Bb2* as their *Pb7*, our *Nd2* as their *pa2*; transferred move a2a3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#664` (ply 73) [structural distance  $d=31$ , QAP obj=15]: the master played Rc8+; under  $\mu$  our *Pa2* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Bb2* as their *Pf7*, our *Nd2* as their *pb3*; transferred move a2a3 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to a3 (summed QAP objective 110, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



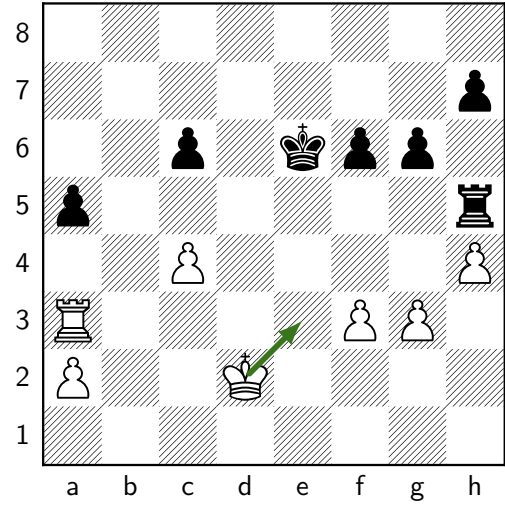
(a)



(b)

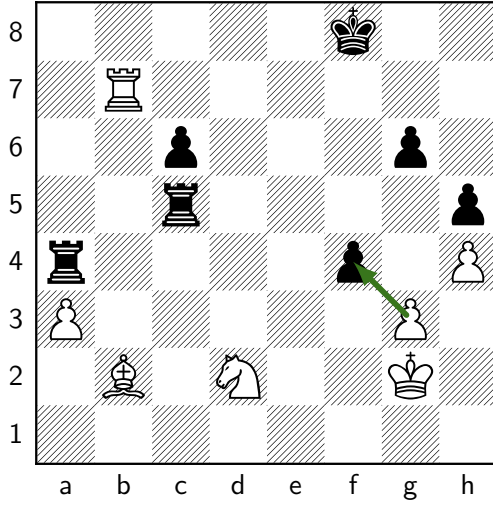


(c)

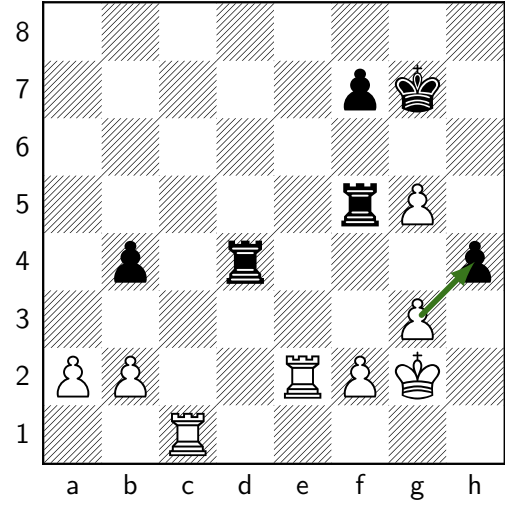


(d)

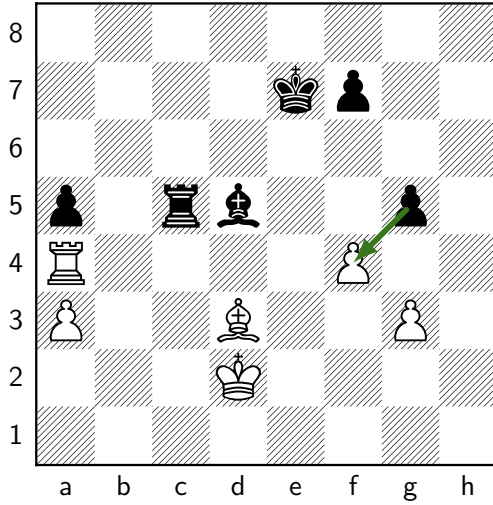
Figure 28: **Move 34, White: Kg2.** (a) Our position; Melanie plays Kg2 (**h2g2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3620` (ply 67) [structural distance  $d=31$ , QAP obj=29]: the master played Kh7; under  $\mu$  our Kh2 plays the role of that game’s Kg1; cross-type roles: our Bb2 as their pa4, our rc5 as their ph3; transferred move h2g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3559` (ply 71) [structural distance  $d=32$ , QAP obj=20]: the master played Kb7; under  $\mu$  our Kh2 plays the role of that game’s Ka2; cross-type roles: our rc5 as their Pf7, our ph5 as their Bf6; transferred move h2g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#932` (ply 76) [structural distance  $d=32$ , QAP obj=20]: the master played Ke3; under  $\mu$  our Kh2 plays the role of that game’s Kd2; cross-type roles: our ph5 as their Ra3, our Nd2 as their Pf3; transferred move h2g2 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Kg2 (summed QAP objective 141, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



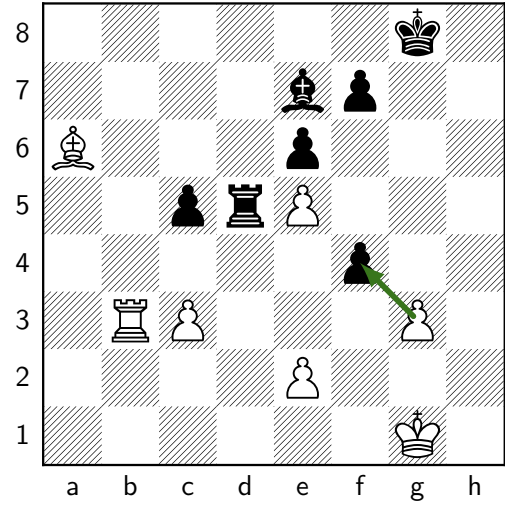
(a)



(b)



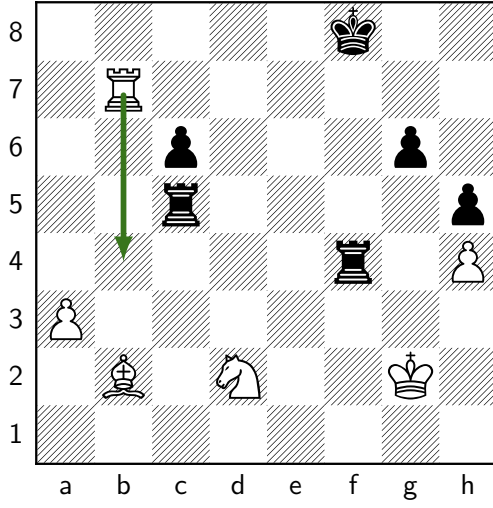
(c)



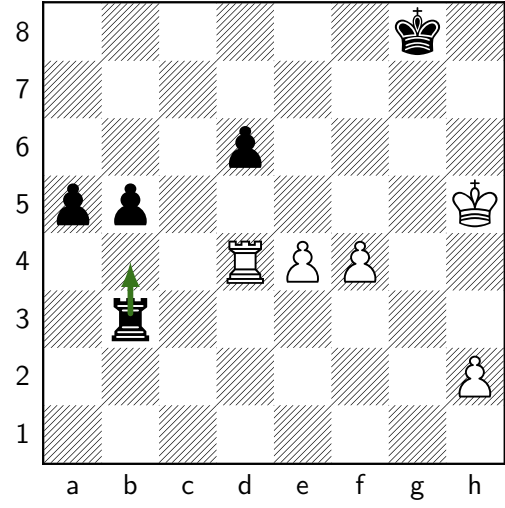
(d)

Figure 29: **Move 35, White: gxf4**. (a) Our position; Melanie plays gxf4 (g3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0002#2766 (ply 62) [structural distance  $d=34$ , QAP obj=61]: the master played gxh4; under  $\mu$  our Pg3 plays the role of that game’s Pg3; cross-type roles: our ph5 as their Rc1, our Nd2 as their Pa2; transferred move g3f4 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0001#2578 (ply 83) [structural distance  $d=34$ , QAP obj=58]: the master played gxf4; under  $\mu$  our Pg3 plays the role of that game’s Pg4; cross-type roles: our Bb2 as their Rc5, our Ph4 as their Bd5; transferred move g3f4 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0003#1215 (ply 58) [structural distance  $d=37$ , QAP obj=55]: the master played gxf4; under  $\mu$  our Pg3 plays the role of that game’s Pg3; cross-type roles: our Nd2 as their Pe2, our ph5 as their Rb3; transferred move g3f4 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to gxf4 (summed QAP objective 371, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.

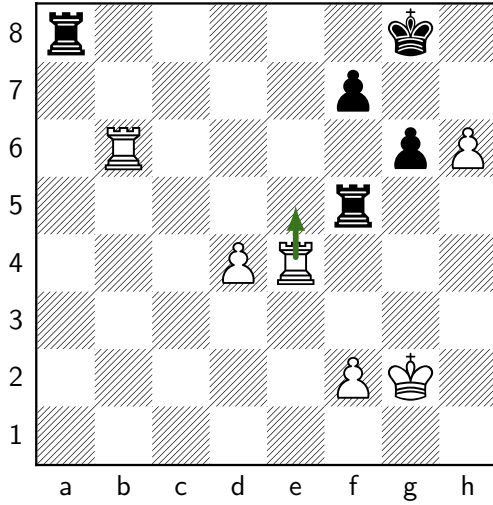




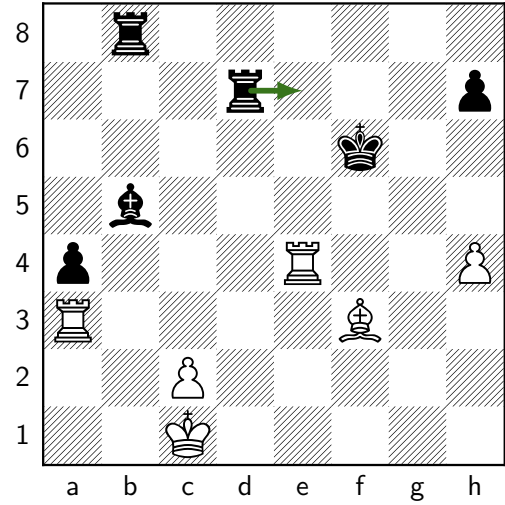
(a)



(b)

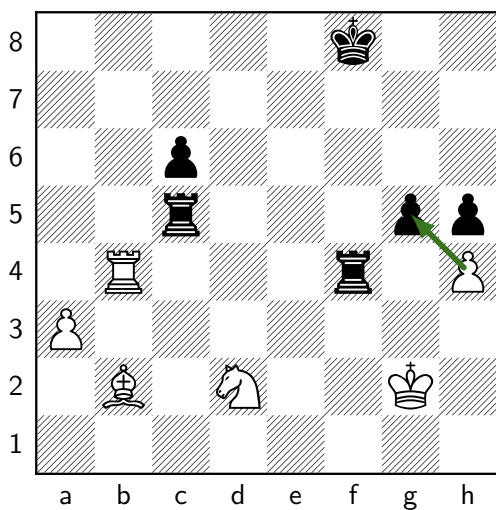


(c)

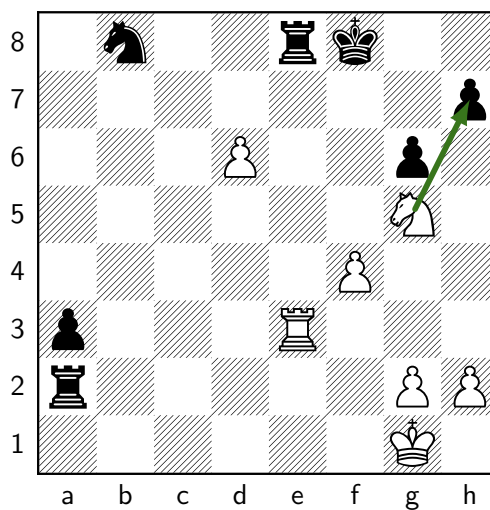


(d)

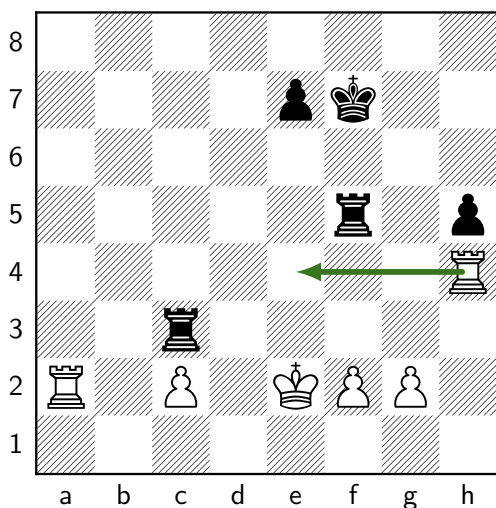
Figure 30: **Move 36, White: Rb4.** (a) Our position; Melanie plays Rb4 (b7b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0000#254 (ply 73) [structural distance  $d=38$ , QAP obj=20]: the master played Rb4; under  $\mu$  our Rb7 plays the role of that game’s Rb6; cross-type roles: our Nd2 as their Pb5; transferred move b7b4 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0000#3345 (ply 74) [structural distance  $d=29$ , QAP obj=13]: the master played Re5; under  $\mu$  our Rb7 plays the role of that game’s Re4; cross-type roles: our Nd2 as their pg6, our Bb2 as their Ph6; transferred move b7b4 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0003#1322 (ply 65) [structural distance  $d=37$ , QAP obj=11]: the master played Re7; under  $\mu$  our Rb7 plays the role of that game’s Rd2; cross-type roles: our Bb2 as their Rb8, our Nd2 as their Bb5; transferred move b7b4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rb4 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



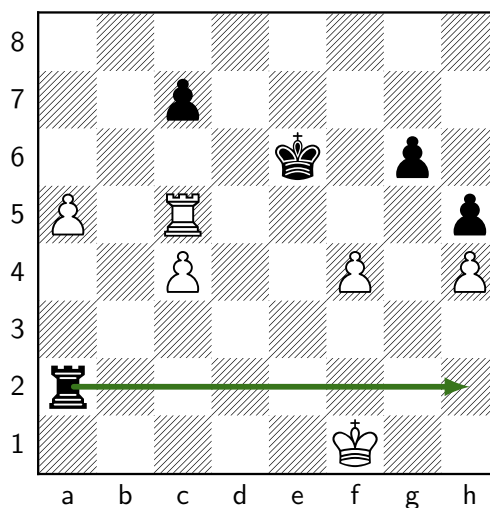
(a)



(b)

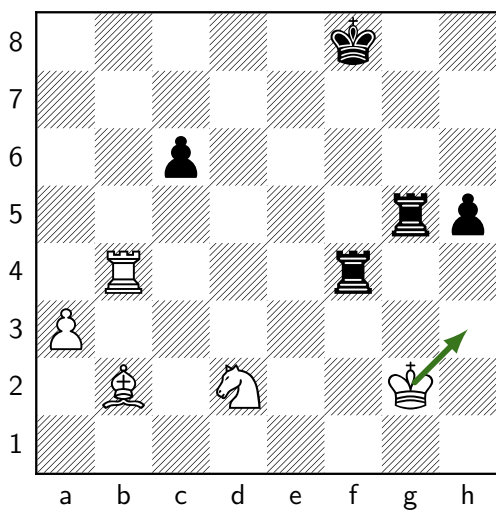


(c)

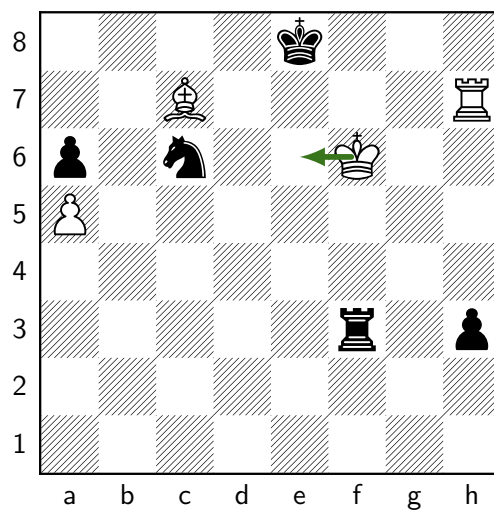


(d)

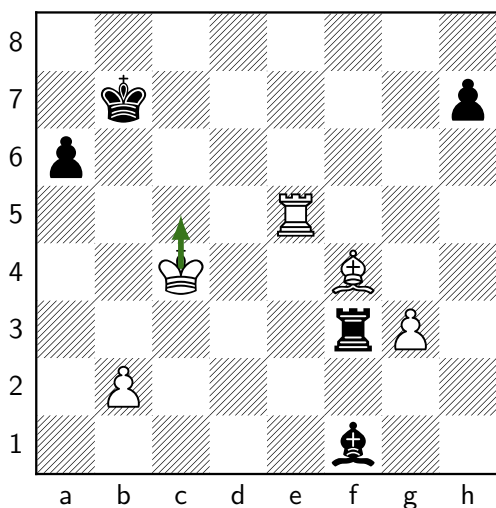
Figure 31: **Move 37, White: h4g5.** (a) Our position; Melanie plays h4g5 (h4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0002#357 (ply 58) [structural distance  $d=40$ , QAP obj=41]: the master played N4h7+; under  $\mu$  our Ph4 plays the role of that game's Ng5 (a **cross-type** correspondence); cross-type roles: our Bb2 as their Ph2, our Nd2 as their Pd6; transferred move h4g5 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0000#1384 (ply 64) [structural distance  $d=37$ , QAP obj=30]: the master played Re4; under  $\mu$  our Ph4 plays the role of that game's Rh4 (a **cross-type** correspondence); cross-type roles: our Pa3 as their Ra2, our Rb4 as their Pc2; transferred move h4g5 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0001#3505 (ply 73) [structural distance  $d=40$ , QAP obj=14]: the master played Rh2; under  $\mu$  our Ph4 plays the role of that game's Ra7 (a **cross-type** correspondence); cross-type roles: our Rb4 as their Pc7, our rc5 as their Ph5; transferred move h4g5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to h4g5 (summed QAP objective 87, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



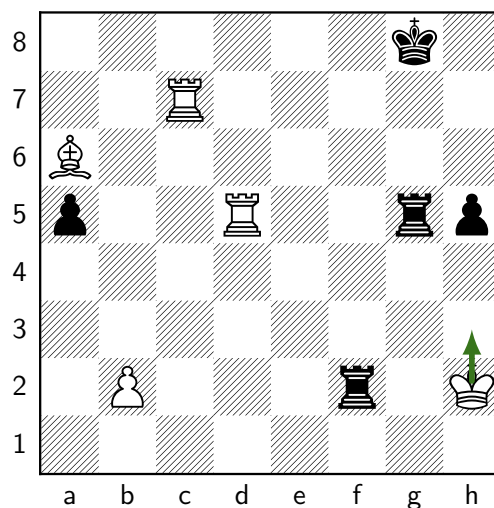
(a)



(b)

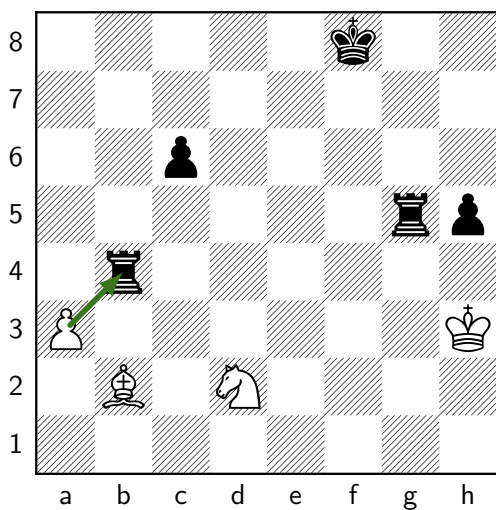


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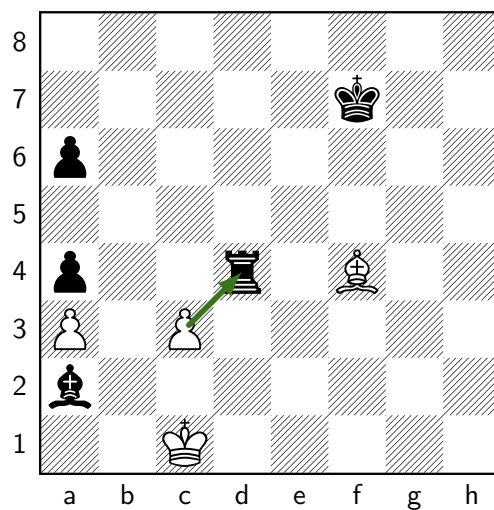


(d)

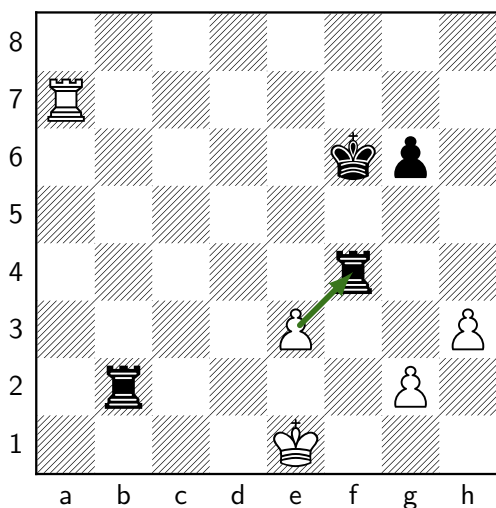
Figure 32: **Move 38, White: Kh3**. (a) Our position; Melanie plays Kh3 (g2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2066` (ply 80) [structural distance  $d=30$ , QAP obj=68]: the master played Ke6; under  $\mu$  our Kg2 plays the role of that game’s Kf6; cross-type roles: our Rb4 as their pa6, our rf4 as their nc6; transferred move g2h3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3678` (ply 80) [structural distance  $d=31$ , QAP obj=67]: the master played Kc5; under  $\mu$  our Kg2 plays the role of that game’s Kc4; cross-type roles: our rg5 as their bf1, our Nd2 as their Pb2; transferred move g2h3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#11` (ply 64) [structural distance  $d=23$ , QAP obj=50]: the master played Kh3; under  $\mu$  our Kg2 plays the role of that game’s Kh2; cross-type roles: our Nd2 as their Pb2, our pc6 as their Ba6; transferred move g2h3 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kh3 (summed QAP objective 417, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



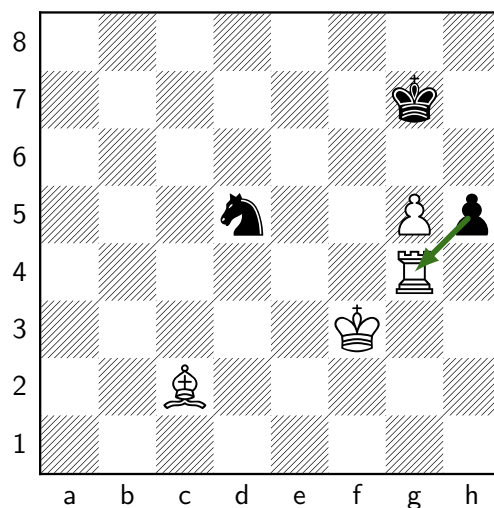
(a)



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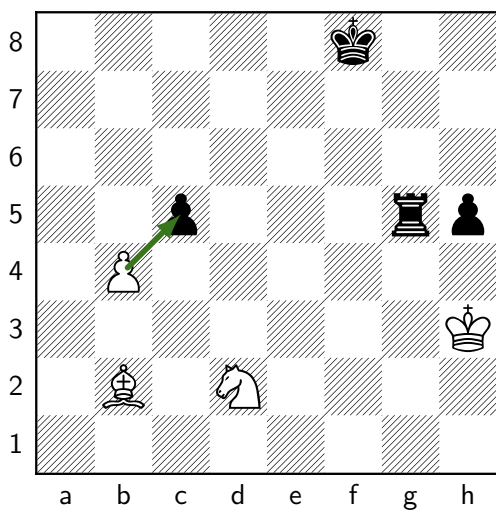


(c)

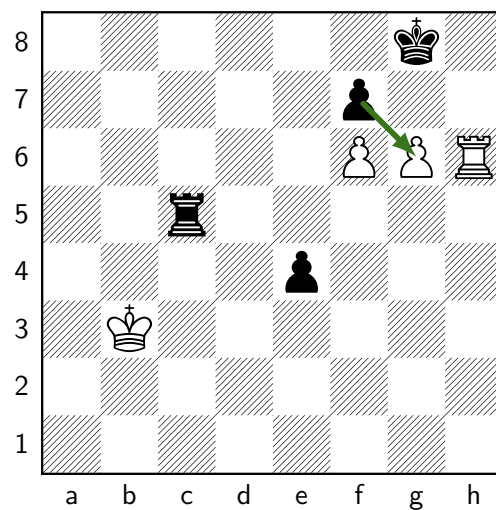


(d)

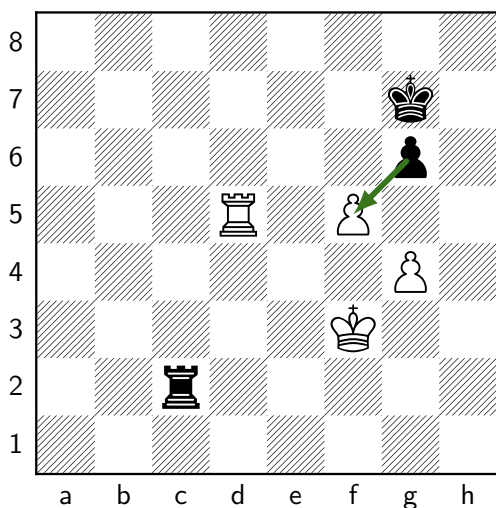
Figure 33: **Move 39, White: axb4**. (a) Our position; Melanie plays axb4 (a3b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0003#1688 (ply 76) [structural distance  $d=18$ , QAP obj=30]: the master played cxd4; under  $\mu$  our  $Pa3$  plays the role of that game’s  $Pc3$ ; cross-type roles: our  $pc6$  as their  $ba2$ , our  $Nd2$  as their  $Pa3$ ; transferred move a3b4 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0001#3300 (ply 86) [structural distance  $d=25$ , QAP obj=20]: the master played exf4; under  $\mu$  our  $Pa3$  plays the role of that game’s  $Pe3$ ; cross-type roles: our  $Bb2$  as their  $Pg2$ , our  $Nd2$  as their  $Ph3$ ; transferred move a3b4 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0000#1058 (ply 93) [structural distance  $d=22$ , QAP obj=19]: the master played hxg4+; under  $\mu$  our  $Pa3$  plays the role of that game’s  $Ph4$ ; cross-type roles: our  $rg5$  as their  $bc2$ ; transferred move a3b4 (**endorses this move**). Synthesis: 15 retrieved analogs transferred to axb4 (summed QAP objective 177, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



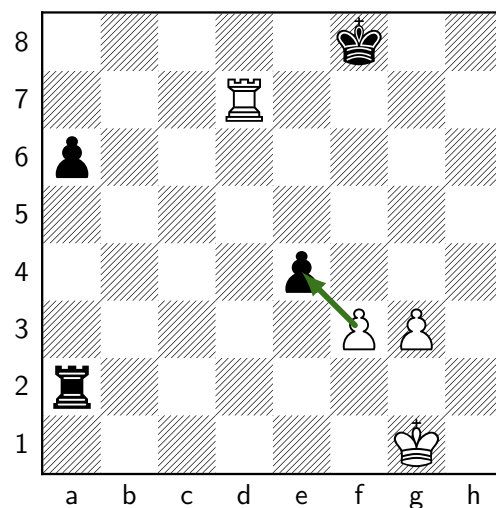
(a)



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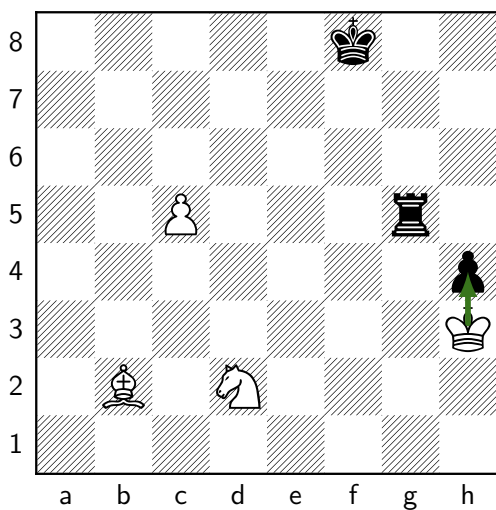


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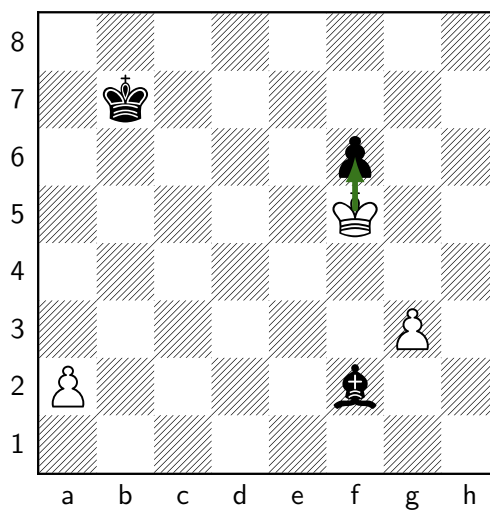


(d)

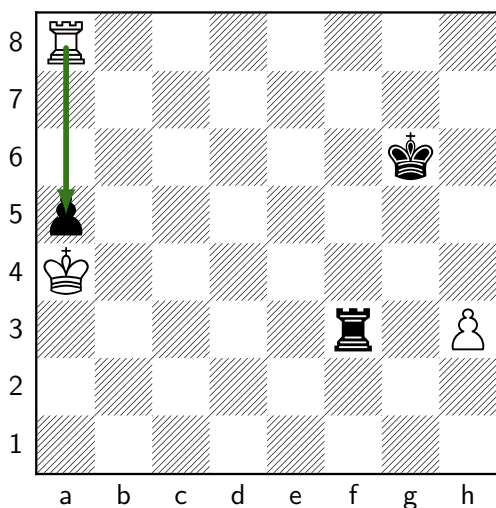
Figure 34: **Move 40, White: bxc5.** (a) Our position; Melanie plays **bxc5** (**b4c5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3597` (ply 89) [structural distance  $d=13$ , QAP obj=39]: the master played **fxg6**; under  $\mu$  our *Pb4* plays the role of that game’s *Pf2*; cross-type roles: our *Bb2* as their *Rc5*, our *Nd2* as their *Pe4*; transferred move **b4c5** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2142` (ply 83) [structural distance  $d=16$ , QAP obj=38]: the master played **gxf5**; under  $\mu$  our *Pb4* plays the role of that game’s *Pg3*; cross-type roles: our *Nd2* as their *pg4*, our *Bb2* as their *Rc2*; transferred move **b4c5** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2918` (ply 94) [structural distance  $d=14$ , QAP obj=38]: the master played **fxe4**; under  $\mu$  our *Pb4* plays the role of that game’s *Pf3*; cross-type roles: our *Nd2* as their *Pg3*, our *Bb2* as their *Rd7*; transferred move **b4c5** (**endorses this move**). Synthesis: 9 retrieved analogs transferred to **bxc5** (summed QAP objective 214, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



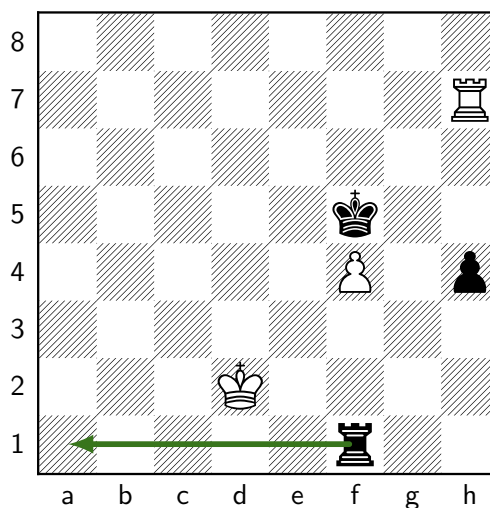
(a)



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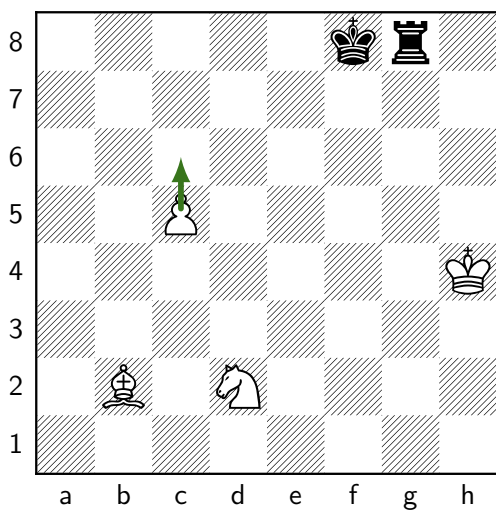


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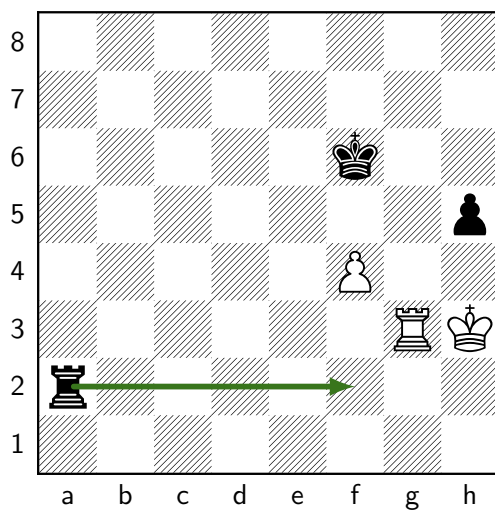


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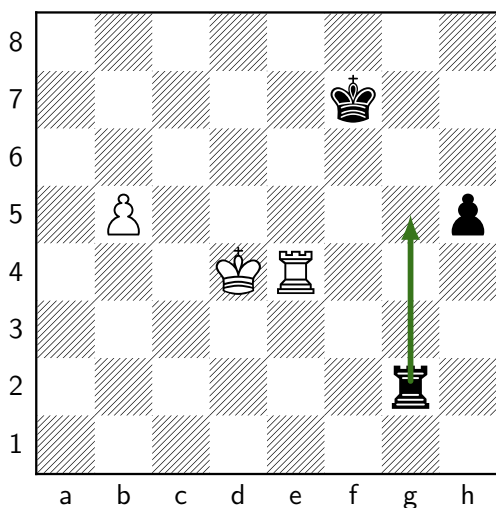
Figure 35: **Move 41, White: Kxh4**. (a) Our position; Melanie plays Kxh4 (h3h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3427` (ply 88) [structural distance  $d=10$ , QAP obj=27]: the master played Kxf6; under  $\mu$  our Kh3 plays the role of that game’s Kf5; cross-type roles: our Bb2 as their Pa2, our Kh3 as their bf2; transferred move h3h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1933` (ply 96) [structural distance  $d=8$ , QAP obj=25]: the master played Rxa5; under  $\mu$  our Kh3 plays the role of that game’s Ra8 (a **cross-type** correspondence); transferred move h3h4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2122` (ply 95) [structural distance  $d=9$ , QAP obj=24]: the master played Ra1; under  $\mu$  our Kh3 plays the role of that game’s Rf8 (a **cross-type** correspondence); transferred move h3h4 (**endorses this move**). Synthesis: 21 retrieved analogs transferred to Kxh4 (summed QAP objective 399, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



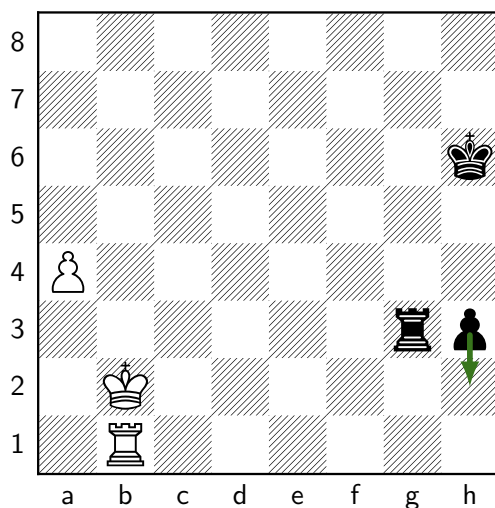
(a)



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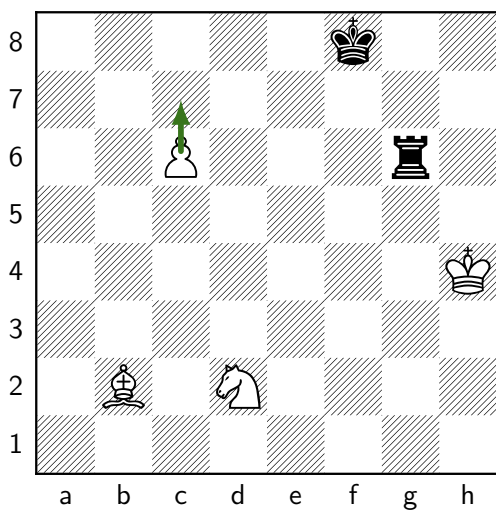


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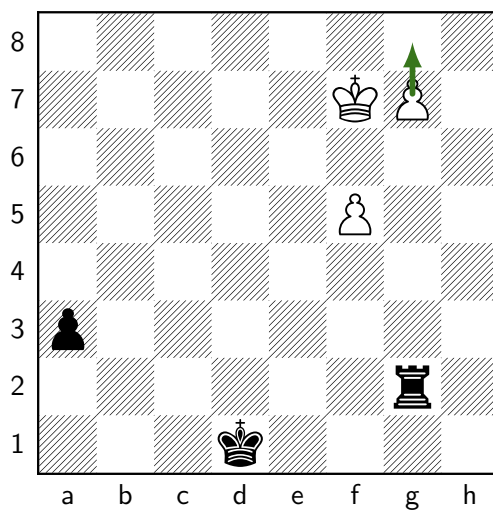


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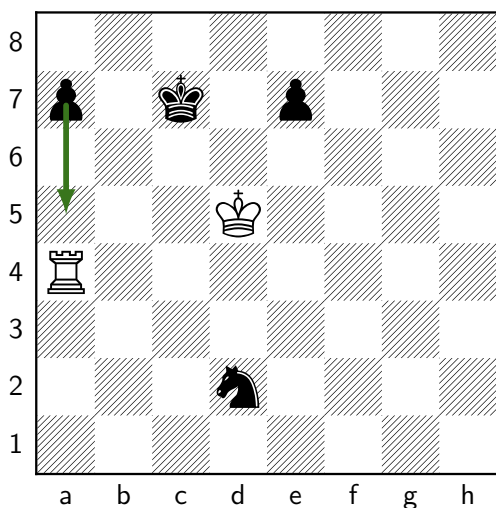
Figure 36: **Move 42, White: c6.** (a) Our position; Melanie plays c6 (c5c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#137` (ply 91) [structural distance  $d=10$ , QAP obj=2]: the master played Rf2; under  $\mu$  our *Pc5* plays the role of that game’s *Ra7* (a **cross-type** correspondence); cross-type roles: our *Bb2* as their *Ph5*, our *Nd2* as their *pf4*; transferred move c5c6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1647` (ply 97) [structural distance  $d=6$ , QAP obj=2]: the master played Rg5; under  $\mu$  our *Pc5* plays the role of that game’s *Rg7* (a **cross-type** correspondence); cross-type roles: our *Nd2* as their *pb5*, our *Bb2* as their *Ph5*; transferred move c5c6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3795` (ply 97) [structural distance  $d=10$ , QAP obj=2]: the master played h2; under  $\mu$  our *Pc5* plays the role of that game’s *Ph6*; cross-type roles: our *Nd2* as their *pa4*, our *Bb2* as their *Rg3*; transferred move c5c6 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to c6 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



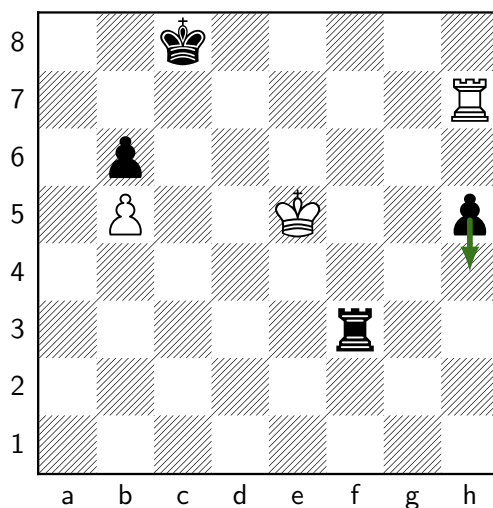
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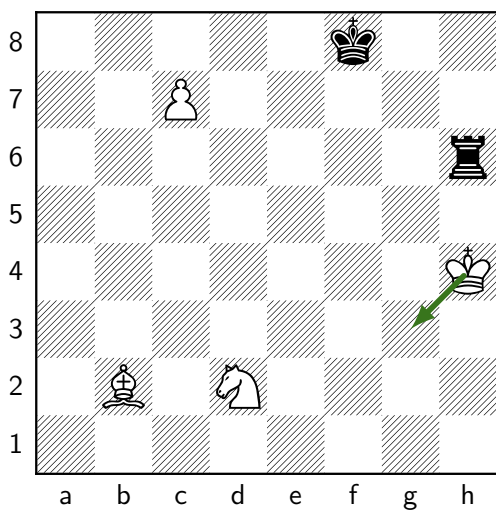
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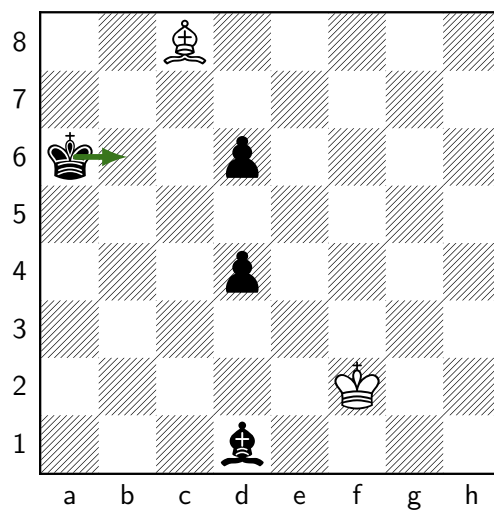
(d)

Figure 37: **Move 43, White: c7**. (a) Our position; Melanie plays c7 (c6c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1270` (ply 98) [structural distance  $d=7$ , QAP obj=15]: the master played g8=R; under  $\mu$  our *Pc6* plays the role of that game’s *Pg7*; cross-type roles: our *Nd2* as their *pa3*, our *Bb2* as their *Pf5*; transferred move c6c7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3112` (ply 101) [structural distance  $d=6$ , QAP obj=14]: the master played a5; under  $\mu$  our *Pc6* plays the role of that game’s *Pa2*; cross-type roles: our *Nd2* as their *Pe7*, our *Bb2* as their *Nd2*; transferred move c6c7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3962` (ply 95) [structural distance  $d=7$ , QAP obj=11]: the master played h4; under  $\mu$  our *Pc6* plays the role of that game’s *Ph4*; cross-type roles: our *Nd2* as their *Pb6*, our *Bb2* as their *Rf3*; transferred move c6c7 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to c7 (summed QAP objective 68, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

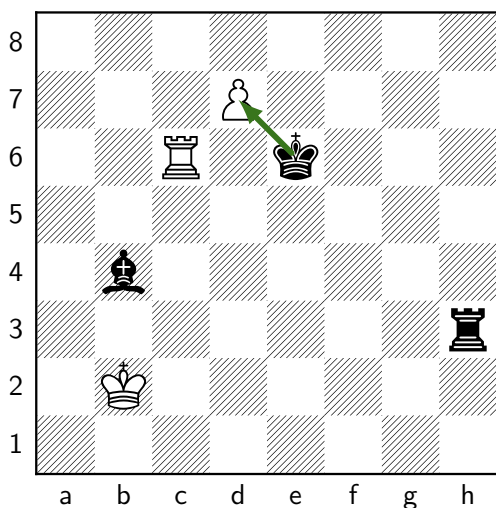




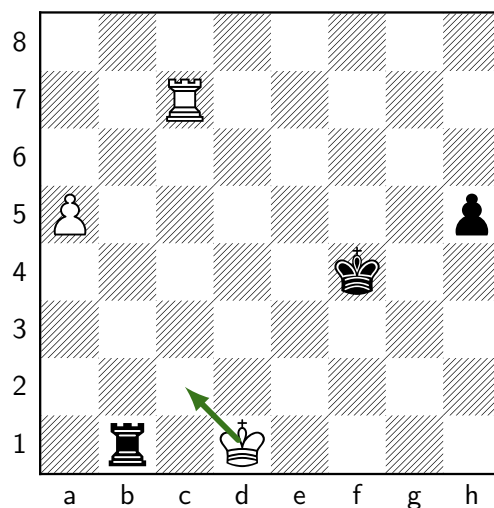
(a)



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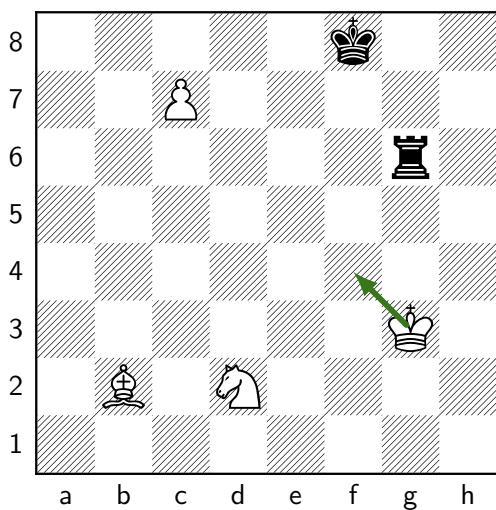


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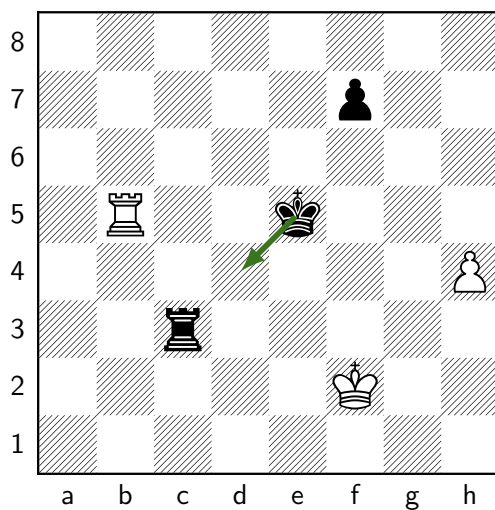


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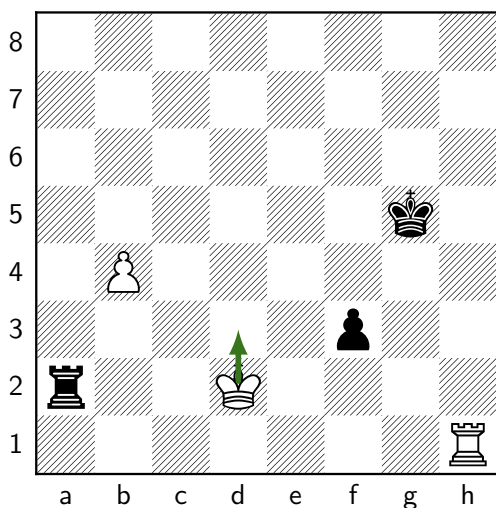
Figure 38: **Move 44, White: Kg3.** (a) Our position; Melanie plays Kg3 (**h4g3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2609` (ply 103) [structural distance  $d=7$ , QAP obj=67]: the master played Kb6; under  $\mu$  our *Kh4* plays the role of that game’s *Ka3*; cross-type roles: our *rh6* as their *bc8*, our *Nd2* as their *Pd6*; transferred move **h4g3 (endorses this move)**. (c) Retrieved from `ccrl_top_p0001#2538` (ply 99) [structural distance  $d=8$ , QAP obj=67]: the master played Kxd7; under  $\mu$  our *Kh4* plays the role of that game’s *Ke3*; cross-type roles: our *Nd2* as their *Rh3*; transferred move **h4g3 (endorses this move)**. (d) Retrieved from `ccrl_top_p0002#1100` (ply 94) [structural distance  $d=5$ , QAP obj=66]: the master played Kc2; under  $\mu$  our *Kh4* plays the role of that game’s *Kd1*; cross-type roles: our *Nd2* as their *ph5*, our *Bb2* as their *Rc7*; transferred move **h4g3 (endorses this move)**. Synthesis: 22 retrieved analogs transferred to Kg3 (summed QAP objective 733, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



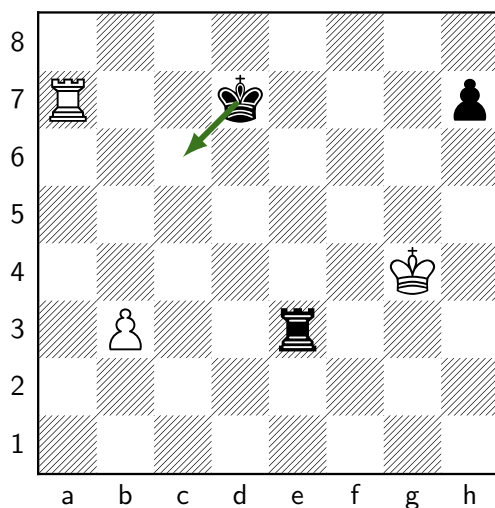
(a)



(b)



(c)



(d)

Figure 39: **Move 45, White: Kf4.** (a) Our position; Melanie plays Kf4 (g3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#854` (ply 81) [structural distance  $d=5$ , QAP obj=67]: the master played Kd4; under  $\mu$  our  $Kg3$  plays the role of that game’s  $Ke4$ ; cross-type roles: our  $Nd2$  as their  $Pf7$ , our  $Bb2$  as their  $Rc3$ ; transferred move g3f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2370` (ply 96) [structural distance  $d=5$ , QAP obj=67]: the master played Kd3; under  $\mu$  our  $Kg3$  plays the role of that game’s  $Kd2$ ; cross-type roles: our  $Bb2$  as their  $Rh1$ , our  $Nd2$  as their  $pf3$ ; transferred move g3f4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#625` (ply 95) [structural distance  $d=8$ , QAP obj=66]: the master played Kc6; under  $\mu$  our  $Kg3$  plays the role of that game’s  $Kd2$ ; cross-type roles: our  $Nd2$  as their  $pb3$ , our  $Bb2$  as their  $Re3$ ; transferred move g3f4 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Kf4 (summed QAP objective 713, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

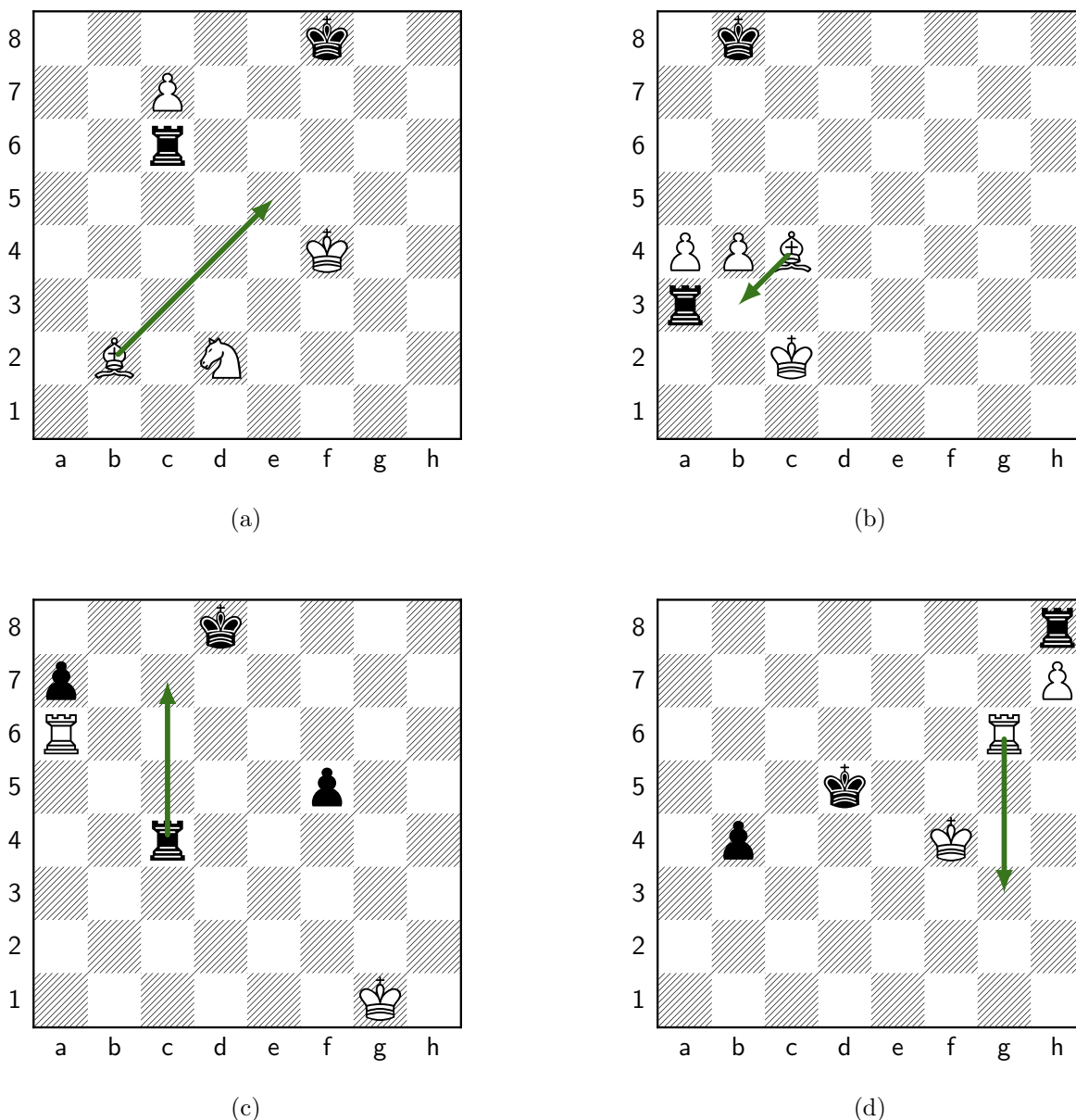


Figure 40: **Move 46, White: Be5.** (a) Our position; Melanie plays **Be5** (**b2e5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1726` (ply 88) [structural distance  $d=2$ , QAP obj=19]: the master played **Bb3**; under  $\mu$  our *Bb2* plays the role of that game’s *Bc4*; cross-type roles: our *Nd2* as their *Pb4*; transferred move **b2e5** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1592` (ply 77) [structural distance  $d=6$ , QAP obj=18]: the master played **Rc7**; under  $\mu$  our *Bb2* plays the role of that game’s *Rc5* (a **cross-type** correspondence); cross-type roles: our *Nd2* as their *Pf5*; transferred move **b2e5** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1769` (ply 106) [structural distance  $d=7$ , QAP obj=18]: the master played **Rg3**; under  $\mu$  our *Bb2* plays the role of that game’s *Rg6* (a **cross-type** correspondence); cross-type roles: our *Nd2* as their *pb4*; transferred move **b2e5** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **Be5** (summed QAP objective 64, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

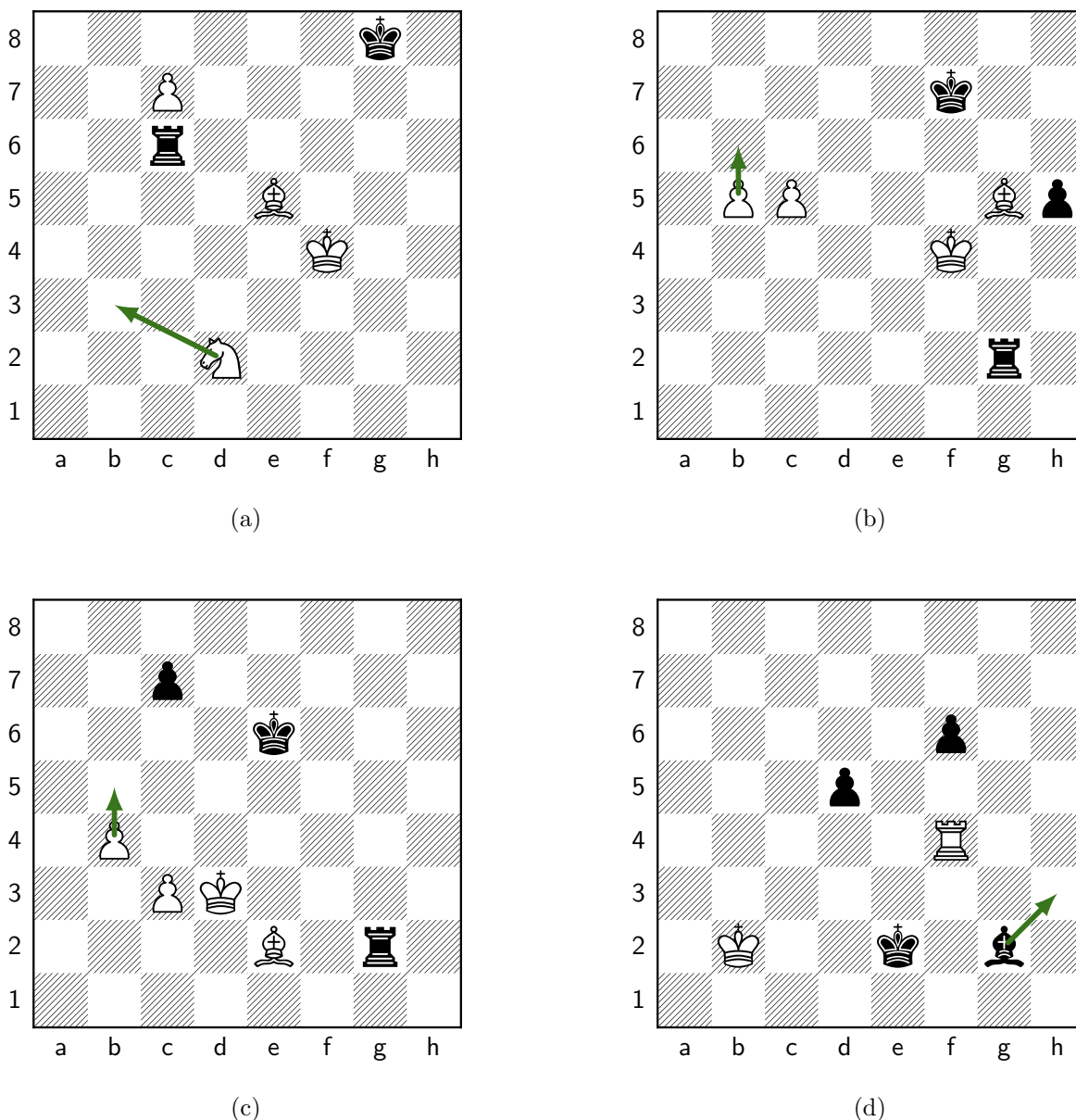
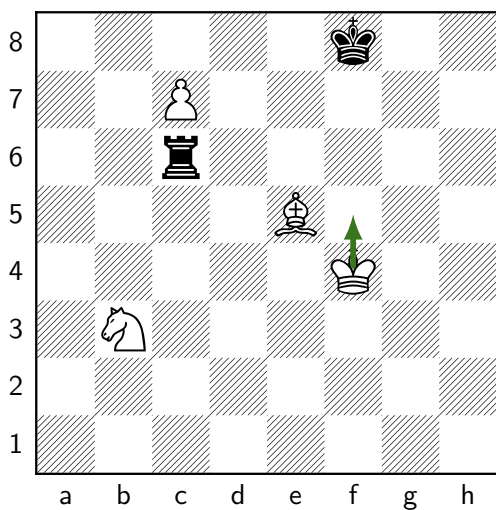
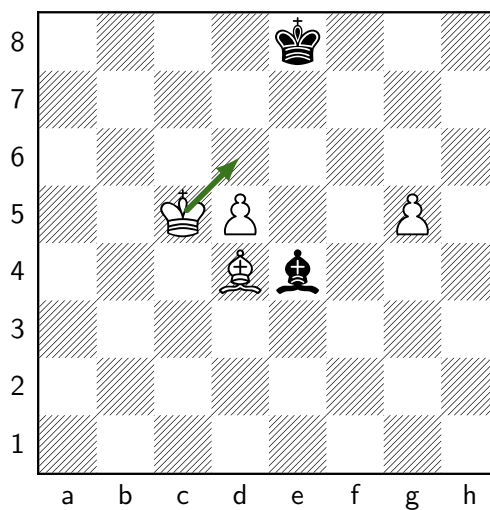


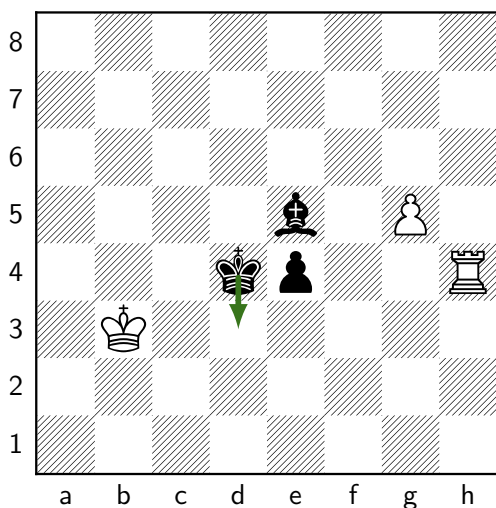
Figure 41: **Move 47, White: Nb3.** (a) Our position; Melanie plays Nb3 (d2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0002#2312 (ply 98) [structural distance  $d=14$ , QAP obj=18]: the master played b6; under  $\mu$  our Nd2 plays the role of that game's Pb5 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Bg5, our Be5 as their ph5; transferred move d2b3 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0000#1850 (ply 102) [structural distance  $d=13$ , QAP obj=12]: the master played b5; under  $\mu$  our Nd2 plays the role of that game's Pb4 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Be2, our Be5 as their Pc3; transferred move d2b3 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0002#287 (ply 105) [structural distance  $d=15$ , QAP obj=9]: the master played Bh3; under  $\mu$  our Nd2 plays the role of that game's Bg7 (a **cross-type** correspondence); cross-type roles: our Be5 as their Pd5; transferred move d2b3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nb3 (summed QAP objective 65, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



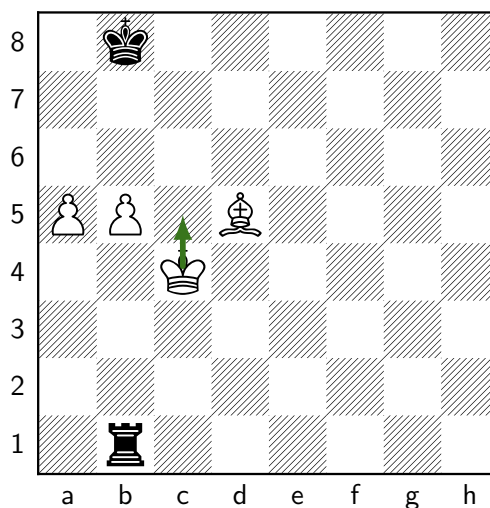
(a)



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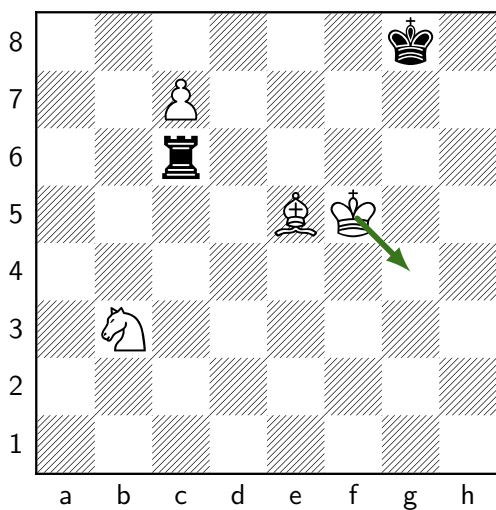


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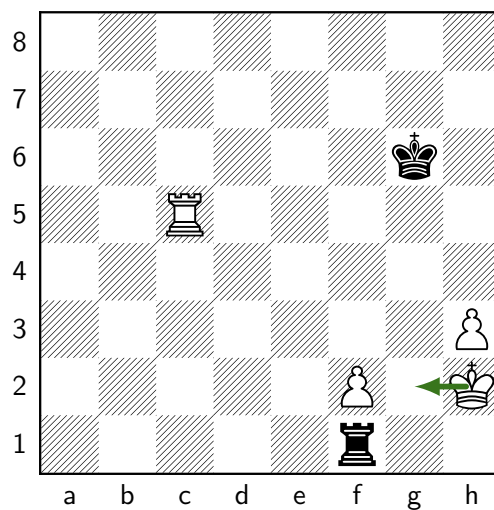


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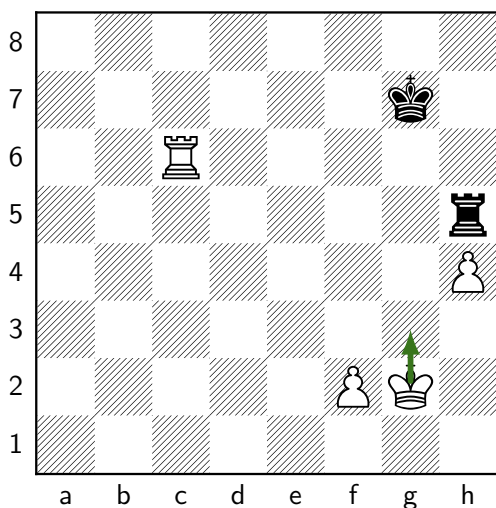
Figure 42: **Move 48, White: Kf5.** (a) Our position; Melanie plays Kf5 (f4f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2899` (ply 106) [structural distance  $d=12$ , QAP obj=21]: the master played Kd6; under  $\mu$  our Kf4 plays the role of that game’s Kc5; cross-type roles: our rc6 as their be4, our Nb3 as their Pg5; transferred move f4f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3998` (ply 101) [structural distance  $d=9$ , QAP obj=9]: the master played Kd3; under  $\mu$  our Kf4 plays the role of that game’s Kd5; cross-type roles: our Nb3 as their pg5; transferred move f4f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1726` (ply 110) [structural distance  $d=7$ , QAP obj=7]: the master played Kc5; under  $\mu$  our Kf4 plays the role of that game’s Kc4; cross-type roles: our Nb3 as their Pa5; transferred move f4f5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kf5 (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



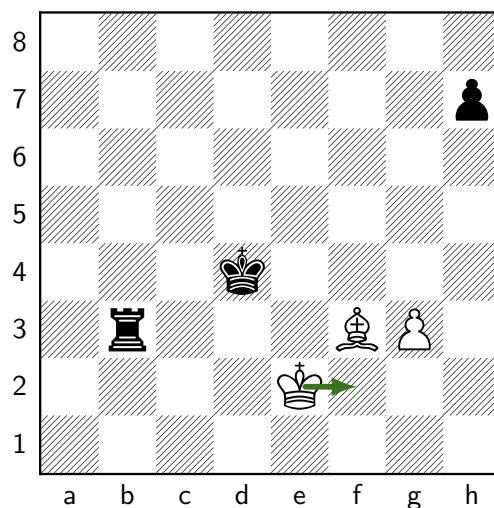
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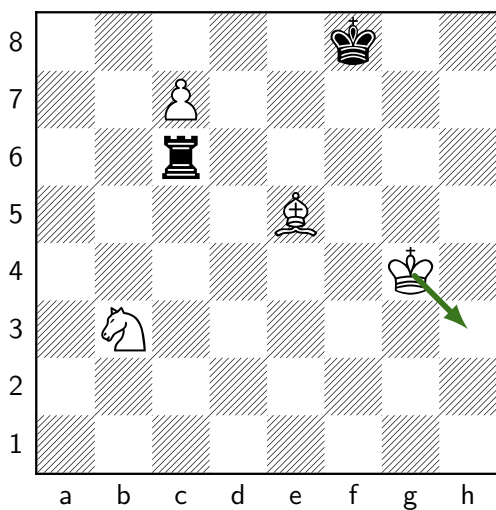


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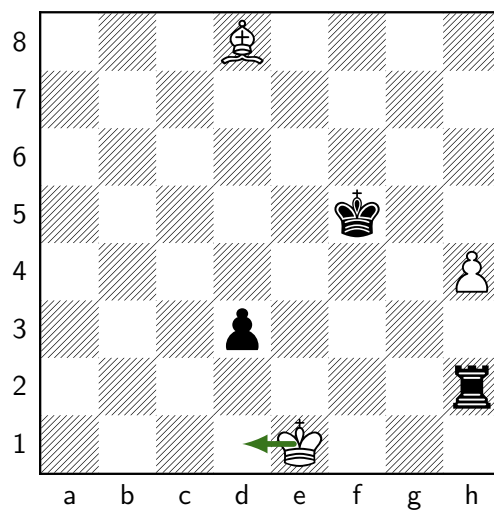


(d)

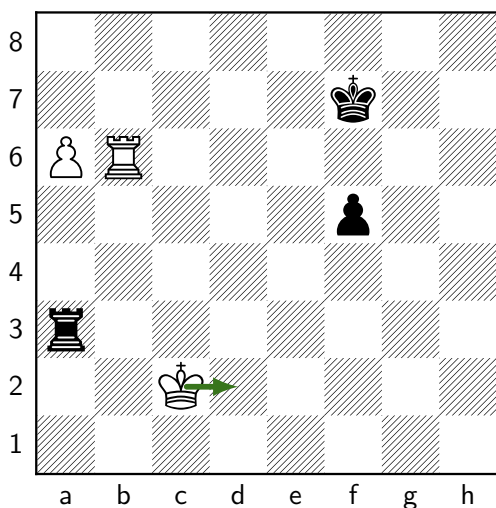
Figure 43: **Move 49, White: Kg4.** (a) Our position; Melanie plays Kg4 (f5g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2499` (ply 90) [structural distance  $d=10$ , QAP obj=20]: the master played Kg2; under  $\mu$  our Kf5 plays the role of that game’s Kh2; cross-type roles: our Be5 as their Ph3, our Nb3 as their Rc5; transferred move f5g4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1040` (ply 108) [structural distance  $d=11$ , QAP obj=19]: the master played Kg3; under  $\mu$  our Kf5 plays the role of that game’s Kg2; cross-type roles: our Be5 as their Pf2, our Nb3 as their Rc6; transferred move f5g4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3664` (ply 94) [structural distance  $d=11$ , QAP obj=14]: the master played Kf2; under  $\mu$  our Kf5 plays the role of that game’s Ke2; cross-type roles: our Pc7 as their Bf3, our Be5 as their Pg3; transferred move f5g4 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Kg4 (summed QAP objective 106, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



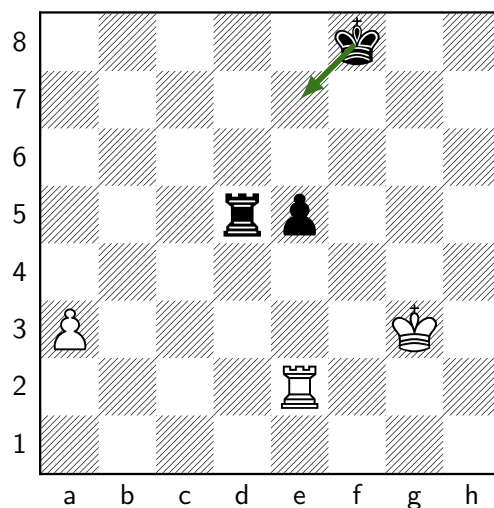
(a)



(b)



(c)



(d)

Figure 44: **Move 50, White: Kh3**. (a) Our position; Melanie plays Kh3 (**g4h3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2581` (ply 110) [structural distance  $d=5$ , QAP obj=11]: the master played Kd1; under  $\mu$  our Kg4 plays the role of that game’s Ke1; cross-type roles: our Nb3 as their pd3; transferred move g4h3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1294` (ply 110) [structural distance  $d=9$ , QAP obj=10]: the master played Kd2; under  $\mu$  our Kg4 plays the role of that game’s Kc2; cross-type roles: our Nb3 as their pf5, our Be5 as their Rb6; transferred move g4h3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#513` (ply 103) [structural distance  $d=9$ , QAP obj=9]: the master played Ke7; under  $\mu$  our Kg4 plays the role of that game’s Kf1; cross-type roles: our Be5 as their Rd5, our Nb3 as their pa3; transferred move g4h3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kh3 (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

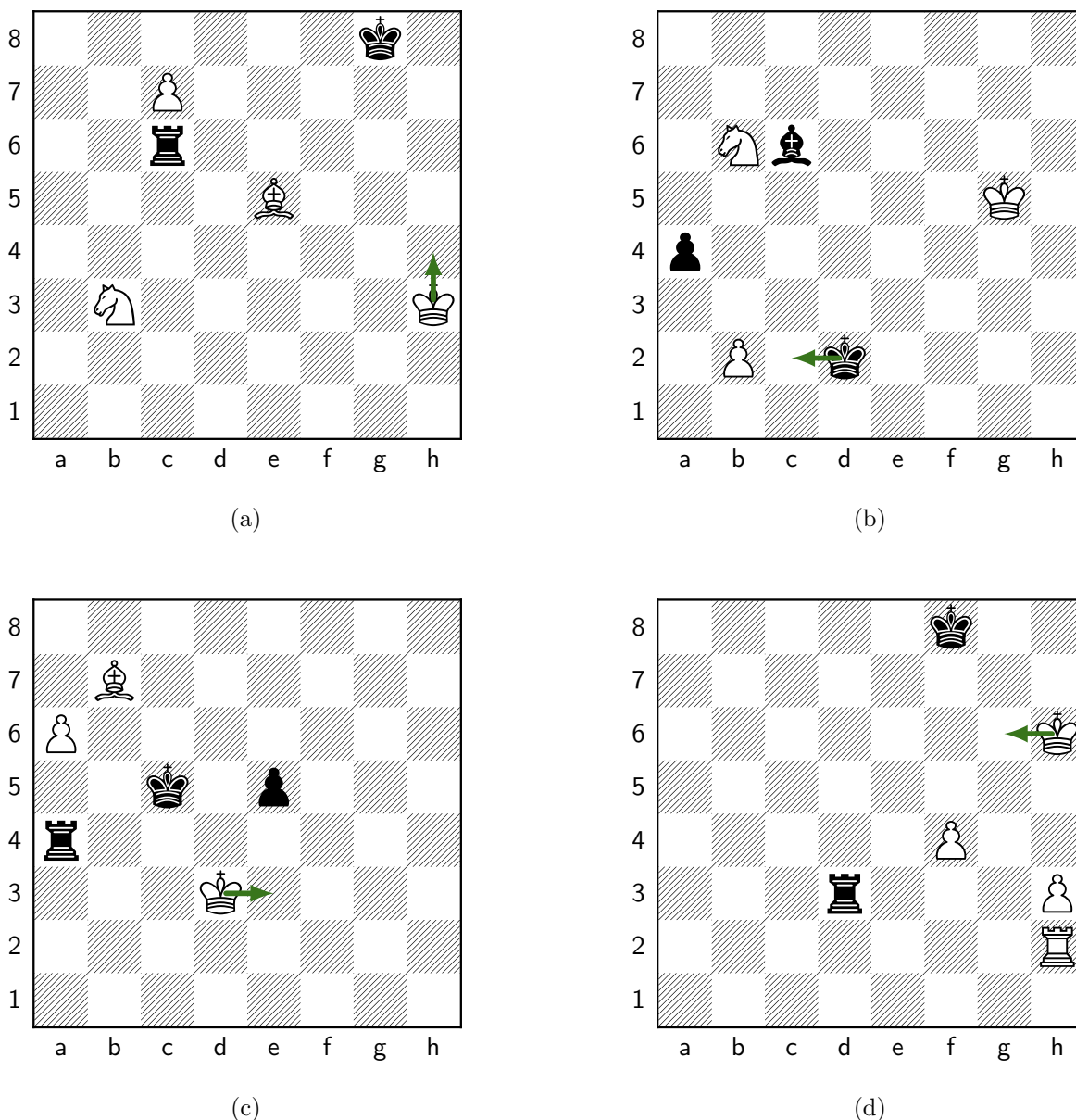
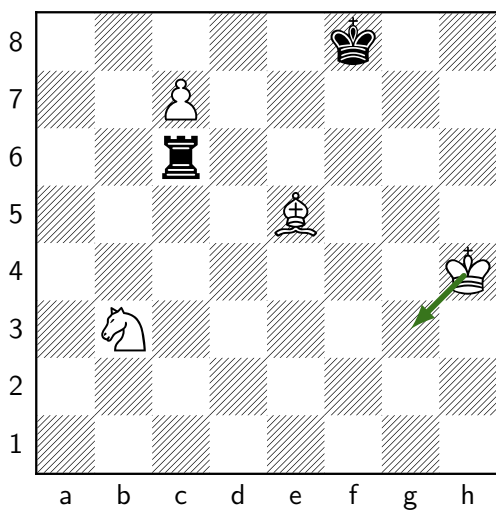
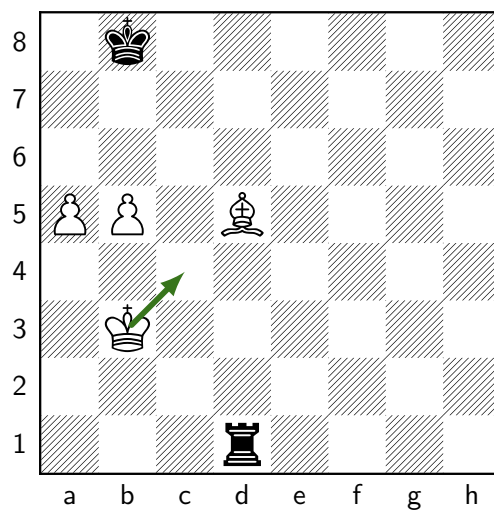


Figure 45: **Move 51, White: Kh4.** (a) Our position; Melanie plays **Kh4** (**h3h4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1518` (ply 117) [structural distance  $d=9$ , QAP obj=13]: the master played **Kc2**; under  $\mu$  our *Kh3* plays the role of that game’s *Kd7*; cross-type roles: our *rc6* as their *nb6*, our *Nb3* as their *pb2*; transferred move **h3h4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#320` (ply 116) [structural distance  $d=8$ , QAP obj=12]: the master played **Ke3**; under  $\mu$  our *Kh3* plays the role of that game’s *Kd3*; cross-type roles: our *Nb3* as their *pe5*; transferred move **h3h4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3816` (ply 100) [structural distance  $d=9$ , QAP obj=8]: the master played **Kg6**; under  $\mu$  our *Kh3* plays the role of that game’s *Kh6*; cross-type roles: our *Be5* as their *Rh2*, our *Nb3* as their *Pf4*; transferred move **h3h4** (**endorses this move**). Synthesis: 18 retrieved analogs transferred to **Kh4** (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

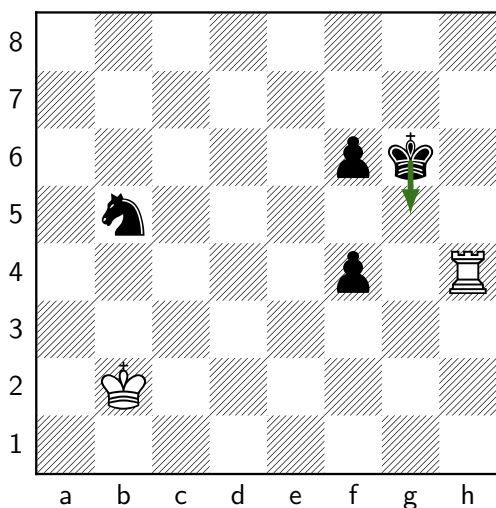




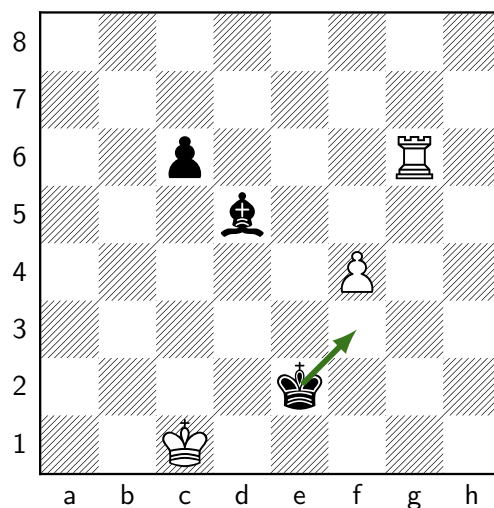
(a)



(b)



(c)



(d)

Figure 46: **Move 52, White: Kg3.** (a) Our position; Melanie plays Kg3 (**h4g3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1726` (ply 108) [structural distance  $d=9$ , QAP obj=13]: the master played Kc4; under  $\mu$  our *Kh4* plays the role of that game’s *Kb3*; cross-type roles: our *Be5* as their *Pa5*, our *Nb3* as their *Pb5*; transferred move **h4g3 (endorses this move)**. (c) Retrieved from `ccr1_top_p0002#1410` (ply 101) [structural distance  $d=7$ , QAP obj=9]: the master played Kg5; under  $\mu$  our *Kh4* plays the role of that game’s *Kg3*; cross-type roles: our *Nb3* as their *Pf6*, our *Be5* as their *Nb5*; transferred move **h4g3 (endorses this move)**. (d) Retrieved from `ccr1_top_p0001#3498` (ply 111) [structural distance  $d=5$ , QAP obj=7]: the master played Kf3; under  $\mu$  our *Kh4* plays the role of that game’s *Ke7*; cross-type roles: our *Nb3* as their *pf4*; transferred move **h4g3 (endorses this move)**. Synthesis: 6 retrieved analogs transferred to Kg3 (summed QAP objective 34, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

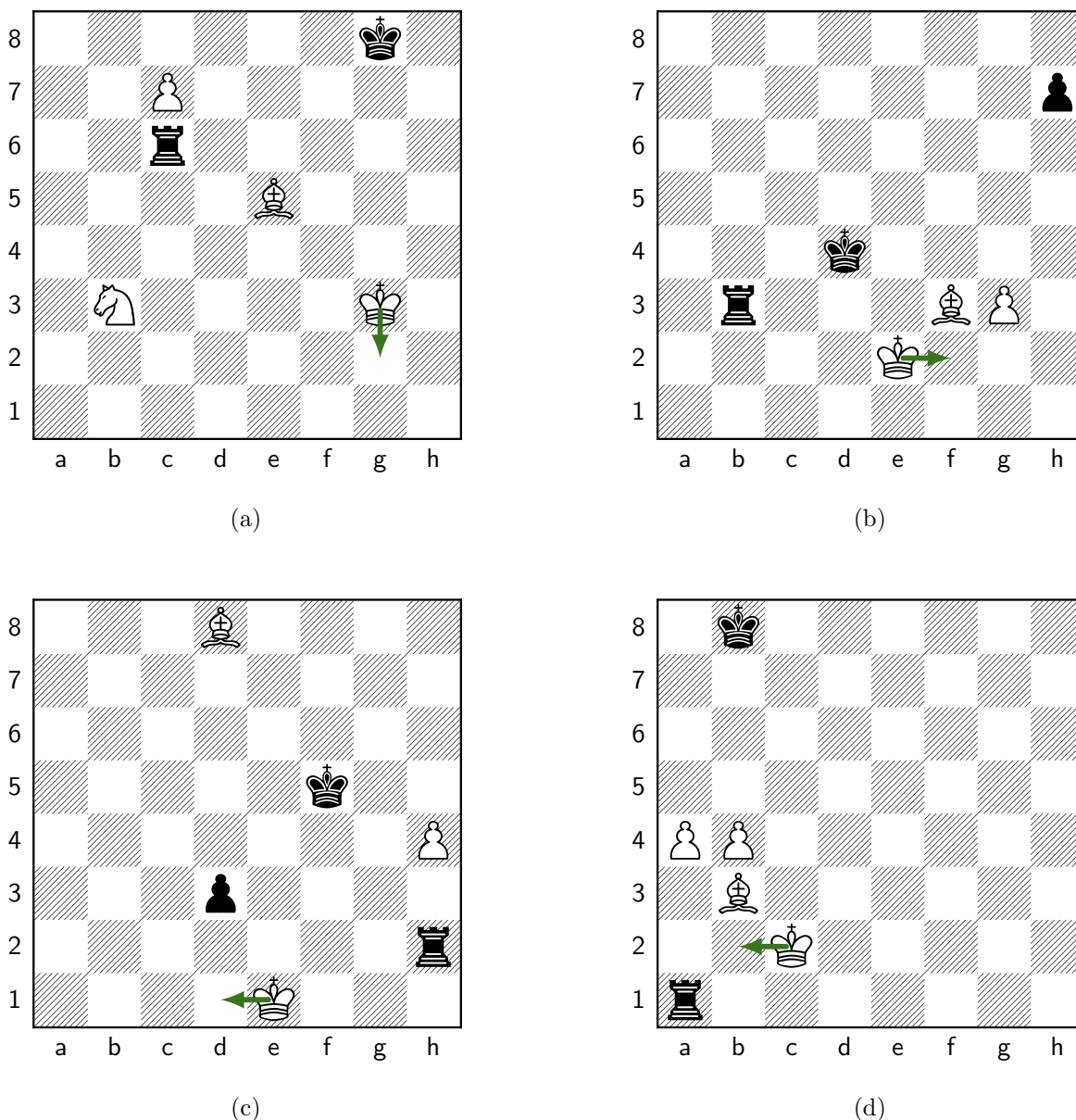


Figure 47: **Move 53, White: Kg2.** (a) Our position; Melanie plays Kg2 (g3g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3664` (ply 94) [structural distance  $d=11$ , QAP obj=14]: the master played Kf2; under  $\mu$  our Kg3 plays the role of that game’s Ke2; cross-type roles: our Pc7 as their Bf3, our Be5 as their Pg3; transferred move g3g2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2581` (ply 110) [structural distance  $d=9$ , QAP obj=10]: the master played Kd1; under  $\mu$  our Kg3 plays the role of that game’s Ke1; cross-type roles: our Nb3 as their pd3; transferred move g3g2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1726` (ply 90) [structural distance  $d=7$ , QAP obj=10]: the master played Kb2; under  $\mu$  our Kg3 plays the role of that game’s Kc2; cross-type roles: our Nb3 as their Pb4; transferred move g3g2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg2 (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

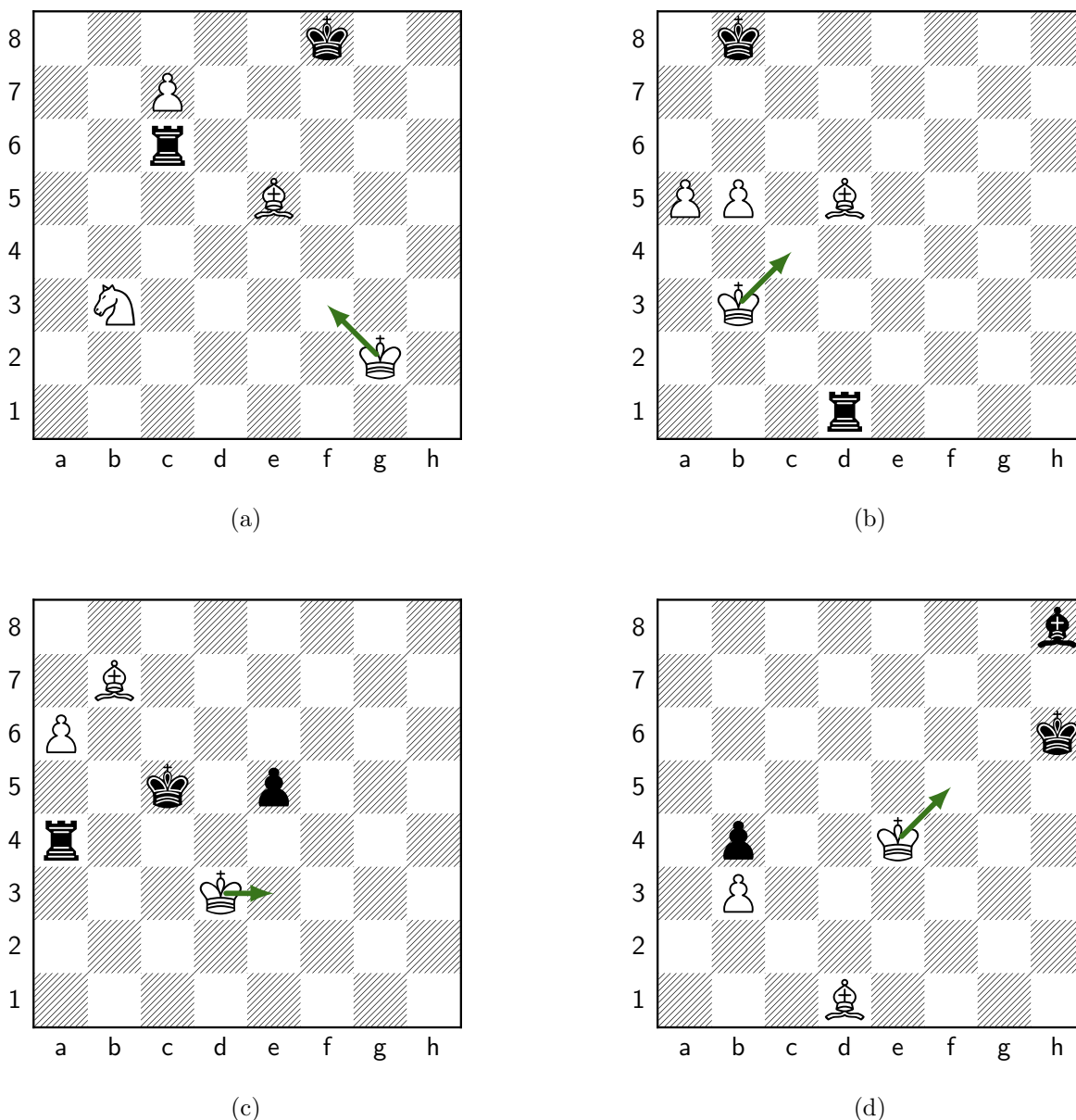
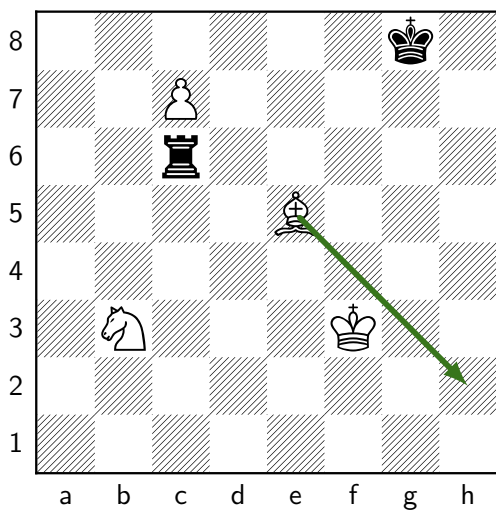
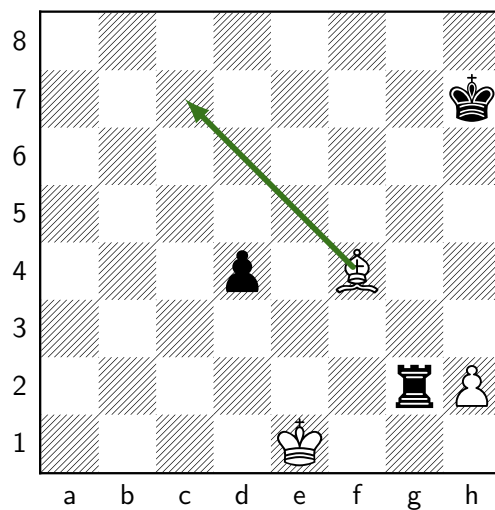


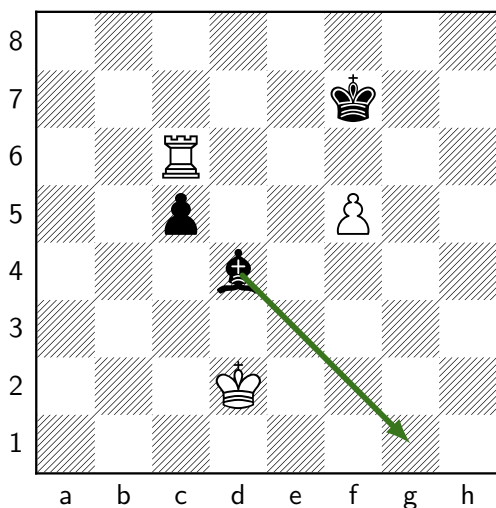
Figure 48: **Move 54, White: Kf3.** (a) Our position; Melanie plays Kf3 (g2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1726` (ply 108) [structural distance  $d=9$ , QAP obj=13]: the master played Kc4; under  $\mu$  our Kg2 plays the role of that game’s Kb3; cross-type roles: our Nb3 as their Pa5, our Be5 as their Pb5; transferred move g2f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#320` (ply 116) [structural distance  $d=8$ , QAP obj=12]: the master played Ke3; under  $\mu$  our Kg2 plays the role of that game’s Kd3; cross-type roles: our Nb3 as their pe5; transferred move g2f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#713` (ply 122) [structural distance  $d=9$ , QAP obj=3]: the master played Kf5; under  $\mu$  our Kg2 plays the role of that game’s Ke4; cross-type roles: our rc6 as their pb4, our Nb3 as their bh8; transferred move g2f3 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Kf3 (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



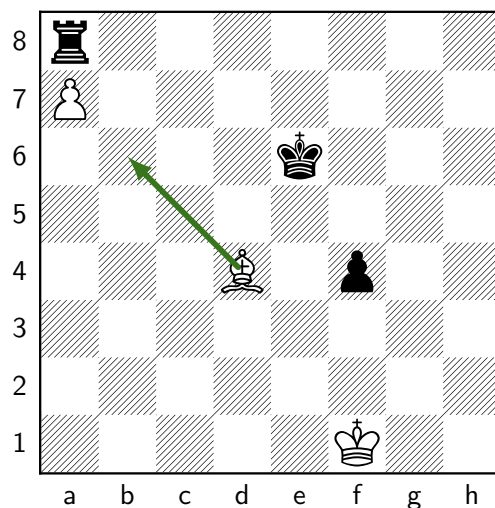
(a)



(b)



(c)



(d)

Figure 49: **Move 55, White: Bh2.** (a) Our position; Melanie plays Bh2 (e5h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2581` (ply 102) [structural distance  $d=4$ , QAP obj=23]: the master played Bc7; under  $\mu$  our Be5 plays the role of that game’s Bf4; cross-type roles: our Nb3 as their pd4; transferred move e5h2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3092` (ply 123) [structural distance  $d=8$ , QAP obj=21]: the master played Bg1; under  $\mu$  our Be5 plays the role of that game’s Bd5; cross-type roles: our Nb3 as their pf5; transferred move e5h2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2824` (ply 122) [structural distance  $d=4$ , QAP obj=20]: the master played Bb6; under  $\mu$  our Be5 plays the role of that game’s Bd4; cross-type roles: our Nb3 as their pf4; transferred move e5h2 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Bh2 (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

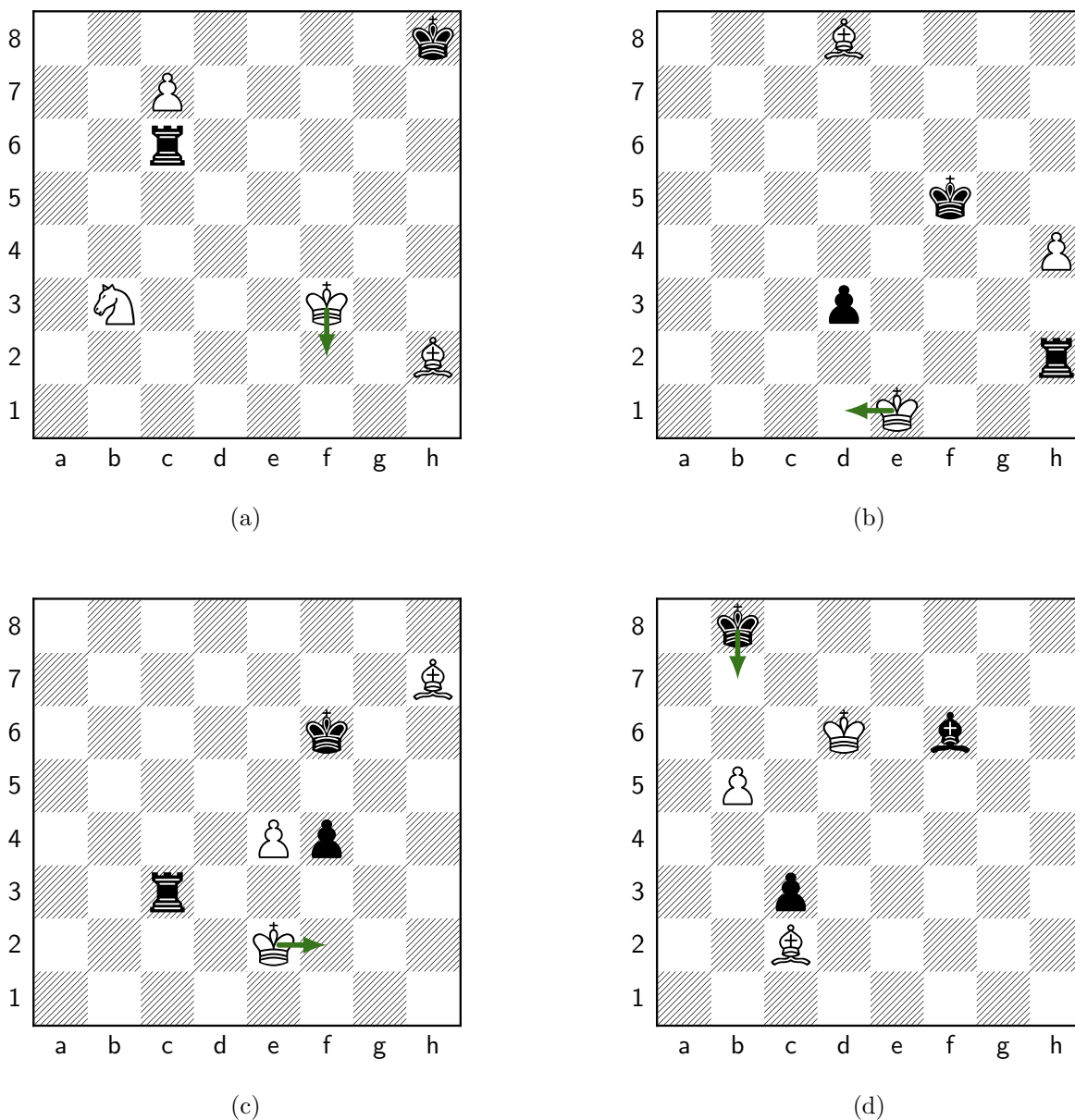
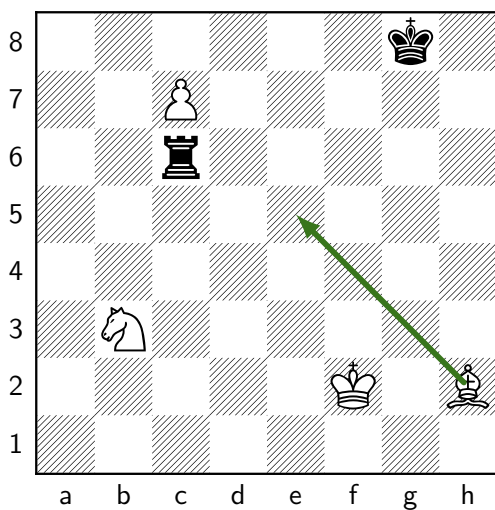
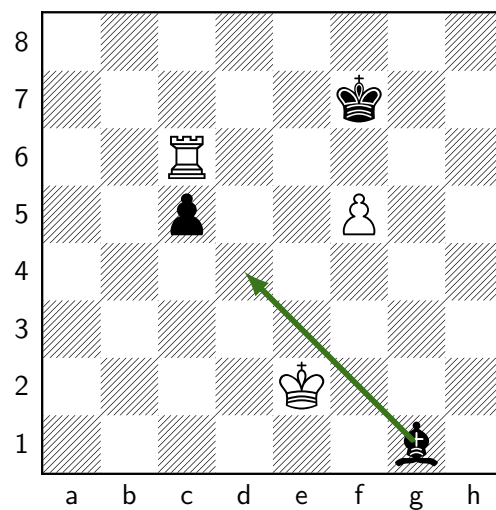


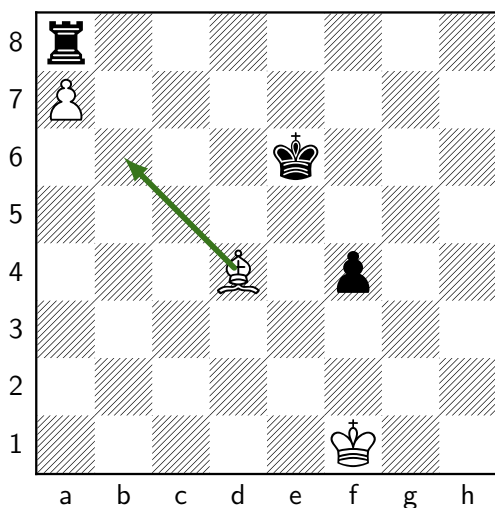
Figure 50: **Move 56, White: Kf2.** (a) Our position; Melanie plays Kf2 (f3f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2581` (ply 110) [structural distance  $d=5$ , QAP obj=12]: the master played Kd1; under  $\mu$  our Kf3 plays the role of that game’s Ke1; cross-type roles: our Nb3 as their pd3; transferred move f3f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1559` (ply 124) [structural distance  $d=8$ , QAP obj=1]: the master played Kf2; under  $\mu$  our Kf3 plays the role of that game’s Ke2; cross-type roles: our Nb3 as their pf4; transferred move f3f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2615` (ply 111) [structural distance  $d=8$ , QAP obj=1]: the master played Kb7; under  $\mu$  our Kf3 plays the role of that game’s Kb1; cross-type roles: our Nb3 as their pb5, our rc6 as their bc2; transferred move f3f2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kf2 (summed QAP objective 27, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



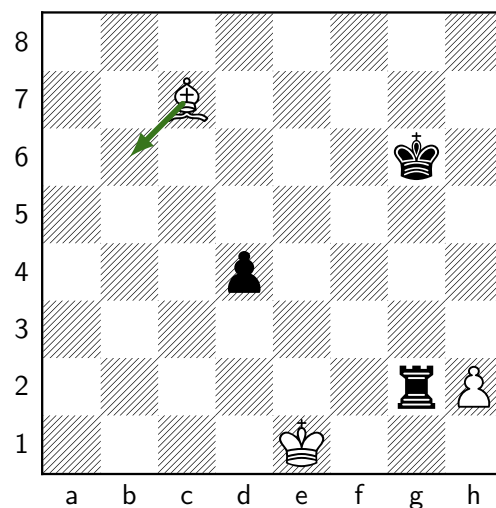
(a)



(b)



(c)



(d)

Figure 51: **Move 57, White: Be5.** (a) Our position; Melanie plays Be5 (h2e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3092` (ply 125) [structural distance  $d=5$ , QAP obj=22]: the master played Bd4; under  $\mu$  our Bh2 plays the role of that game’s Bg8; cross-type roles: our Nb3 as their pf5; transferred move h2e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2824` (ply 122) [structural distance  $d=4$ , QAP obj=20]: the master played Bb6; under  $\mu$  our Bh2 plays the role of that game’s Bd4; cross-type roles: our Nb3 as their pf4; transferred move h2e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2581` (ply 104) [structural distance  $d=7$ , QAP obj=20]: the master played Bb6; under  $\mu$  our Bh2 plays the role of that game’s Bc7; cross-type roles: our Nb3 as their pd4; transferred move h2e5 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Be5 (summed QAP objective 123, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

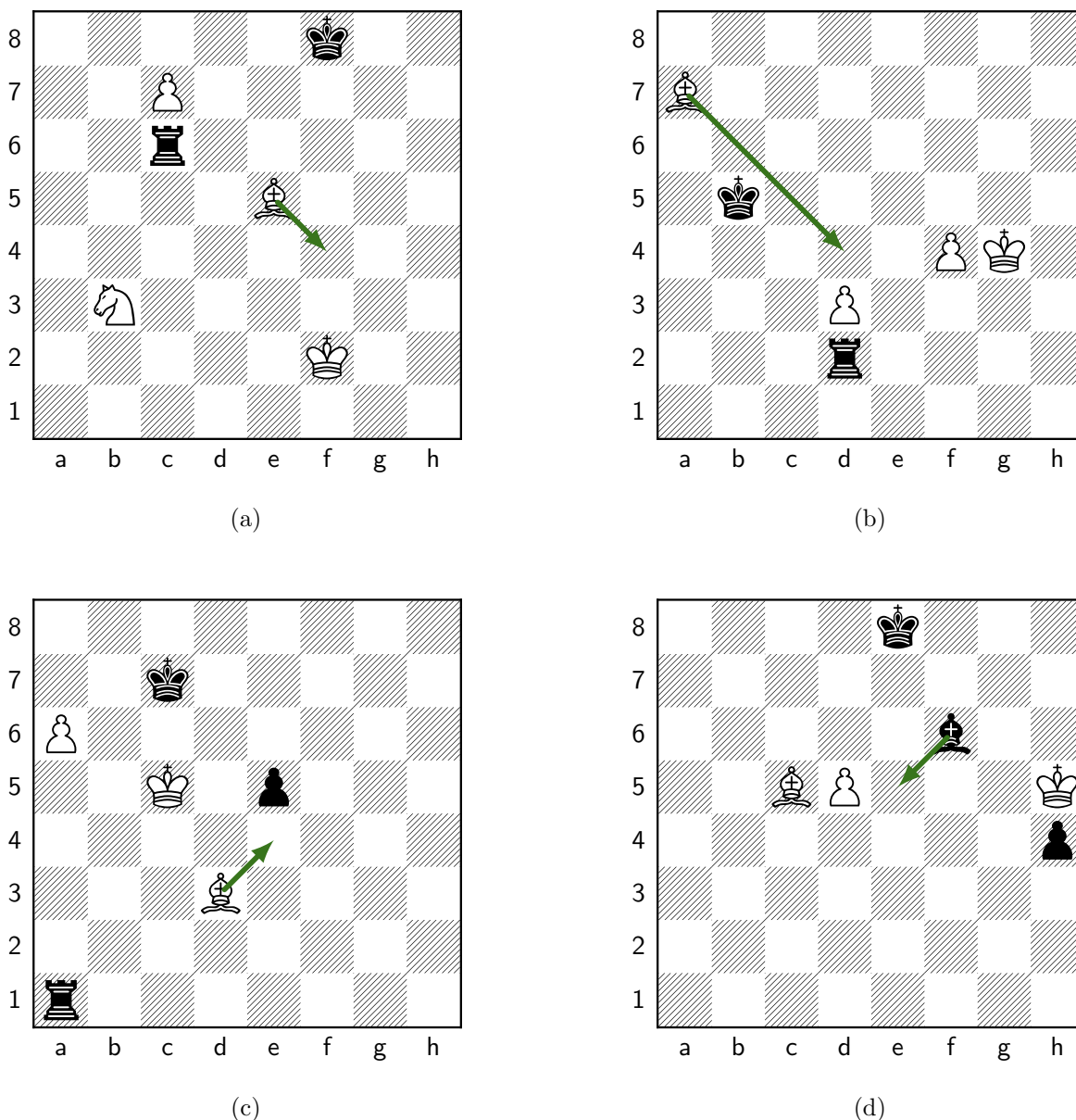
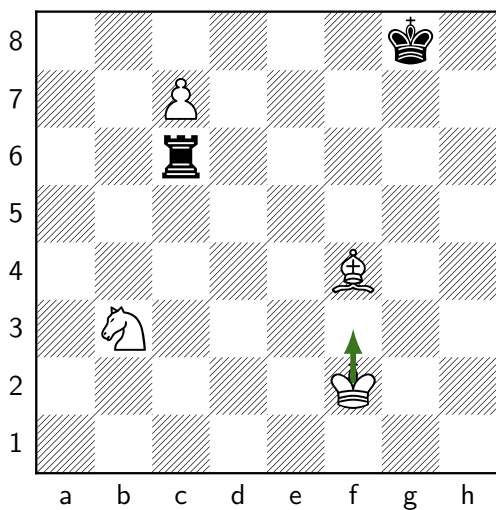
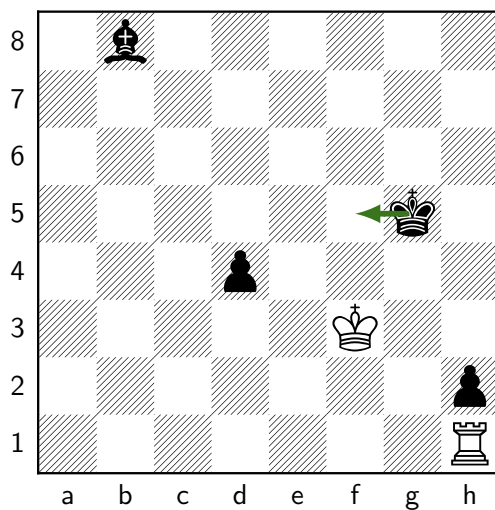


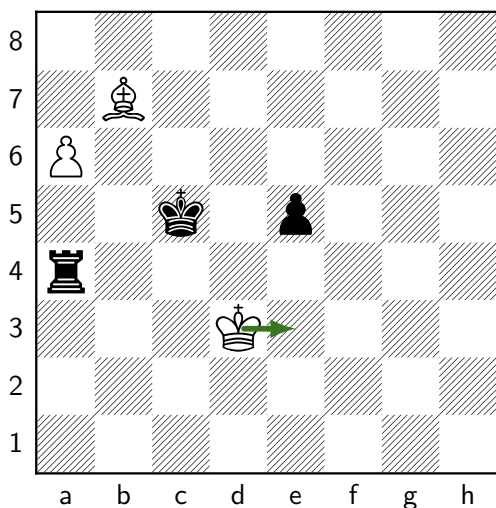
Figure 52: **Move 58, White: Bf4.** (a) Our position; Melanie plays Bf4 (e5f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2655` (ply 106) [structural distance  $d=7$ , QAP obj=19]: the master played Bd4; under  $\mu$  our Be5 plays the role of that game’s Ba7; cross-type roles: our Nb3 as their Pf4; transferred move e5f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#320` (ply 108) [structural distance  $d=5$ , QAP obj=3]: the master played Be4; under  $\mu$  our Be5 plays the role of that game’s Bd3; cross-type roles: our Nb3 as their pe5; transferred move e5f4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#857` (ply 127) [structural distance  $d=8$ , QAP obj=3]: the master played Be5; under  $\mu$  our Be5 plays the role of that game’s Bf3; cross-type roles: our rc6 as their bc5, our Nb3 as their pd5; transferred move e5f4 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Bf4 (summed QAP objective 37, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



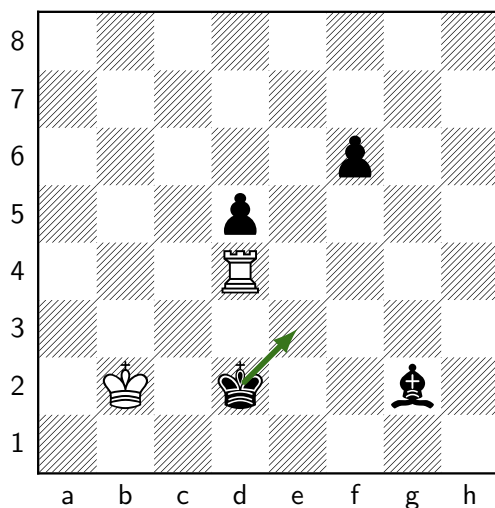
(a)



(b)



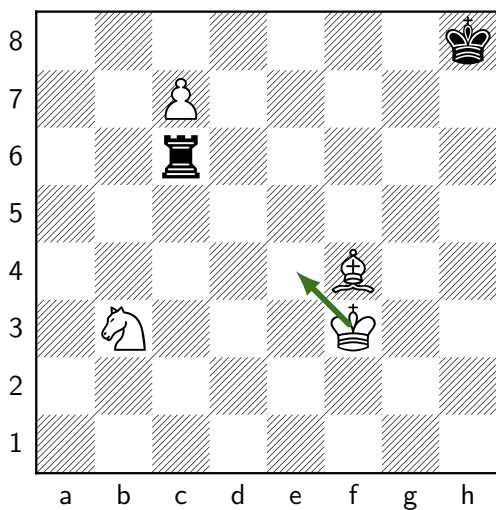
(c)



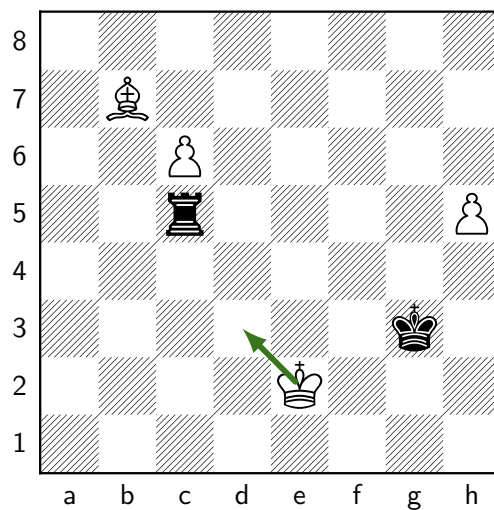
(d)

Figure 53: **Move 59, White: Kf3**. (a) Our position; Melanie plays Kf3 (f2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3412` (ply 133) [structural distance  $d=6$ , QAP obj=20]: the master played Kf5; under  $\mu$  our Kf2 plays the role of that game’s Kg4; cross-type roles: our Nb3 as their Pd4; transferred move f2f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#320` (ply 116) [structural distance  $d=9$ , QAP obj=12]: the master played Ke3; under  $\mu$  our Kf2 plays the role of that game’s Kd3; cross-type roles: our Nb3 as their pe5; transferred move f2f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#287` (ply 101) [structural distance  $d=8$ , QAP obj=4]: the master played Ke3; under  $\mu$  our Kf2 plays the role of that game’s Kd7; cross-type roles: our Nb3 as their Pf6; transferred move f2f3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kf3 (summed QAP objective 39, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

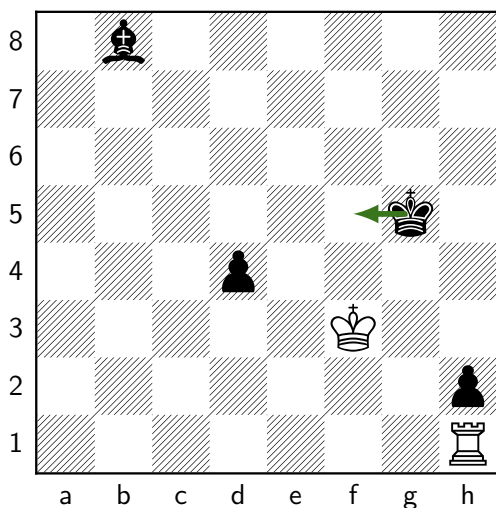




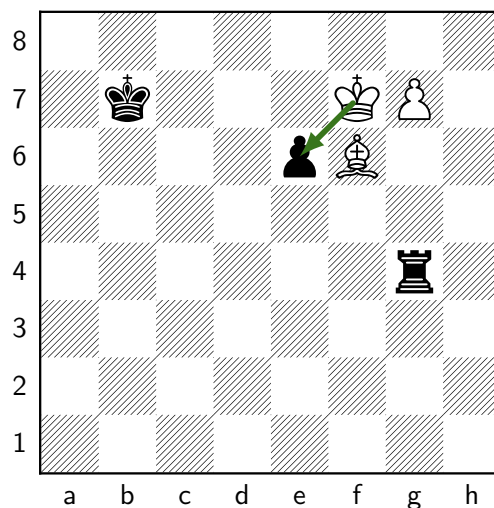
(a)



(b)

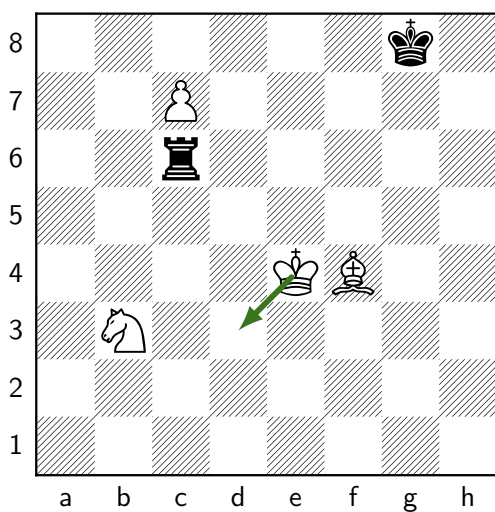


(c)

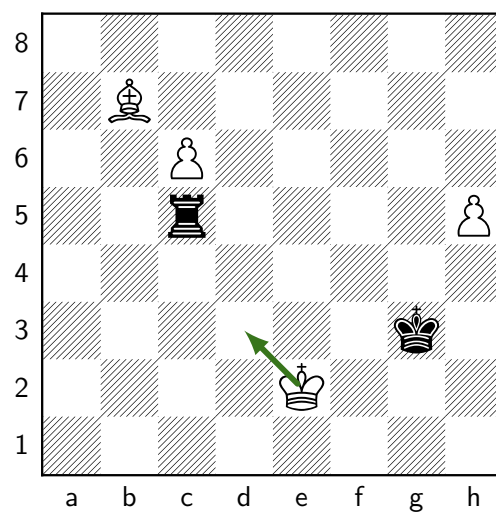


(d)

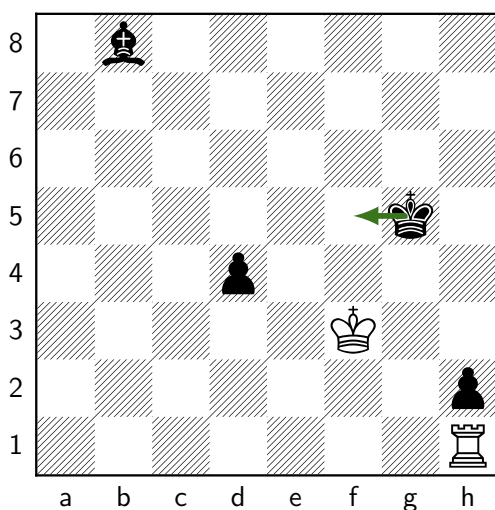
Figure 54: **Move 60, White: Ke4**. (a) Our position; Melanie plays Ke4 (f3e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0001#3006 (ply 134) [structural distance  $d=10$ , QAP obj=20]: the master played Kd3; under  $\mu$  our Kf3 plays the role of that game’s Ke2; cross-type roles: our Nb3 as their Ph5; transferred move f3e4 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0002#3412 (ply 133) [structural distance  $d=10$ , QAP obj=19]: the master played Kf5; under  $\mu$  our Kf3 plays the role of that game’s Kg4; cross-type roles: our Nb3 as their Pd4; transferred move f3e4 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0000#2462 (ply 114) [structural distance  $d=8$ , QAP obj=10]: the master played Kxe6; under  $\mu$  our Kf3 plays the role of that game’s Kf7; cross-type roles: our Nb3 as their pe6; transferred move f3e4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ke4 (summed QAP objective 56, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



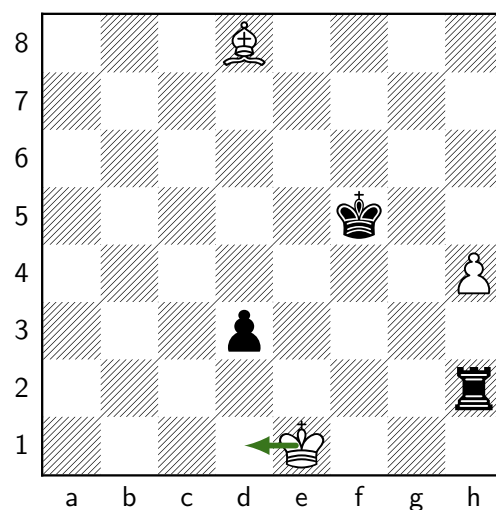
(a)



(b)

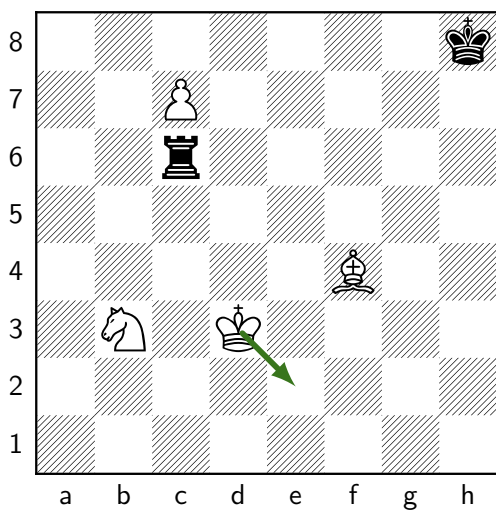


(c)

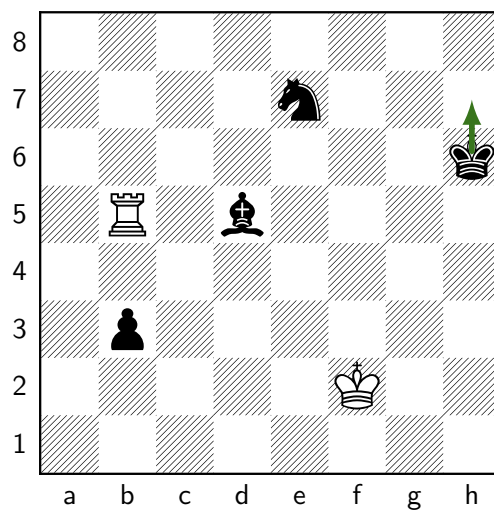


(d)

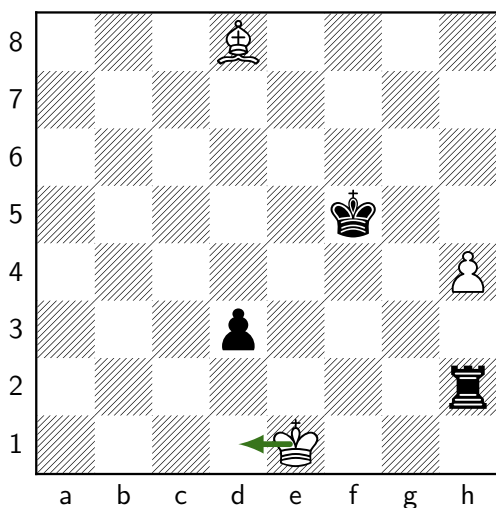
Figure 55: **Move 61, White: Kd3.** (a) Our position; Melanie plays Kd3 (e4d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3006` (ply 134) [structural distance  $d=10$ , QAP obj=20]: the master played Kd3; under  $\mu$  our  $Ke4$  plays the role of that game's  $Ke2$ ; cross-type roles: our  $Nb3$  as their  $Ph5$ ; transferred move e4d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3412` (ply 133) [structural distance  $d=10$ , QAP obj=19]: the master played Kf5; under  $\mu$  our  $Ke4$  plays the role of that game's  $Kg4$ ; cross-type roles: our  $Nb3$  as their  $Pd4$ ; transferred move e4d3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2581` (ply 110) [structural distance  $d=9$ , QAP obj=10]: the master played Kd1; under  $\mu$  our  $Ke4$  plays the role of that game's  $Ke1$ ; cross-type roles: our  $Nb3$  as their  $pd3$ ; transferred move e4d3 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kd3 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



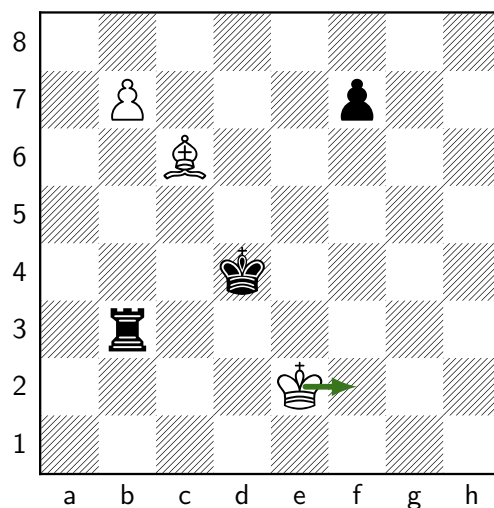
(a)



(b)



(c)



(d)

Figure 56: **Move 62, White: Ke2.** (a) Our position; Melanie plays Ke2 (d3e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#282` (ply 129) [structural distance  $d=8$ , QAP obj=16]: the master played Kh7; under  $\mu$  our  $Kd3$  plays the role of that game’s  $Kh3$ ; cross-type roles: our  $Bf4$  as their  $Ne7$ , our  $Pc7$  as their  $Bd5$ ; transferred move d3e2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2581` (ply 110) [structural distance  $d=5$ , QAP obj=12]: the master played Kd1; under  $\mu$  our  $Kd3$  plays the role of that game’s  $Ke1$ ; cross-type roles: our  $Nb3$  as their  $pd3$ ; transferred move d3e2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#487` (ply 130) [structural distance  $d=3$ , QAP obj=9]: the master played Kf2; under  $\mu$  our  $Kd3$  plays the role of that game’s  $Ke2$ ; cross-type roles: our  $Nb3$  as their  $pf7$ ; transferred move d3e2 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Ke2 (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

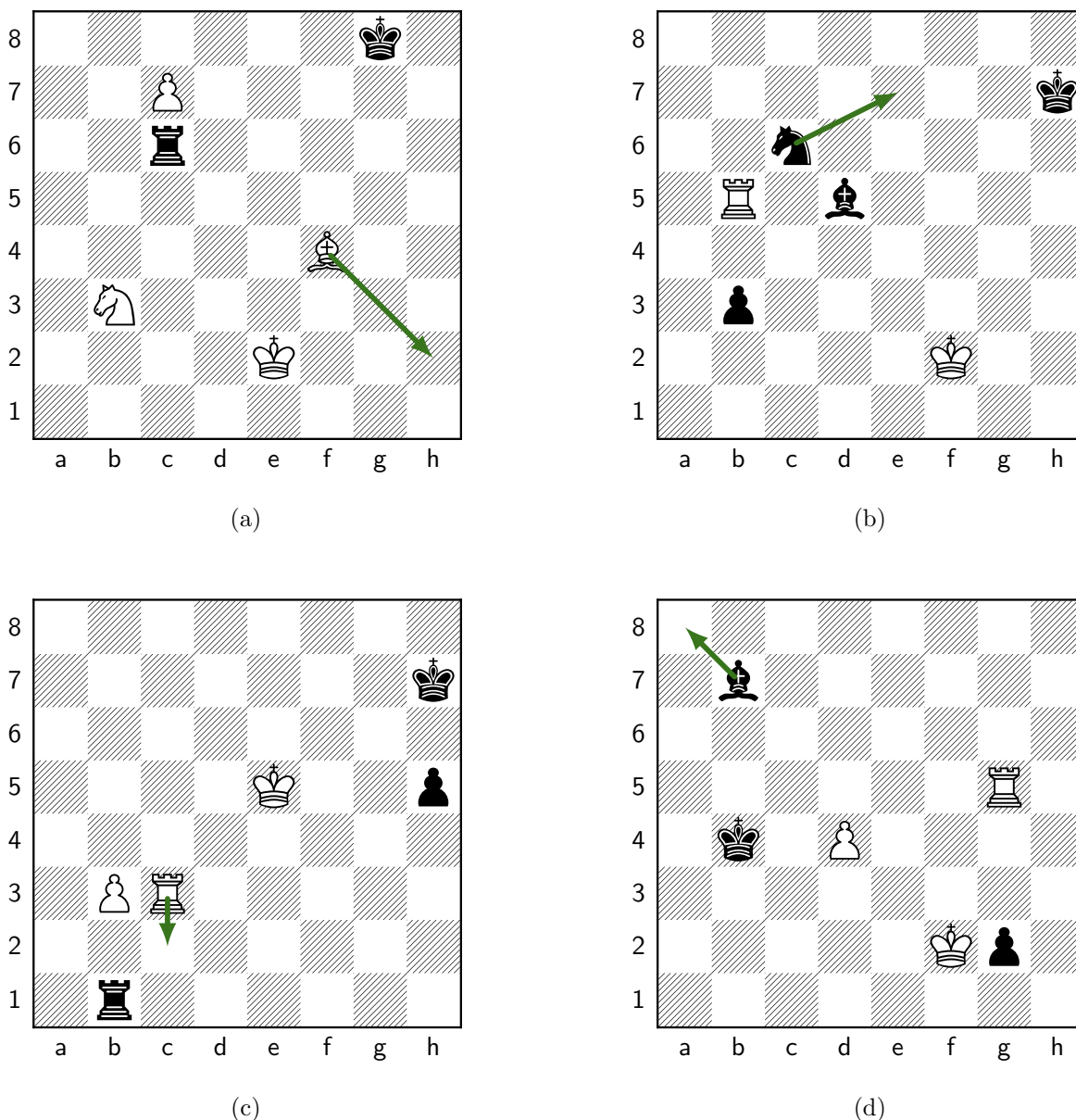
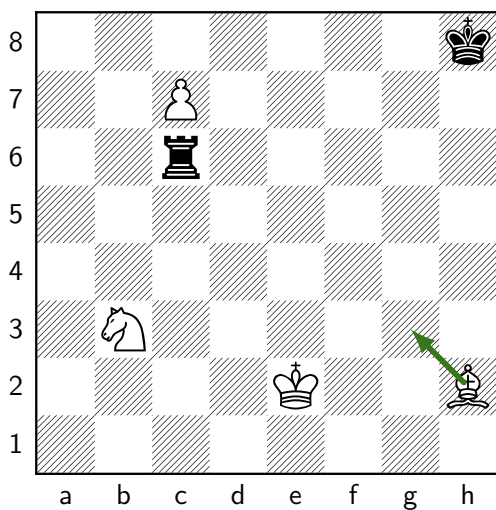
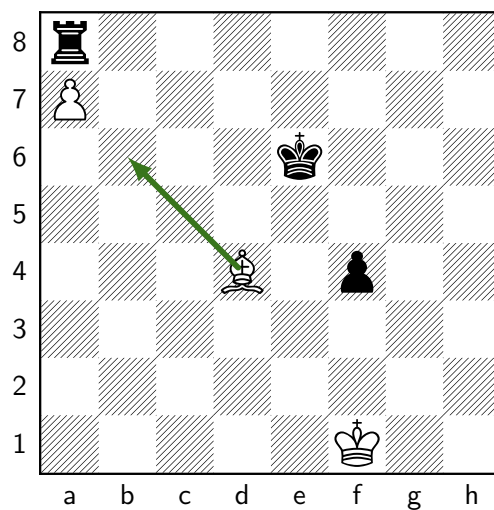


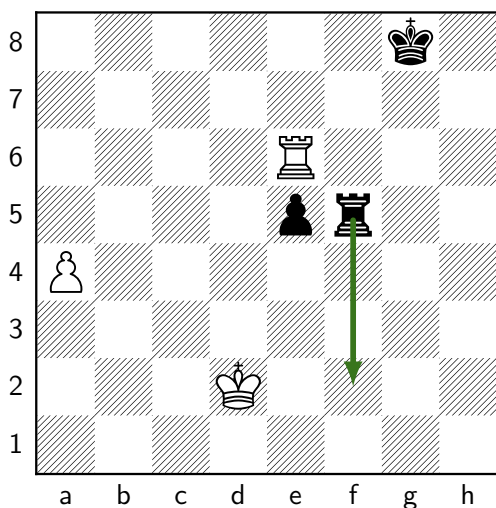
Figure 57: **Move 63, White: Bh2.** (a) Our position; Melanie plays Bh2 (f4h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#282` (ply 137) [structural distance  $d=7$ , QAP obj=14]: the master played Ne7; under  $\mu$  our Bf4 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Bd5; transferred move f4h2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3417` (ply 110) [structural distance  $d=9$ , QAP obj=12]: the master played Rc2; under  $\mu$  our Bf4 plays the role of that game’s Rc3 (a **cross-type** correspondence); cross-type roles: our Nb3 as their ph5; transferred move f4h2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#144` (ply 133) [structural distance  $d=8$ , QAP obj=8]: the master played Ba8; under  $\mu$  our Bf4 plays the role of that game’s Bb2; cross-type roles: our Nb3 as their pd4; transferred move f4h2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bh2 (summed QAP objective 47, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



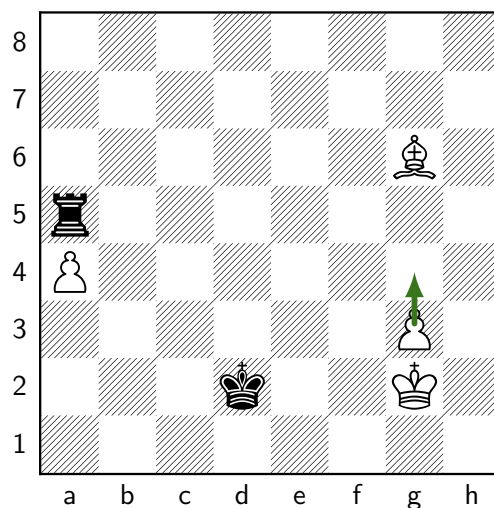
(a)



(b)

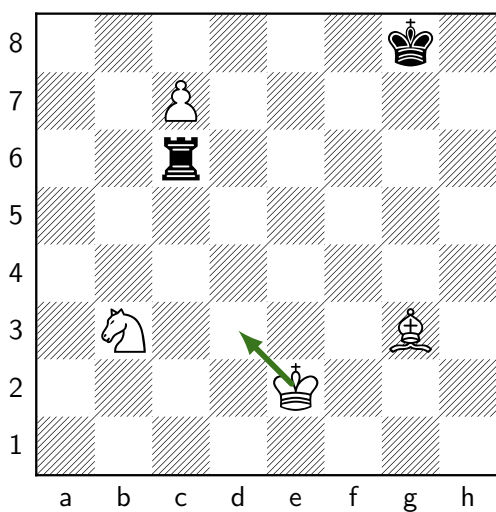


(c)

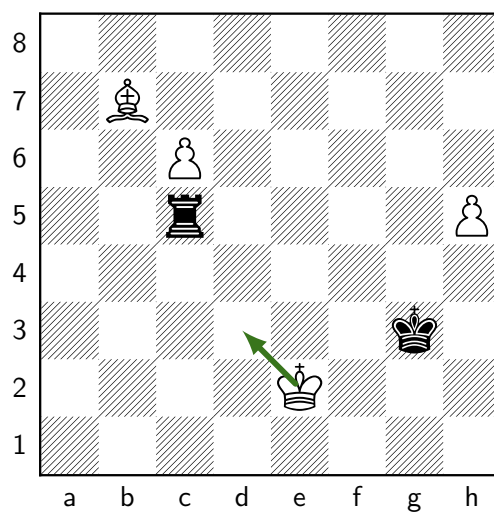


(d)

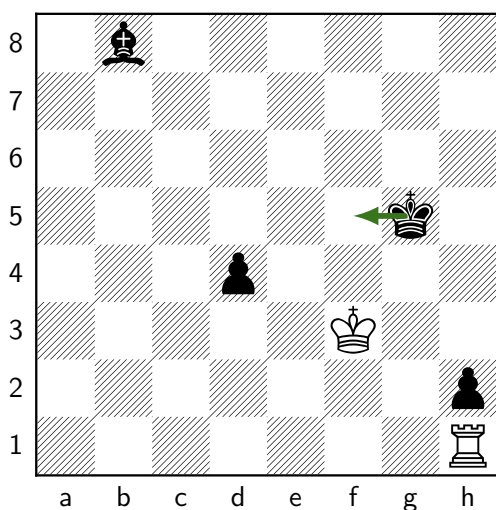
Figure 58: **Move 64, White: Bg3**. (a) Our position; Melanie plays Bg3 (h2g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0002#2824 (ply 122) [structural distance  $d=4$ , QAP obj=20]: the master played Bb6; under  $\mu$  our Bh2 plays the role of that game’s Bd4; cross-type roles: our Nb3 as their pf4; transferred move h2g3 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0000#2559 (ply 127) [structural distance  $d=9$ , QAP obj=19]: the master played Rf2+; under  $\mu$  our Bh2 plays the role of that game’s Rf4 (a **cross-type** correspondence); cross-type roles: our Nb3 as their pa4; transferred move h2g3 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0002#1292 (ply 134) [structural distance  $d=8$ , QAP obj=18]: the master played g4; under  $\mu$  our Bh2 plays the role of that game’s Pg3 (a **cross-type** correspondence); cross-type roles: our Nb3 as their Bg6; transferred move h2g3 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Bg3 (summed QAP objective 89, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



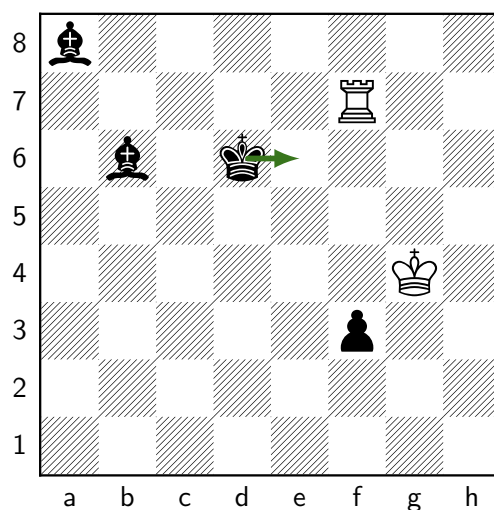
(a)



(b)

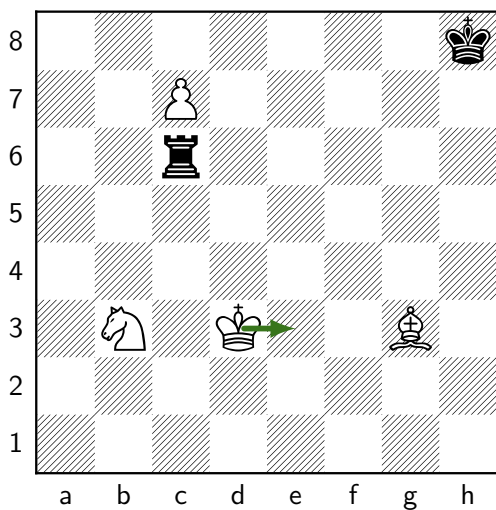


(c)

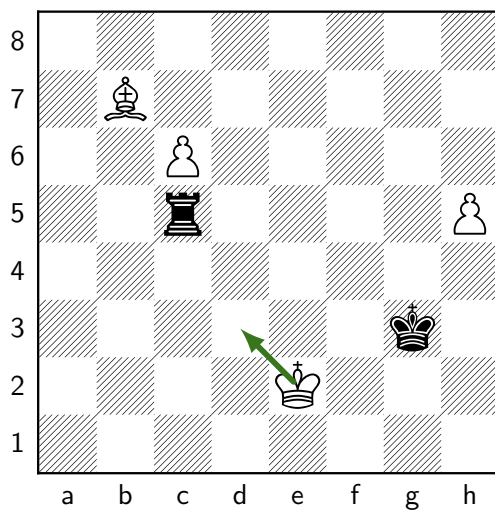


(d)

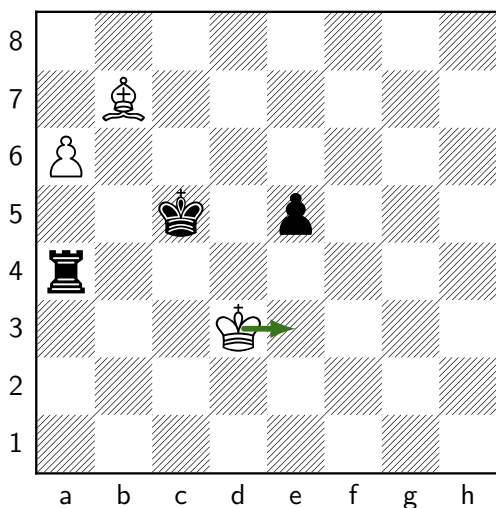
Figure 59: **Move 65, White: Kd3**. (a) Our position; Melanie plays Kd3 (e2d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3006` (ply 134) [structural distance  $d=8$ , QAP obj=20]: the master played Kd3; under  $\mu$  our *Ke2* plays the role of that game’s *Ke2*; cross-type roles: our *Nb3* as their *Ph5*; transferred move e2d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3412` (ply 133) [structural distance  $d=6$ , QAP obj=20]: the master played Kf5; under  $\mu$  our *Ke2* plays the role of that game’s *Kg4*; cross-type roles: our *Nb3* as their *Pd4*; transferred move e2d3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#486` (ply 137) [structural distance  $d=6$ , QAP obj=8]: the master played Ke6; under  $\mu$  our *Ke2* plays the role of that game’s *Kd3*; cross-type roles: our *Nb3* as their *Bb6*; transferred move e2d3 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kd3 (summed QAP objective 68, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



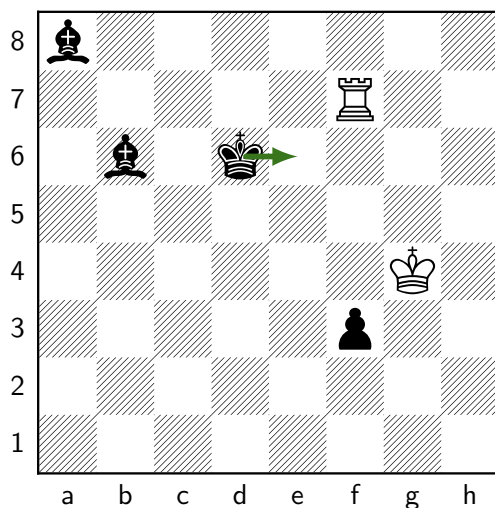
(a)



(b)

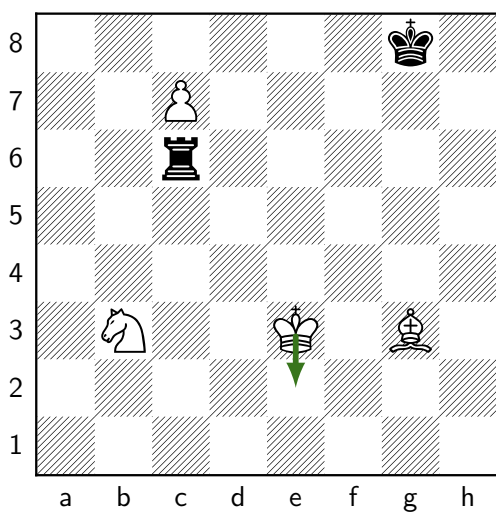


(c)

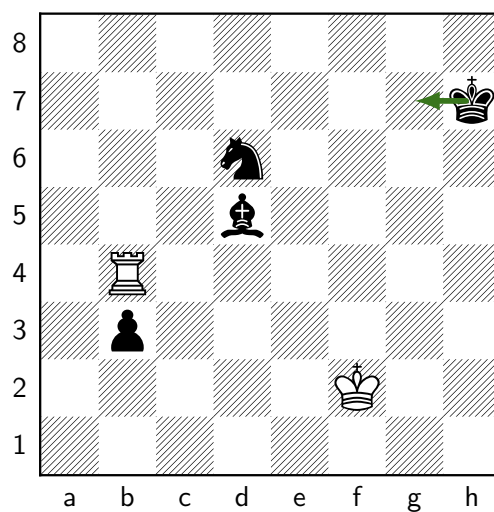


(d)

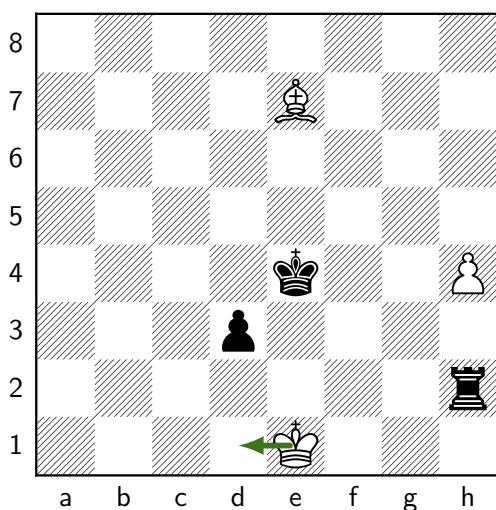
Figure 60: **Move 66, White: Ke3.** (a) Our position; Melanie plays Ke3 (d3e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3006` (ply 134) [structural distance  $d=8$ , QAP obj=20]: the master played Kd3; under  $\mu$  our  $Kd3$  plays the role of that game’s  $Ke2$ ; cross-type roles: our  $Nb3$  as their  $Ph5$ ; transferred move d3e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#320` (ply 116) [structural distance  $d=8$ , QAP obj=12]: the master played Ke3; under  $\mu$  our  $Kd3$  plays the role of that game’s  $Kd3$ ; cross-type roles: our  $Nb3$  as their  $pe5$ ; transferred move d3e3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#486` (ply 137) [structural distance  $d=6$ , QAP obj=10]: the master played Ke6; under  $\mu$  our  $Kd3$  plays the role of that game’s  $Kd3$ ; cross-type roles: our  $Nb3$  as their  $Bb6$ ; transferred move d3e3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ke3 (summed QAP objective 51, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



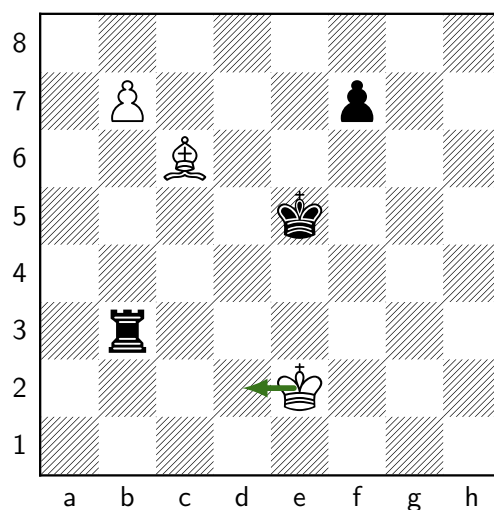
(a)



(b)



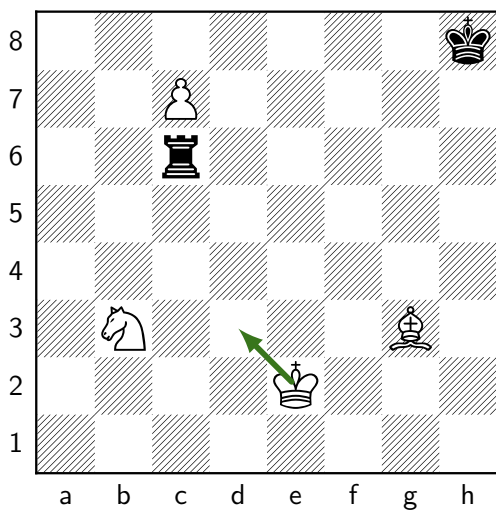
(c)



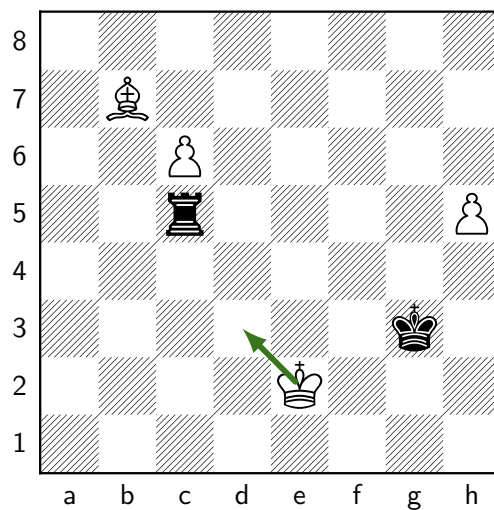
(d)

Figure 61: **Move 67, White: Ke2**. (a) Our position; Melanie plays Ke2 (e3e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#282` (ply 143) [structural distance  $d=0$ , QAP obj=21]: the master played Kg7; under  $\mu$  our Ke3 plays the role of that game's Kh2; transferred move e3e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2581` (ply 124) [structural distance  $d=8$ , QAP obj=11]: the master played Kd1; under  $\mu$  our Ke3 plays the role of that game's Ke1; cross-type roles: our Nb3 as their pd3; transferred move e3e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#487` (ply 148) [structural distance  $d=3$ , QAP obj=7]: the master played Kd2; under  $\mu$  our Ke3 plays the role of that game's Ke2; cross-type roles: our Nb3 as their pf7; transferred move e3e2 (**endorses this move**). Synthesis: 15 retrieved analogs transferred to Ke2 (summed QAP objective 81, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

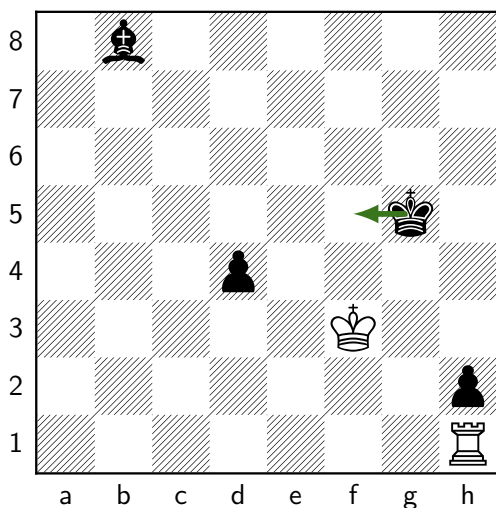




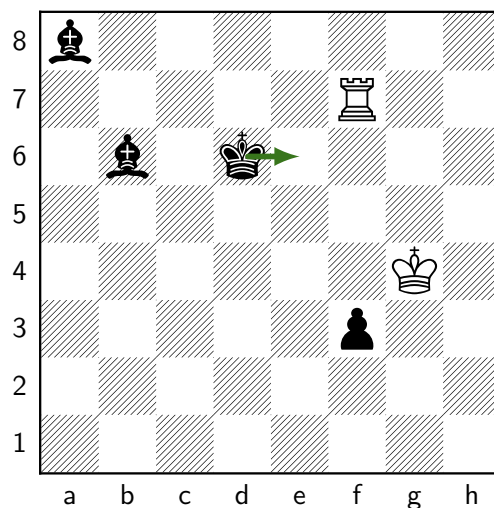
(a)



(b)

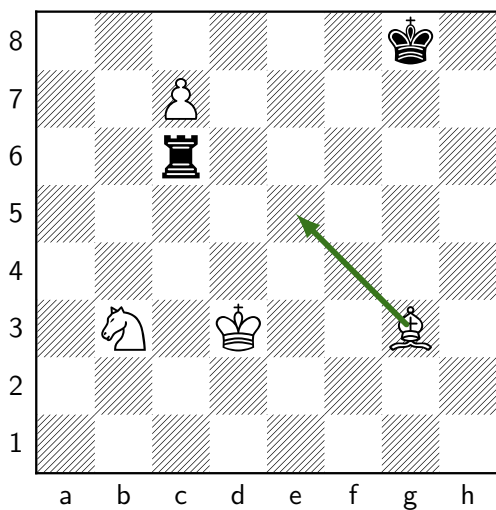


(c)

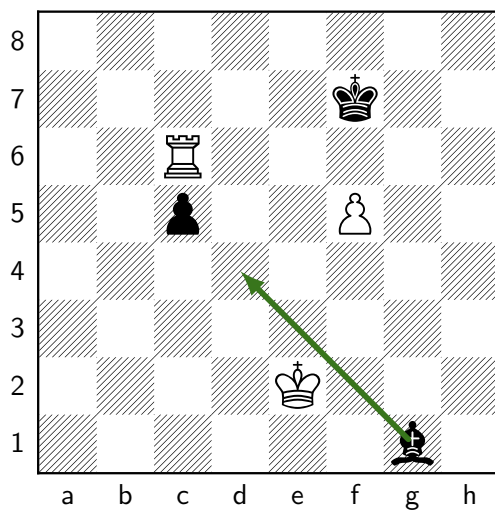


(d)

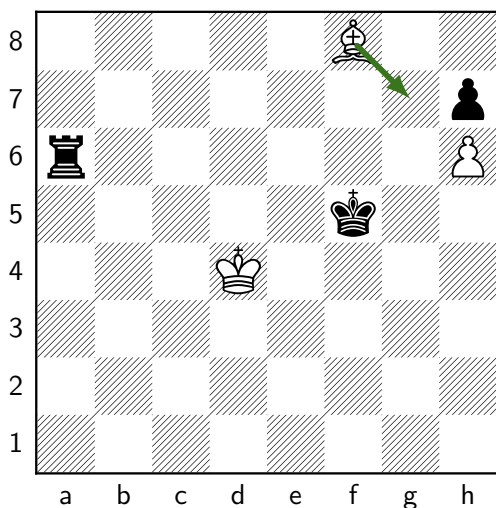
Figure 62: **Move 68, White: Kd3.** (a) Our position; Melanie plays Kd3 (e2d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3006` (ply 134) [structural distance  $d=8$ , QAP obj=20]: the master played Kd3; under  $\mu$  our *Ke2* plays the role of that game's *Ke2*; cross-type roles: our *Nb3* as their *Ph5*; transferred move e2d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3412` (ply 133) [structural distance  $d=6$ , QAP obj=20]: the master played Kf5; under  $\mu$  our *Ke2* plays the role of that game's *Kg4*; cross-type roles: our *Nb3* as their *Pd4*; transferred move e2d3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#486` (ply 137) [structural distance  $d=6$ , QAP obj=8]: the master played Ke6; under  $\mu$  our *Ke2* plays the role of that game's *Kd3*; cross-type roles: our *Nb3* as their *Bb6*; transferred move e2d3 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kd3 (summed QAP objective 74, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



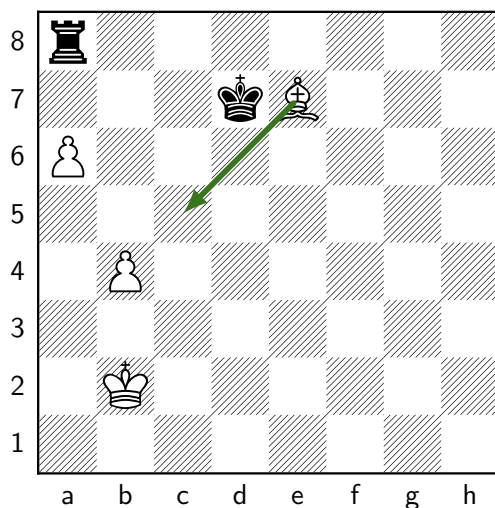
(a)



(b)

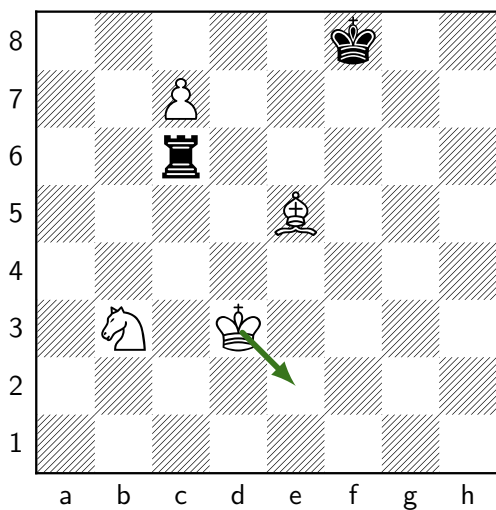


(c)

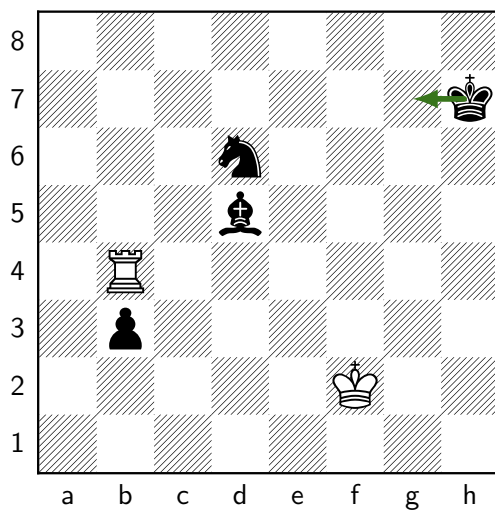


(d)

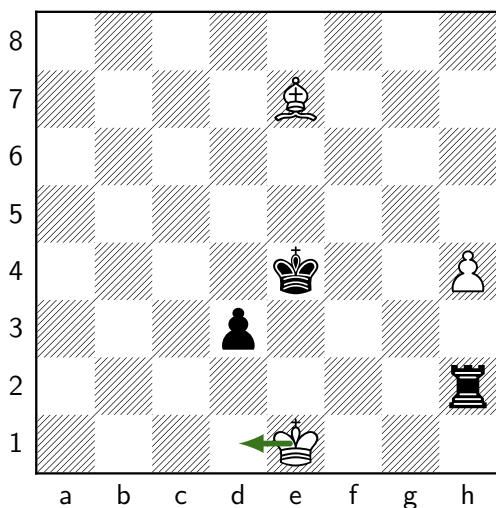
Figure 63: **Move 69, White: Be5**. (a) Our position; Melanie plays Be5 (g3e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3092` (ply 125) [structural distance  $d=5$ , QAP obj=22]: the master played Bd4; under  $\mu$  our Bg3 plays the role of that game’s Bg8; cross-type roles: our Nb3 as their pf5; transferred move g3e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2574` (ply 152) [structural distance  $d=5$ , QAP obj=3]: the master played Bg7; under  $\mu$  our Bg3 plays the role of that game’s Bf8; cross-type roles: our Nb3 as their ph7; transferred move g3e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2870` (ply 132) [structural distance  $d=6$ , QAP obj=2]: the master played Bc5; under  $\mu$  our Bg3 plays the role of that game’s Be7; cross-type roles: our Nb3 as their Pa6; transferred move g3e5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Be5 (summed QAP objective 69, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



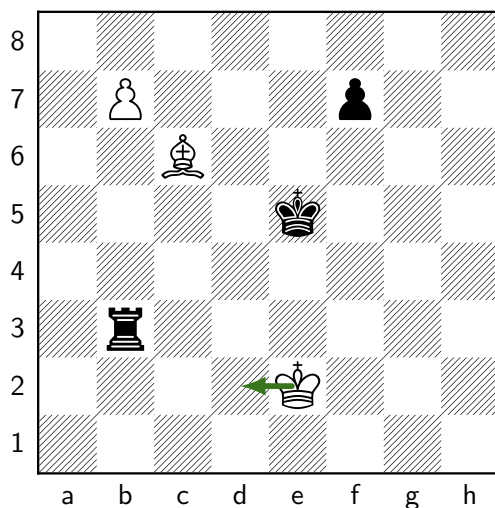
(a)



(b)



(c)



(d)

Figure 64: **Move 70, White: Ke2**. (a) Our position; Melanie plays Ke2 (d3e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#282` (ply 143) [structural distance  $d=0$ , QAP obj=24]: the master played Kg7; under  $\mu$  our *Kd3* plays the role of that game’s *Kh2*; transferred move d3e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2581` (ply 124) [structural distance  $d=8$ , QAP obj=11]: the master played Kd1; under  $\mu$  our *Kd3* plays the role of that game’s *Ke1*; cross-type roles: our *Nb3* as their *pd3*; transferred move d3e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#487` (ply 148) [structural distance  $d=2$ , QAP obj=9]: the master played Kd2; under  $\mu$  our *Kd3* plays the role of that game’s *Ke2*; cross-type roles: our *Nb3* as their *pf7*; transferred move d3e2 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Ke2 (summed QAP objective 77, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

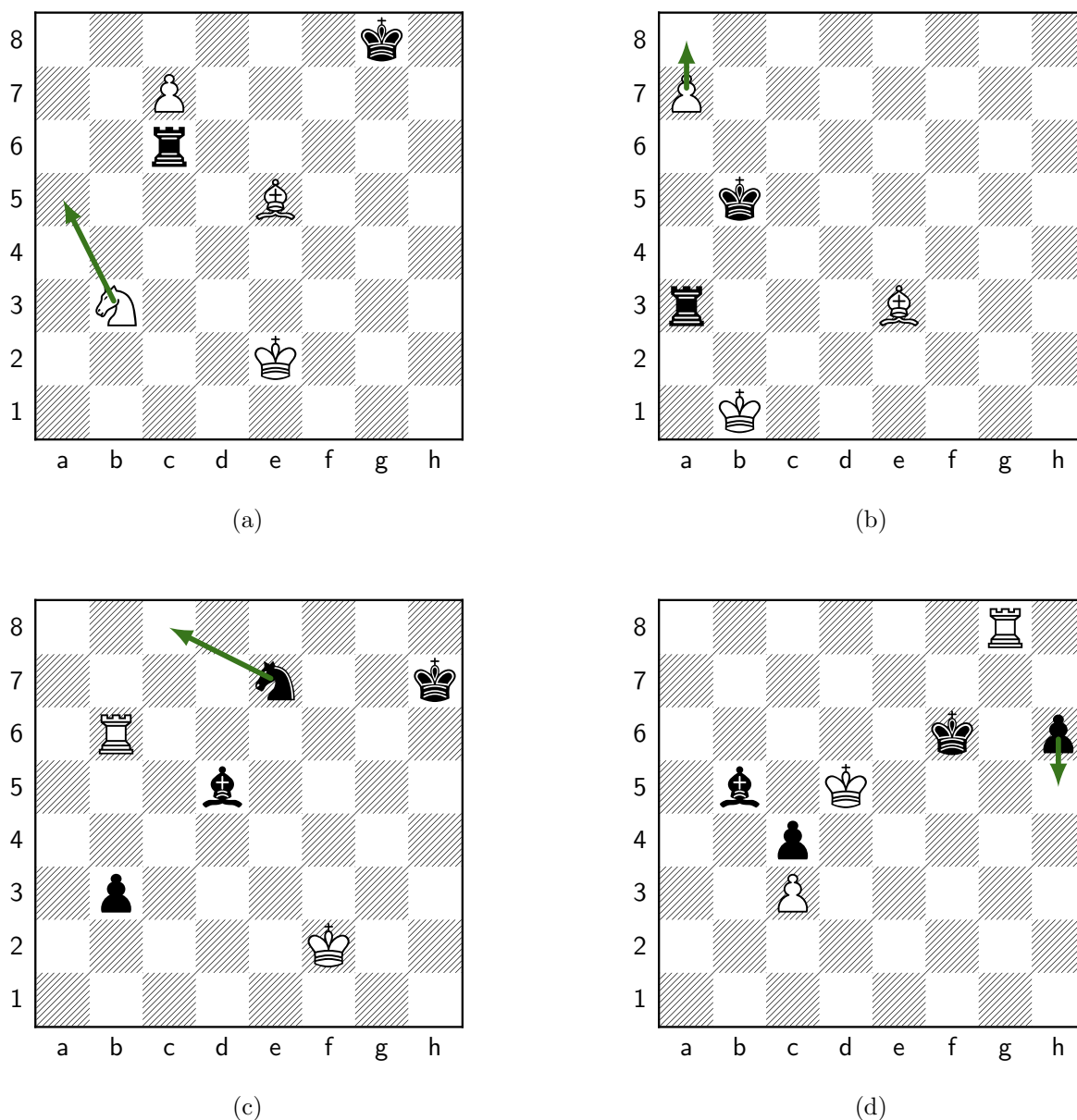
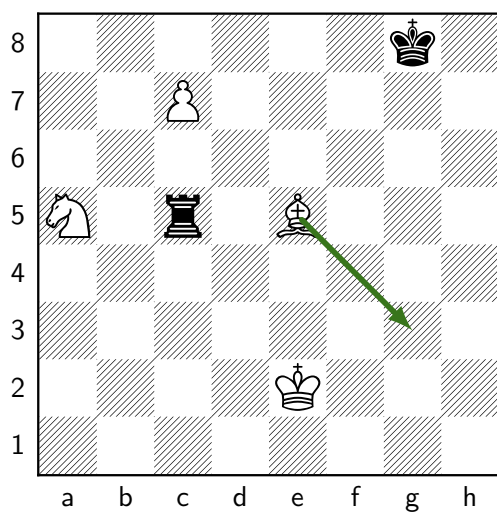
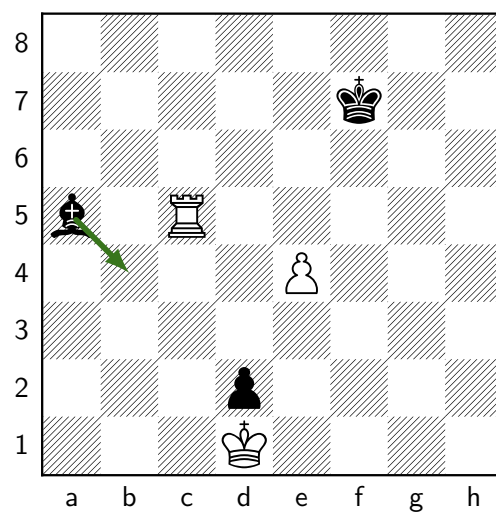


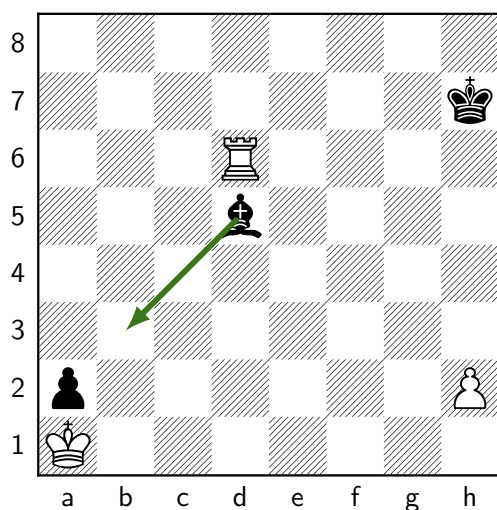
Figure 65: **Move 71, White: Na5.** (a) Our position; Melanie plays Na5 (b3a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#929` (ply 146) [structural distance  $d=5$ , QAP obj=13]: the master played a8=Q; under  $\mu$  our Nb3 plays the role of that game’s Pa7 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Be3; transferred move b3a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#282` (ply 139) [structural distance  $d=4$ , QAP obj=11]: the master played Nc8; under  $\mu$  our Nb3 plays the role of that game’s Ne2; transferred move b3a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1555` (ply 133) [structural distance  $d=9$ , QAP obj=2]: the master played h5; under  $\mu$  our Nb3 plays the role of that game’s Ph3 (a **cross-type** correspondence); transferred move b3a5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Na5 (summed QAP objective 28, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



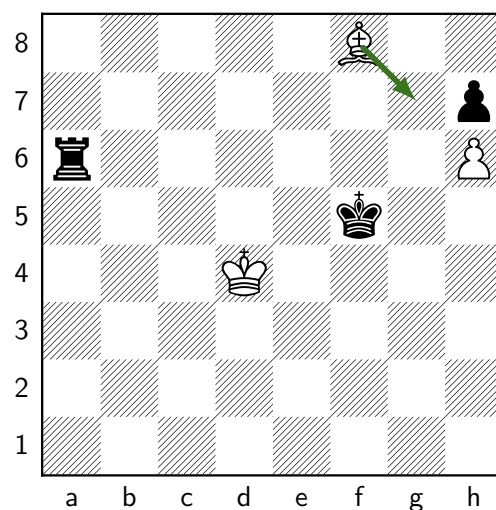
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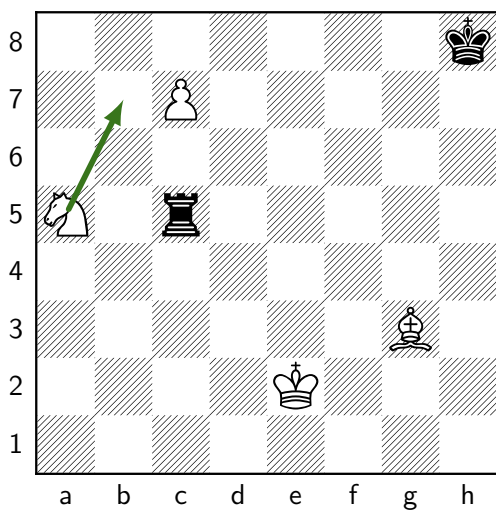


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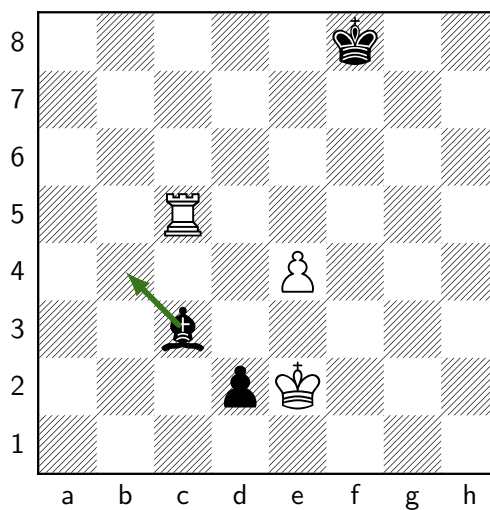


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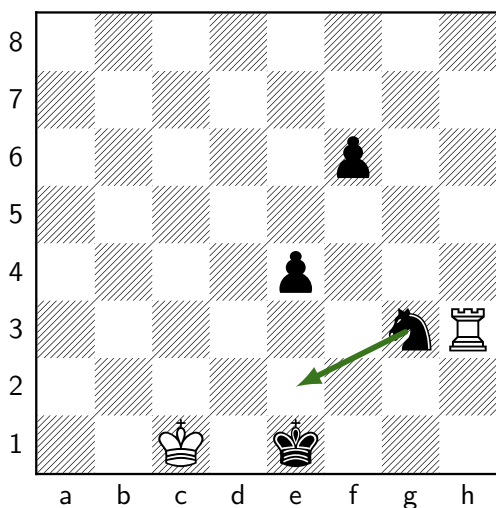
Figure 66: **Move 72, White: Bg3 (e5g3)**. (a) Our position; Melanie plays Bg3 (e5g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1034` (ply 133) [structural distance  $d=13$ , QAP obj=20]: the master played Bb4; under  $\mu$  our Be5 plays the role of that game’s Ba4; cross-type roles: our Na5 as their pe4; transferred move e5g3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1646` (ply 139) [structural distance  $d=12$ , QAP obj=11]: the master played Bb3; under  $\mu$  our Be5 plays the role of that game’s Bd4; cross-type roles: our Na5 as their ph2; transferred move e5g3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2574` (ply 152) [structural distance  $d=14$ , QAP obj=3]: the master played Bg7; under  $\mu$  our Be5 plays the role of that game’s Bf8; cross-type roles: our Na5 as their ph7; transferred move e5g3 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Bg3 (summed QAP objective 100, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



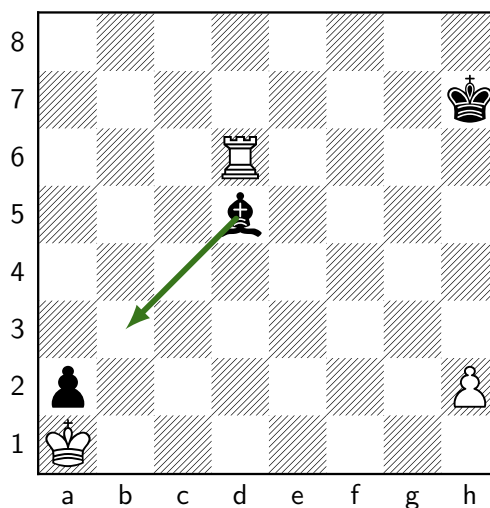
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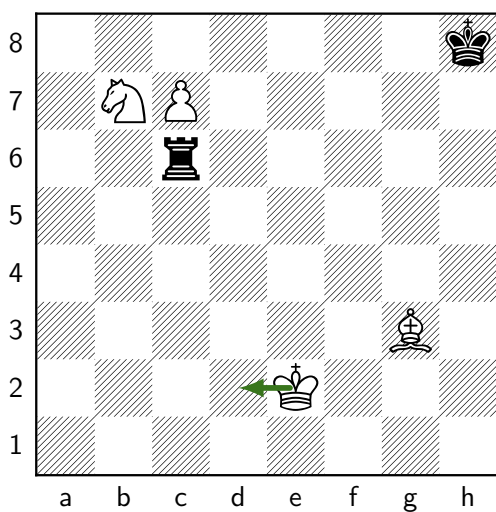


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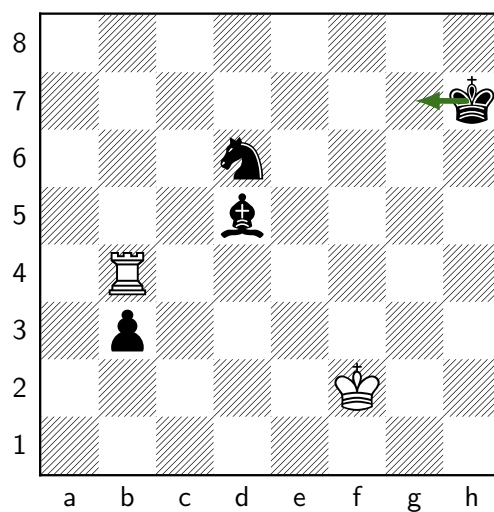


(d)

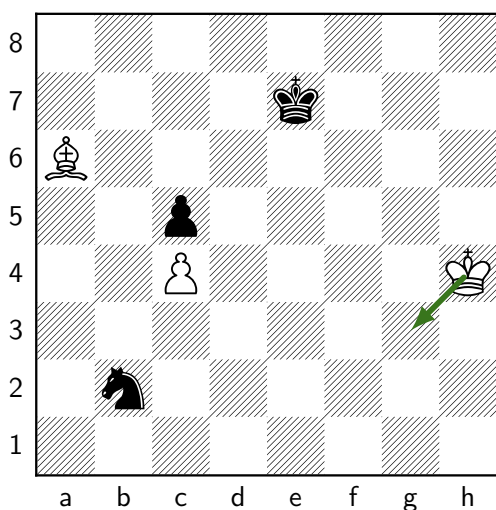
Figure 67: **Move 73, White: Nb7 (a5b7)**. (a) Our position; Melanie plays Nb7 (a5b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1034` (ply 161) [structural distance  $d=8$ , QAP obj=18]: the master played Bb4; under  $\mu$  our Na5 plays the role of that game’s Bc6 (a **cross-type** correspondence); cross-type roles: our Bg3 as their pe4; transferred move a5b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#391` (ply 139) [structural distance  $d=10$ , QAP obj=10]: the master played Ne2+; under  $\mu$  our Na5 plays the role of that game’s Ng6; cross-type roles: our Bg3 as their Pe4; transferred move a5b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1646` (ply 139) [structural distance  $d=10$ , QAP obj=9]: the master played Bb3; under  $\mu$  our Na5 plays the role of that game’s Bd4 (a **cross-type** correspondence); cross-type roles: our Bg3 as their ph2; transferred move a5b7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Nb7 (summed QAP objective 93, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



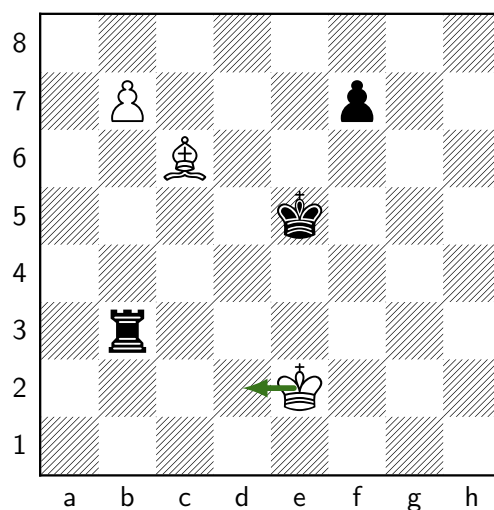
(a)



(b)



(c)



(d)

Figure 68: **Move 74, White: Kd2**. (a) Our position; Melanie plays Kd2 (e2d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#282` (ply 143) [structural distance  $d=0$ , QAP obj=21]: the master played Kg7; under  $\mu$  our *Ke2* plays the role of that game’s *Kh2*; transferred move e2d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1102` (ply 156) [structural distance  $d=8$ , QAP obj=12]: the master played Kg3; under  $\mu$  our *Ke2* plays the role of that game’s *Kh4*; cross-type roles: our *rc6* as their *nb2*, our *Nb7* as their *pc5*; transferred move e2d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#487` (ply 148) [structural distance  $d=3$ , QAP obj=7]: the master played Kd2; under  $\mu$  our *Ke2* plays the role of that game’s *Ke2*; cross-type roles: our *Nb7* as their *pf7*; transferred move e2d2 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kd2 (summed QAP objective 63, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

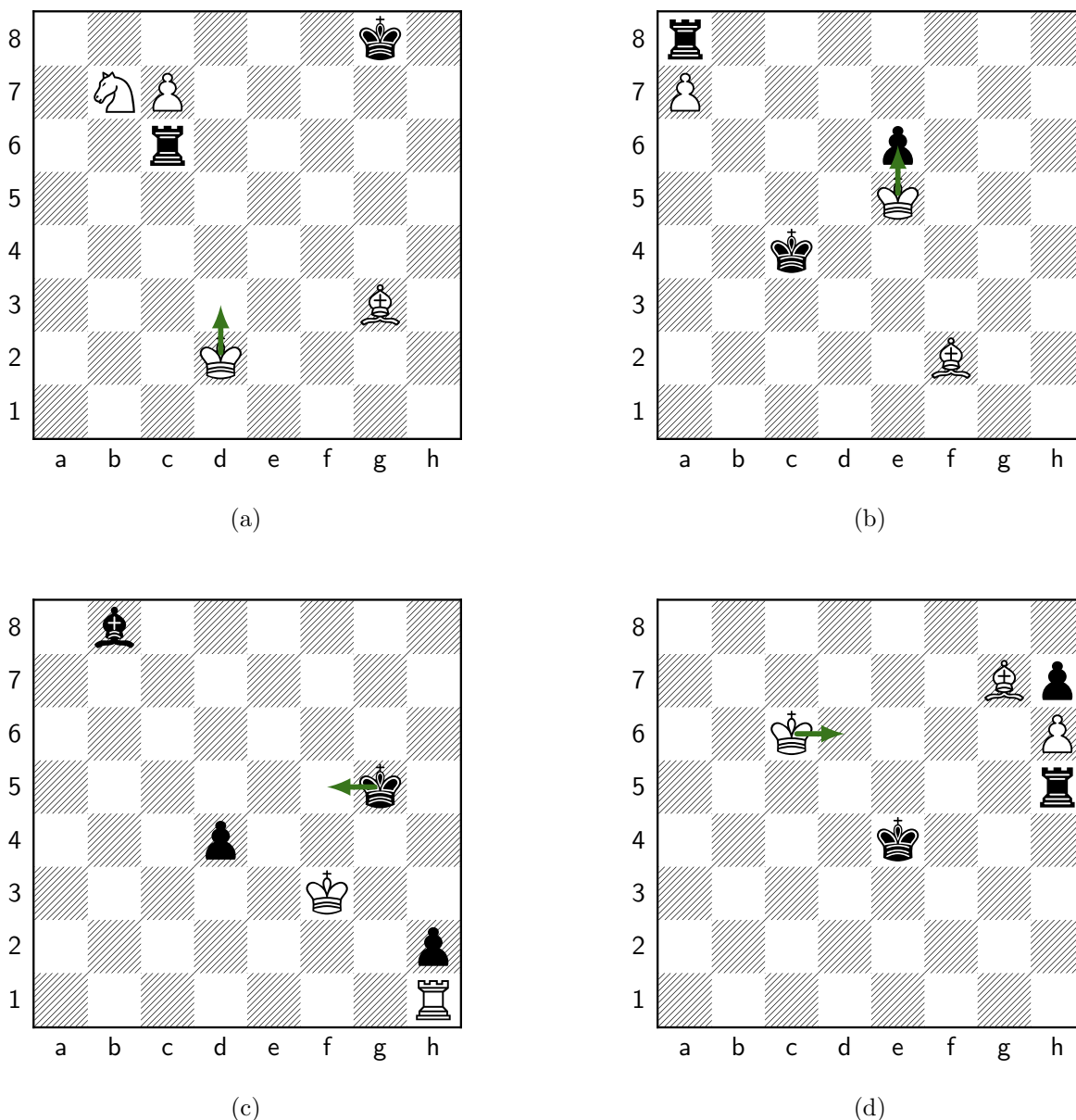
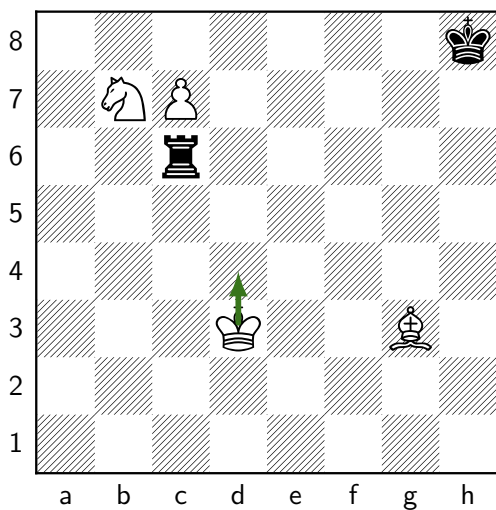
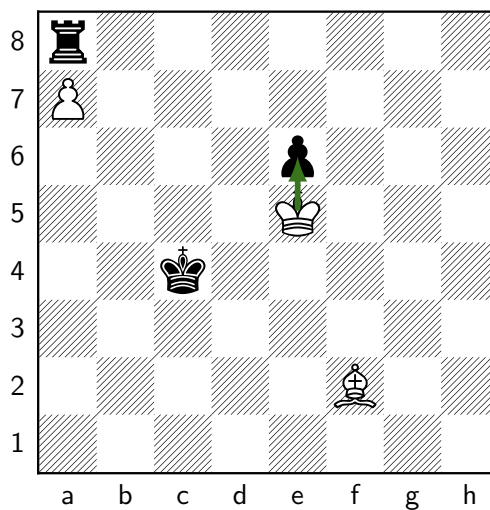


Figure 69: **Move 75, White: Kd3.** (a) Our position; Melanie plays Kd3 (d2d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2062` (ply 152) [structural distance  $d=9$ , QAP obj=21]: the master played Kxe6; under  $\mu$  our Kd2 plays the role of that game’s Ke5; cross-type roles: our Nb7 as their pe6; transferred move d2d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3412` (ply 133) [structural distance  $d=5$ , QAP obj=20]: the master played Kf5; under  $\mu$  our Kd2 plays the role of that game’s Kg4; cross-type roles: our Nb7 as their Pd4; transferred move d2d3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2574` (ply 164) [structural distance  $d=9$ , QAP obj=20]: the master played Kd6; under  $\mu$  our Kd2 plays the role of that game’s Kc6; cross-type roles: our Nb7 as their ph7; transferred move d2d3 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Kd3 (summed QAP objective 116, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

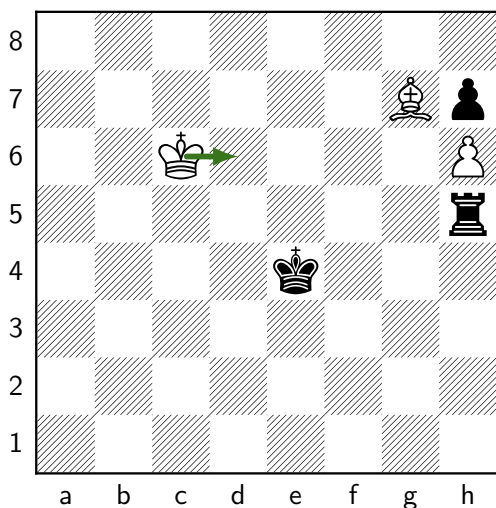




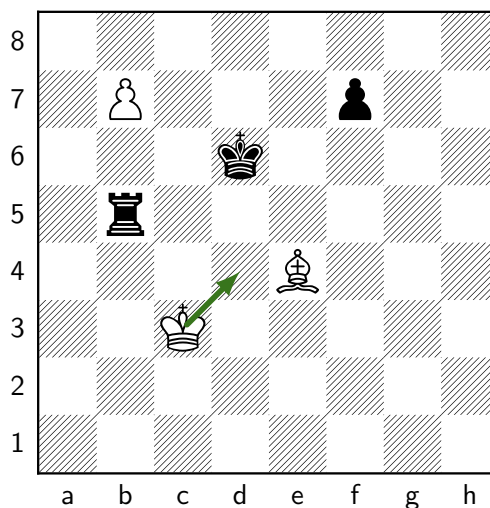
(a)



(b)

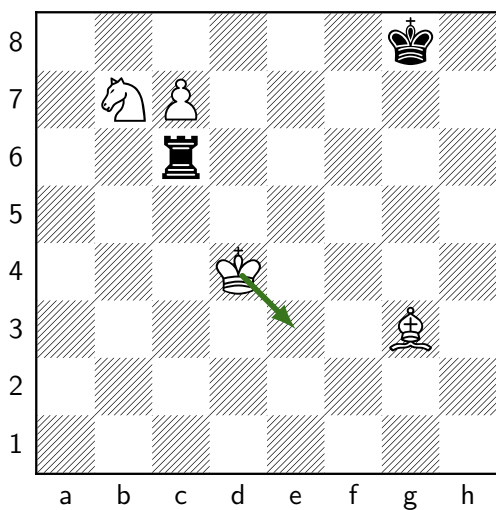


(c)

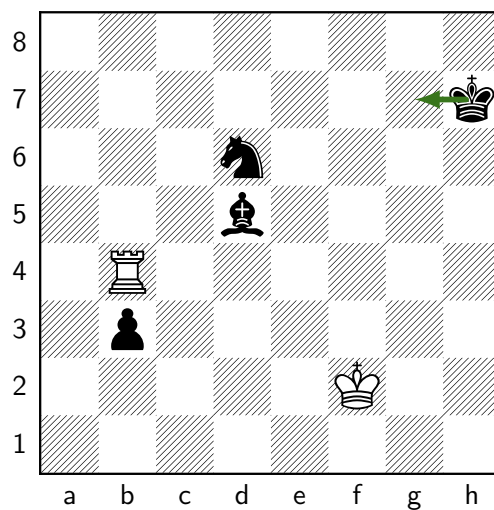


(d)

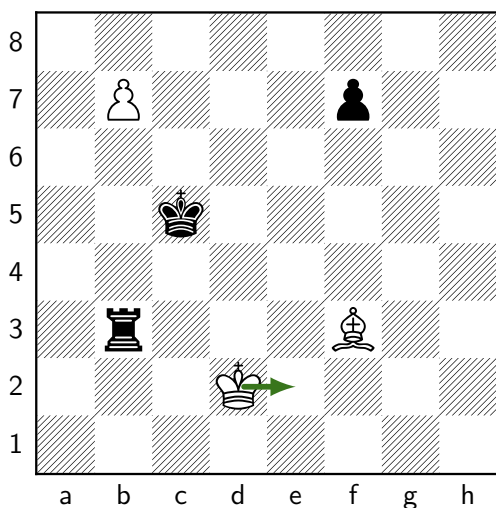
Figure 70: **Move 76, White: Kd4**. (a) Our position; Melanie plays Kd4 (d3d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top-p0000#2062` (ply 152) [structural distance  $d=9$ , QAP obj=21]: the master played Kxe6; under  $\mu$  our *Kd3* plays the role of that game’s *Ke5*; cross-type roles: our *Nb7* as their *pe6*; transferred move d3d4 (**endorses this move**). (c) Retrieved from `ccr1_top-p0001#2574` (ply 164) [structural distance  $d=9$ , QAP obj=20]: the master played Kd6; under  $\mu$  our *Kd3* plays the role of that game’s *Kc6*; cross-type roles: our *Nb7* as their *ph7*; transferred move d3d4 (**endorses this move**). (d) Retrieved from `ccr1_top-p0003#487` (ply 160) [structural distance  $d=3$ , QAP obj=13]: the master played Kd4; under  $\mu$  our *Kd3* plays the role of that game’s *Kc3*; cross-type roles: our *Nb7* as their *pf7*; transferred move d3d4 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kd4 (summed QAP objective 96, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



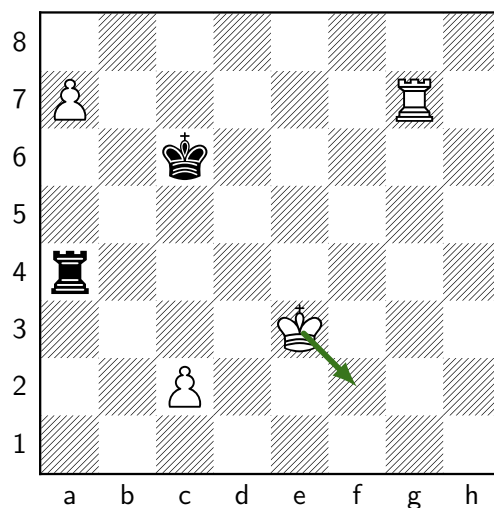
(a)



(b)

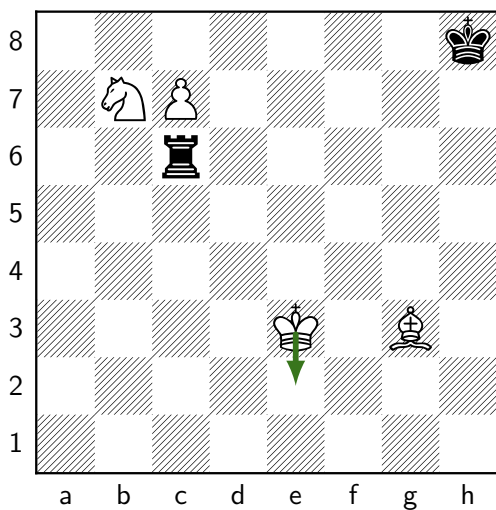


(c)

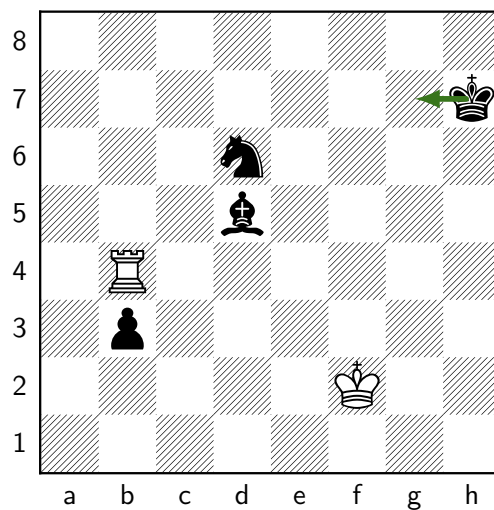


(d)

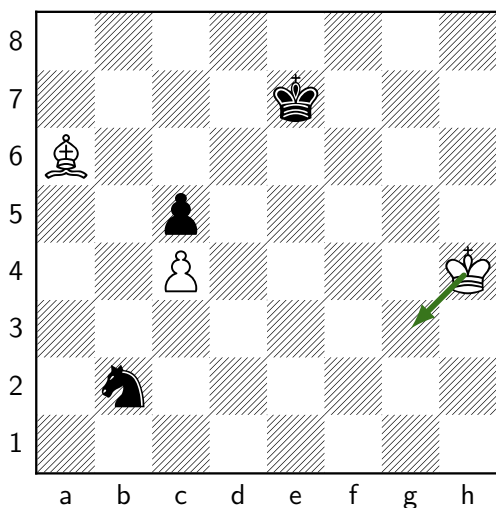
Figure 71: **Move 77, White: Ke3**. (a) Our position; Melanie plays Ke3 (d4e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#282` (ply 143) [structural distance  $d=1$ , QAP obj=21]: the master played Kg7; under  $\mu$  our  $Kd4$  plays the role of that game’s  $Kh2$ ; transferred move d4e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#487` (ply 144) [structural distance  $d=6$ , QAP obj=14]: the master played Ke2; under  $\mu$  our  $Kd4$  plays the role of that game’s  $Kd2$ ; cross-type roles: our  $Pc7$  as their  $Bf3$ , our  $Bg3$  as their  $Pb7$ ; transferred move d4e3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2307` (ply 154) [structural distance  $d=9$ , QAP obj=10]: the master played Kf2; under  $\mu$  our  $Kd4$  plays the role of that game’s  $Ke3$ ; cross-type roles: our  $Nb7$  as their  $Pc2$ , our  $Bg3$  as their  $Rg7$ ; transferred move d4e3 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Ke3 (summed QAP objective 67, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



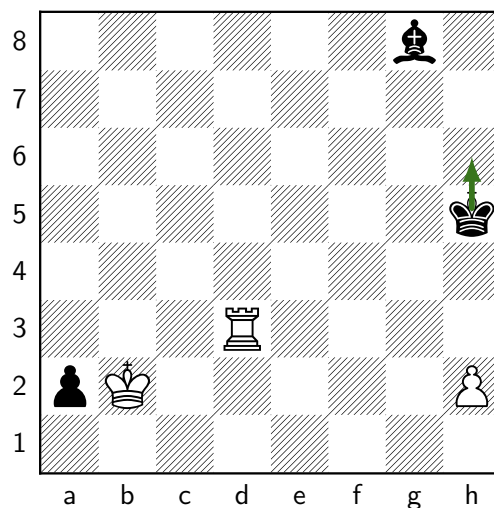
(a)



(b)

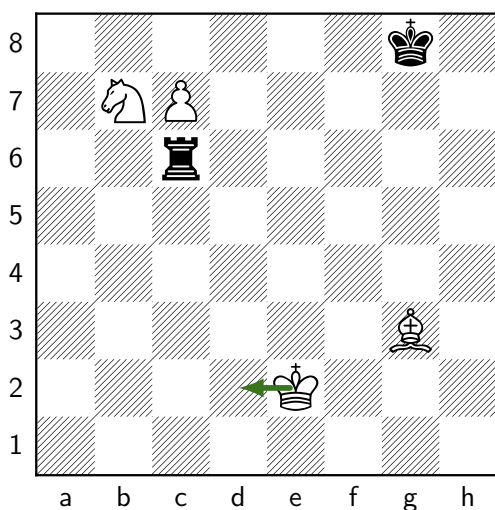


(c)

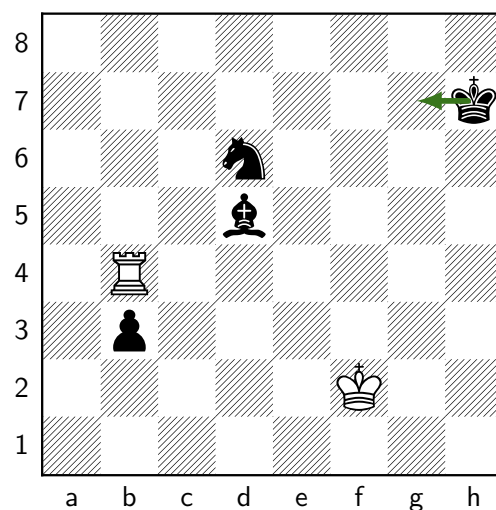


(d)

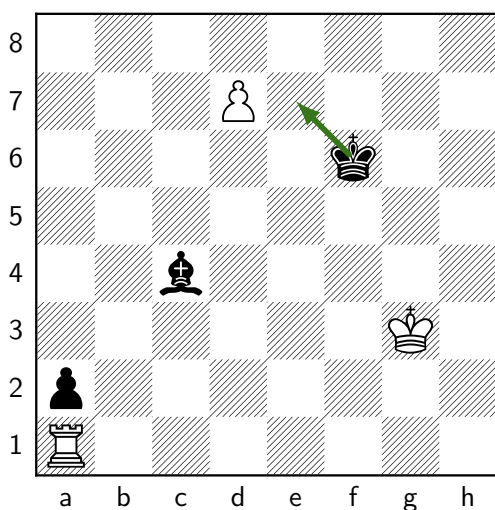
Figure 72: **Move 78, White: Ke2**. (a) Our position; Melanie plays Ke2 (e3e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#282` (ply 143) [structural distance  $d=0$ , QAP obj=21]: the master played Kg7; under  $\mu$  our Ke3 plays the role of that game’s Kh2; transferred move e3e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1102` (ply 156) [structural distance  $d=8$ , QAP obj=12]: the master played Kg3; under  $\mu$  our Ke3 plays the role of that game’s Kh4; cross-type roles: our rc6 as their nb2, our Nb7 as their pc5; transferred move e3e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1646` (ply 149) [structural distance  $d=8$ , QAP obj=1]: the master played Kh6; under  $\mu$  our Ke3 plays the role of that game’s Kh4; cross-type roles: our Nb7 as their ph2; transferred move e3e2 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Ke2 (summed QAP objective 39, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



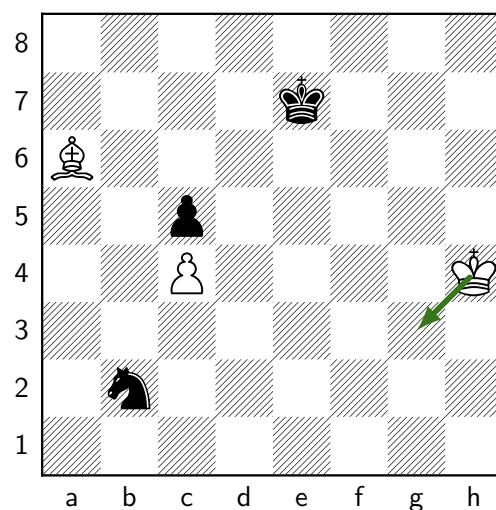
(a)



(b)

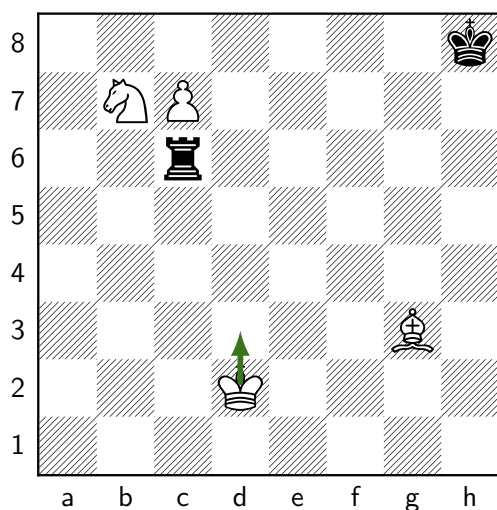


(c)

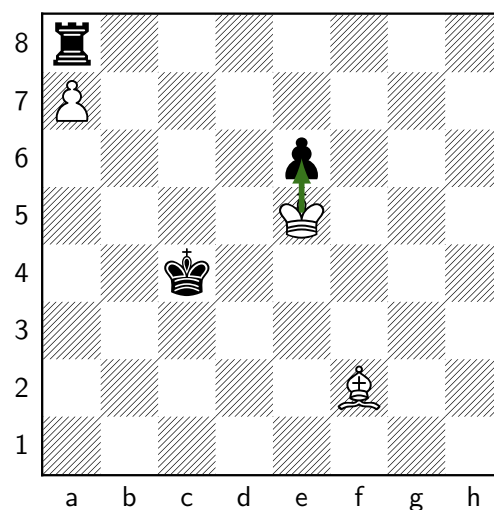


(d)

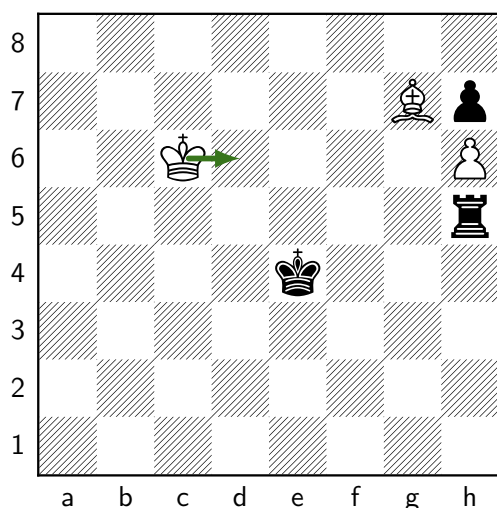
Figure 73: **Move 79, White: Kd2.** (a) Our position; Melanie plays Kd2 (e2d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#282` (ply 143) [structural distance  $d=0$ , QAP obj=21]: the master played Kg7; under  $\mu$  our *Ke2* plays the role of that game’s *Kh2*; transferred move e2d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3771` (ply 173) [structural distance  $d=3$ , QAP obj=20]: the master played Ke7; under  $\mu$  our *Ke2* plays the role of that game’s *Kf3*; cross-type roles: our *Nb7* as their *pd7*; transferred move e2d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1102` (ply 156) [structural distance  $d=8$ , QAP obj=12]: the master played Kg3; under  $\mu$  our *Ke2* plays the role of that game’s *Kh4*; cross-type roles: our *rc6* as their *nb2*, our *Nb7* as their *pc5*; transferred move e2d2 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Kd2 (summed QAP objective 93, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



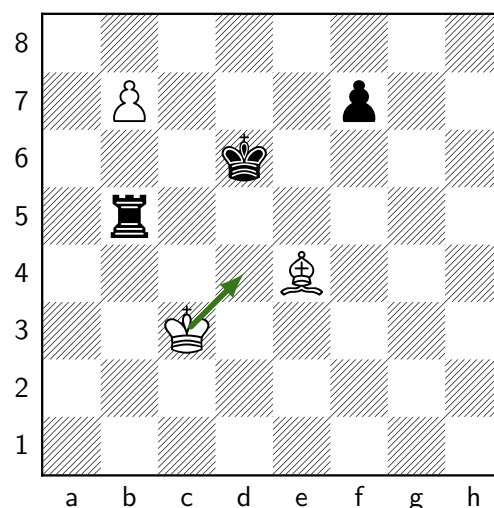
(a)



(b)



(c)



(d)

Figure 74: **Move 80, White: Kd3.** (a) Our position; Melanie plays Kd3 (d2d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2062` (ply 152) [structural distance  $d=9$ , QAP obj=21]: the master played Kxe6; under  $\mu$  our *Kd2* plays the role of that game’s *Ke5*; cross-type roles: our *Nb7* as their *pe6*; transferred move d2d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2574` (ply 164) [structural distance  $d=9$ , QAP obj=20]: the master played Kd6; under  $\mu$  our *Kd2* plays the role of that game’s *Kc6*; cross-type roles: our *Nb7* as their *ph7*; transferred move d2d3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#487` (ply 160) [structural distance  $d=3$ , QAP obj=13]: the master played Kd4; under  $\mu$  our *Kd2* plays the role of that game’s *Kc3*; cross-type roles: our *Nb7* as their *pf7*; transferred move d2d3 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Kd3 (summed QAP objective 85, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

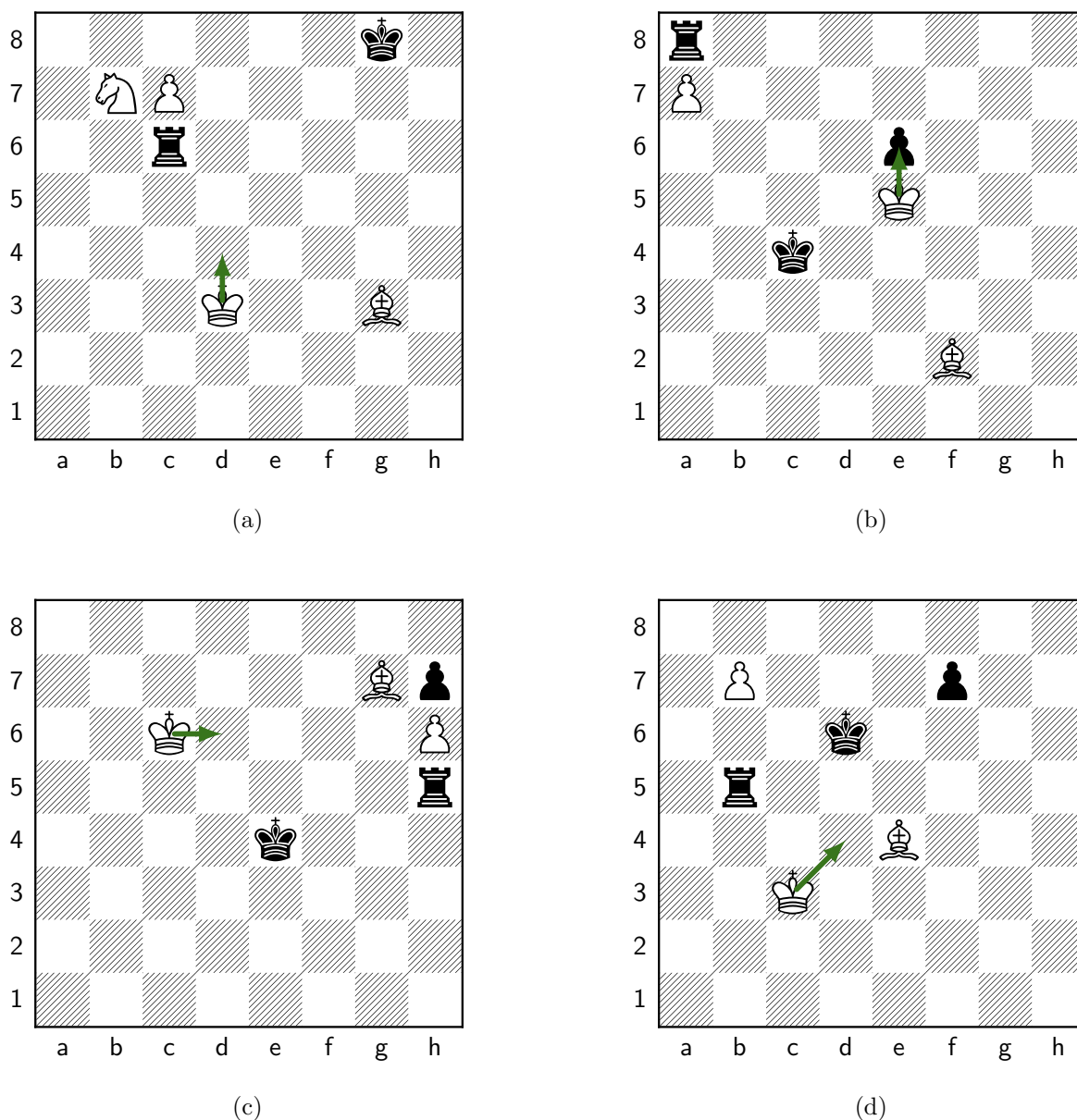
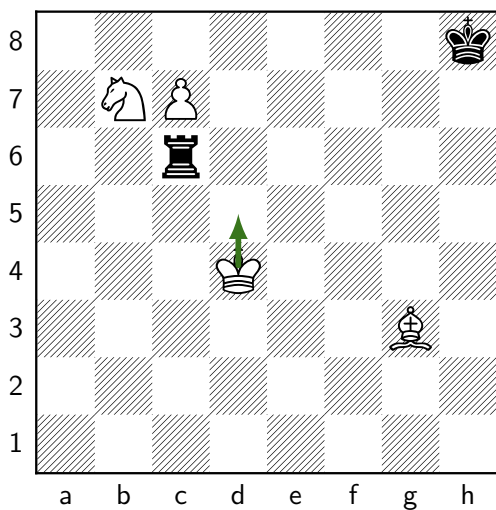
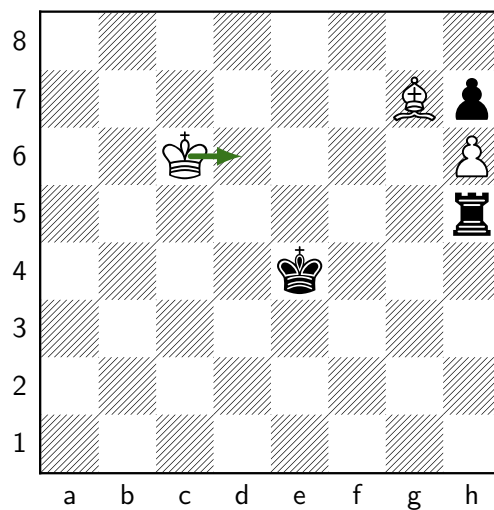


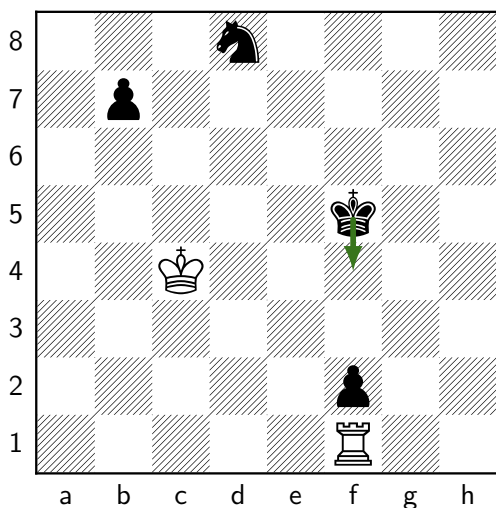
Figure 75: **Move 81, White: Kd4.** (a) Our position; Melanie plays Kd4 (d3d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2062` (ply 152) [structural distance  $d=9$ , QAP obj=21]: the master played Kxe6; under  $\mu$  our Kd3 plays the role of that game’s Ke5; cross-type roles: our Nb7 as their pe6; transferred move d3d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2574` (ply 164) [structural distance  $d=9$ , QAP obj=20]: the master played Kd6; under  $\mu$  our Kd3 plays the role of that game’s Kc6; cross-type roles: our Nb7 as their ph7; transferred move d3d4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#487` (ply 160) [structural distance  $d=3$ , QAP obj=13]: the master played Kd4; under  $\mu$  our Kd3 plays the role of that game’s Kc3; cross-type roles: our Nb7 as their pf7; transferred move d3d4 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kd4 (summed QAP objective 80, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



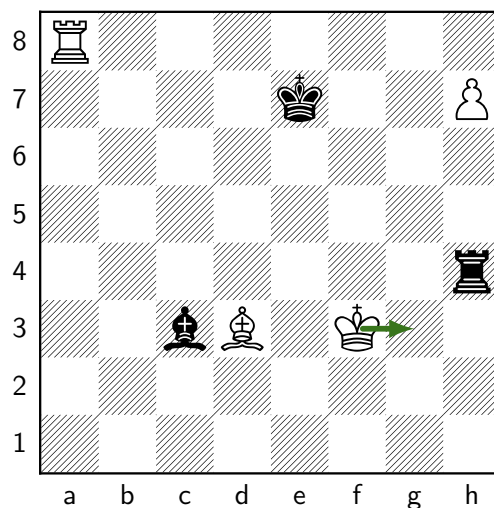
(a)



(b)

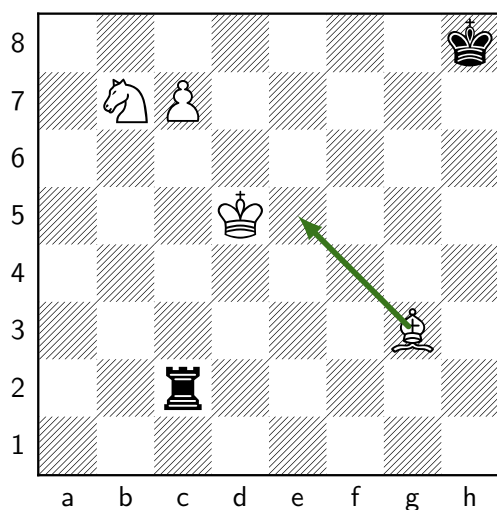


(c)

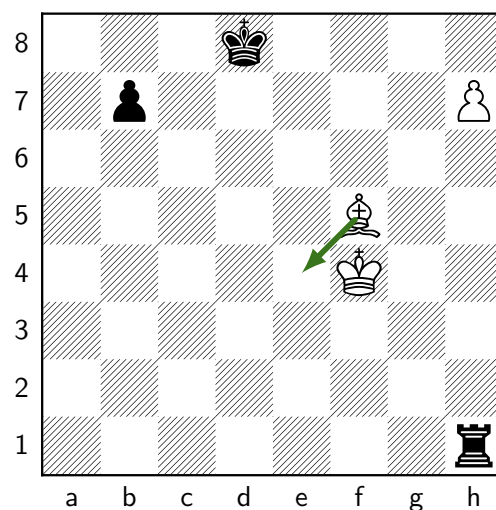


(d)

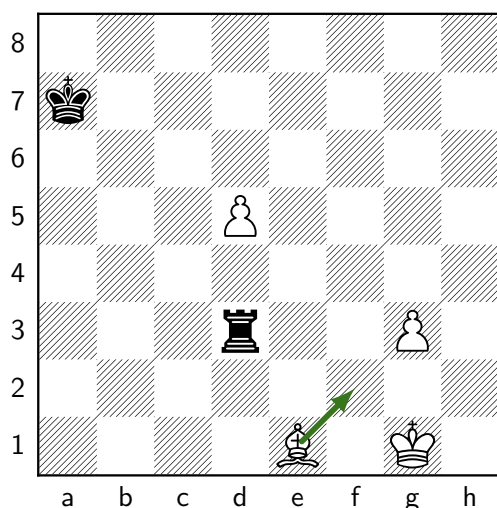
Figure 76: **Move 82, White: Kd5**. (a) Our position; Melanie plays Kd5 (d4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top.p0001#2574` (ply 164) [structural distance  $d=9$ , QAP obj=20]: the master played Kd6; under  $\mu$  our  $Kd4$  plays the role of that game's  $Kc6$ ; cross-type roles: our  $Nb7$  as their  $ph7$ ; transferred move d4d5 (**endorses this move**). (c) Retrieved from `ccl1_top.p0000#796` (ply 171) [structural distance  $d=9$ , QAP obj=18]: the master played Kf4; under  $\mu$  our  $Kd4$  plays the role of that game's  $Kf4$ ; cross-type roles: our  $Bg3$  as their  $Pb7$ ; transferred move d4d5 (**endorses this move**). (d) Retrieved from `ccl1_top.p0001#76` (ply 166) [structural distance  $d=6$ , QAP obj=11]: the master played Kg3; under  $\mu$  our  $Kd4$  plays the role of that game's  $Kf3$ ; cross-type roles: our  $Nb7$  as their  $Ra8$ ; transferred move d4d5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kd5 (summed QAP objective 64, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



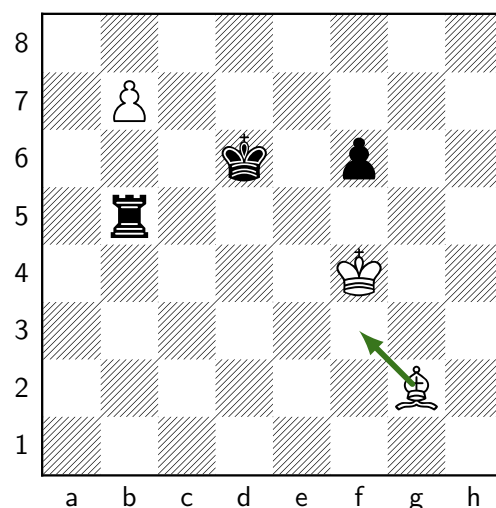
(a)



(b)



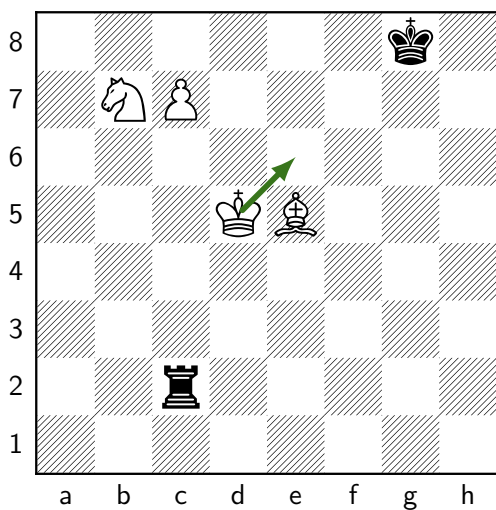
(c)



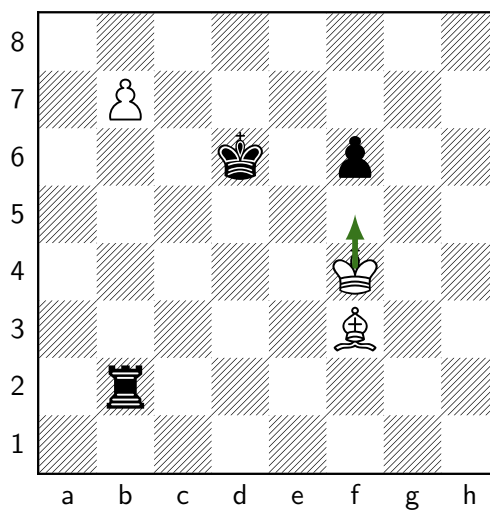
(d)

Figure 77: **Move 83, White: Be5+ (g3e5).** (a) Our position; Melanie plays Be5+ (g3e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0002#726 (ply 148) [structural distance  $d=7$ , QAP obj=18]: the master played Be4; under  $\mu$  our Bg3 plays the role of that game’s Bf5; cross-type roles: our Nb7 as their pb7; transferred move g3e5 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0001#2092 (ply 170) [structural distance  $d=6$ , QAP obj=13]: the master played Bf2+; under  $\mu$  our Bg3 plays the role of that game’s Be1; transferred move g3e5 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0003#487 (ply 176) [structural distance  $d=5$ , QAP obj=11]: the master played Bf3; under  $\mu$  our Bg3 plays the role of that game’s Bg2; cross-type roles: our Nb7 as their pf6; transferred move g3e5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Be5+ (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

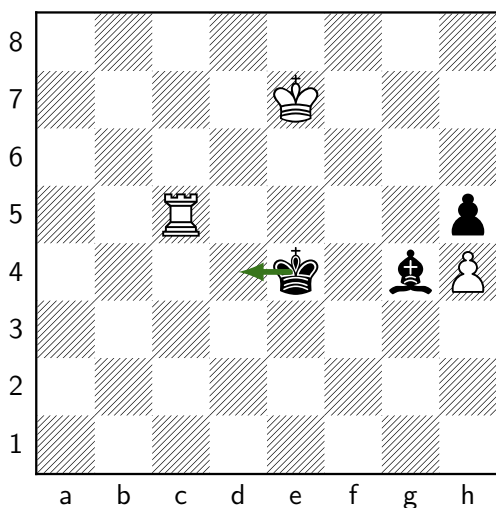




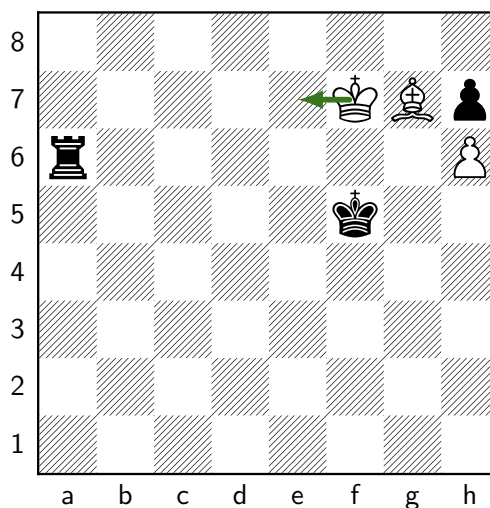
(a)



(b)

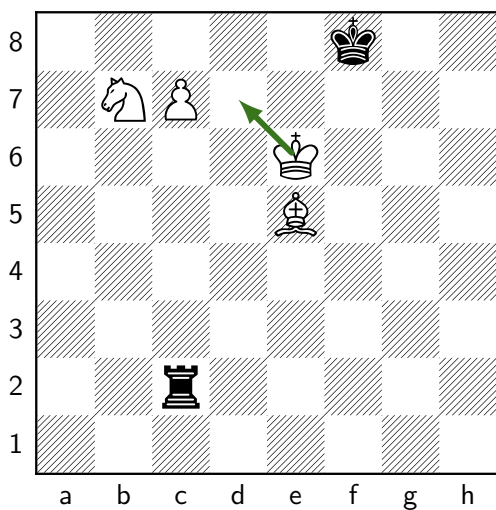


(c)

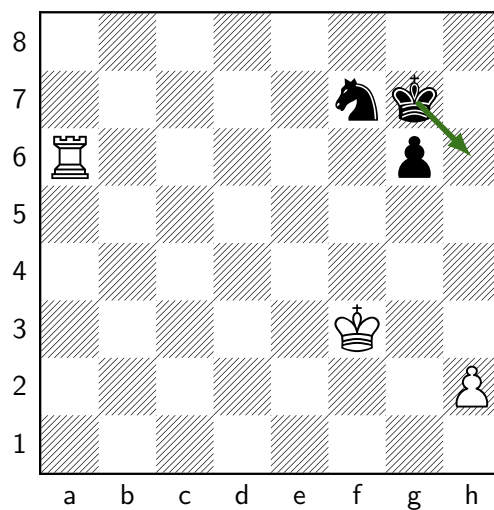


(d)

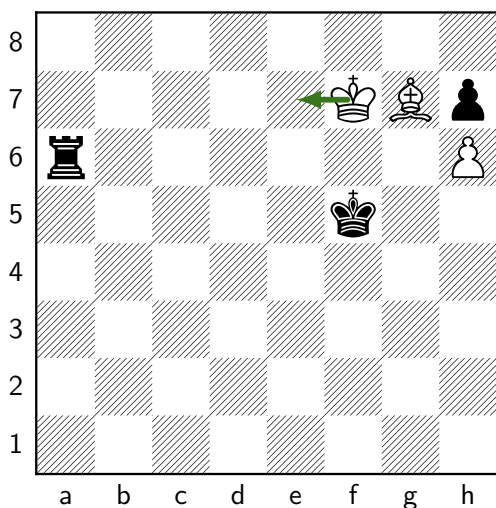
Figure 78: **Move 84, White: Ke6**. (a) Our position; Melanie plays **Ke6** (**d5e6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#487` (ply 182) [structural distance  $d=5$ , QAP obj=22]: the master played **Kf5**; under  $\mu$  our **Kd5** plays the role of that game’s **Kf4**; cross-type roles: our **Nb7** as their **pf6**; transferred move **d5e6** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#729` (ply 171) [structural distance  $d=9$ , QAP obj=21]: the master played **Kd4**; under  $\mu$  our **Kd5** plays the role of that game’s **Ke5**; cross-type roles: our **Nb7** as their **ph4**; transferred move **d5e6** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2574` (ply 180) [structural distance  $d=10$ , QAP obj=18]: the master played **Ke7**; under  $\mu$  our **Kd5** plays the role of that game’s **Kf7**; cross-type roles: our **Nb7** as their **ph7**; transferred move **d5e6** (**endorses this move**). Synthesis: 13 retrieved analogs transferred to **Ke6** (summed QAP objective 120, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



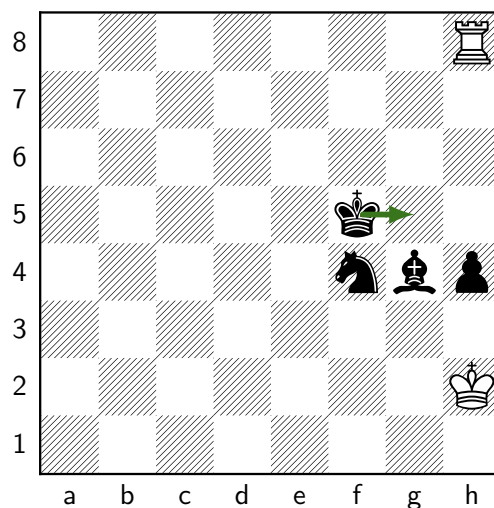
(a)



(b)

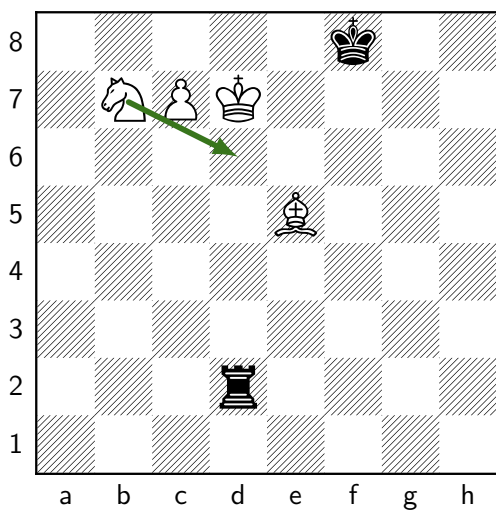


(c)

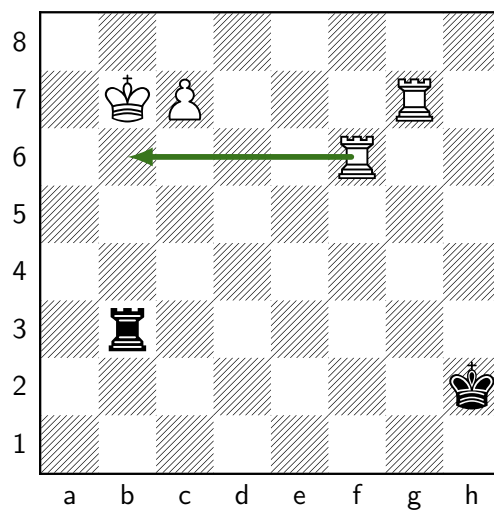


(d)

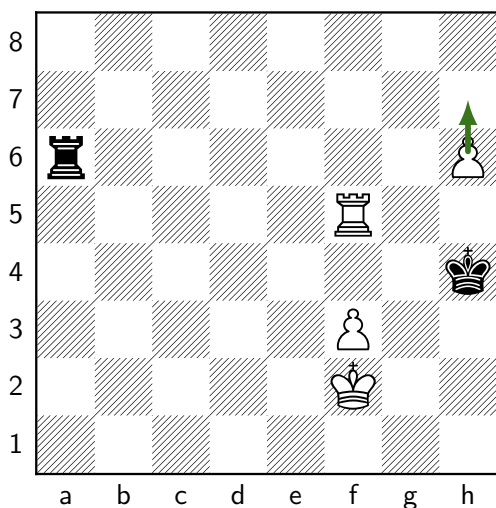
Figure 79: **Move 85, White: Kd7.** (a) Our position; Melanie plays Kd7 (e6d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1777` (ply 165) [structural distance  $d=11$ , QAP obj=18]: the master played Kh6; under  $\mu$  our Ke6 plays the role of that game’s Kg2; cross-type roles: our Be5 as their Nf7, our Nb7 as their ph2; transferred move e6d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2574` (ply 180) [structural distance  $d=10$ , QAP obj=17]: the master played Ke7; under  $\mu$  our Ke6 plays the role of that game’s Kf7; cross-type roles: our Nb7 as their ph7; transferred move e6d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1023` (ply 165) [structural distance  $d=11$ , QAP obj=15]: the master played Kg5; under  $\mu$  our Ke6 plays the role of that game’s Kf4; cross-type roles: our Be5 as their Nf4, our Nb7 as their Bg4; transferred move e6d7 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Kd7 (summed QAP objective 81, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



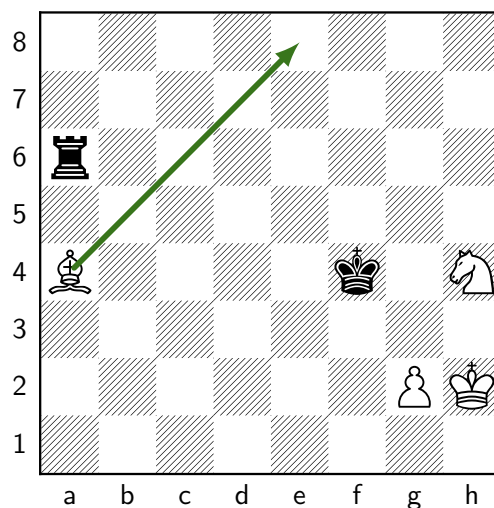
(a)



(b)

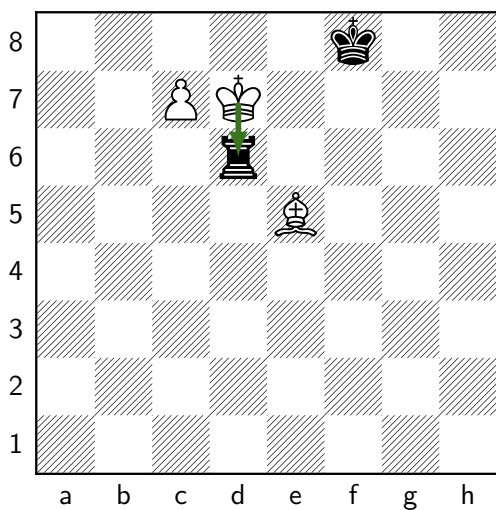


(c)

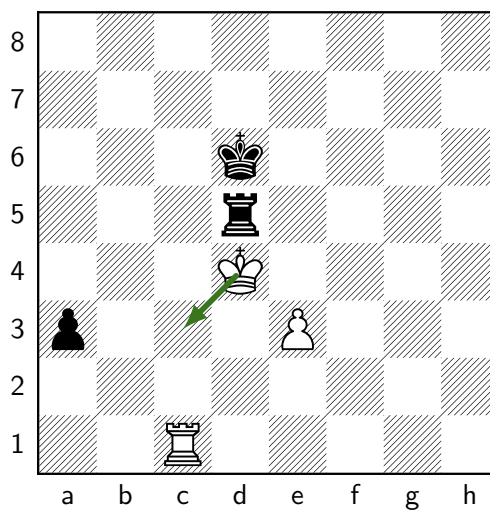


(d)

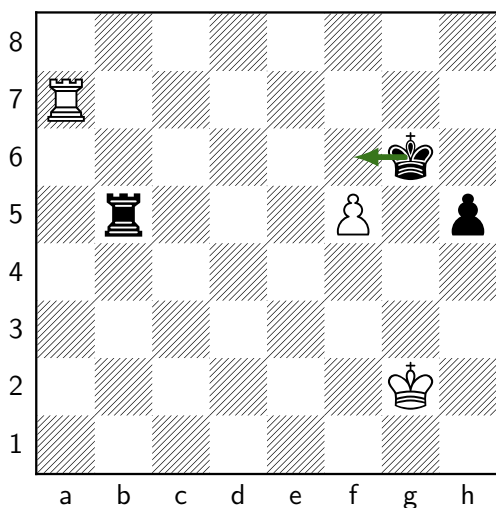
Figure 80: **Move 86, White: Nd6**. (a) Our position; Melanie plays Nd6 (b7d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#905` (ply 158) [structural distance  $d=12$ , QAP obj=59]: the master played Rb6; under  $\mu$  our Nb7 plays the role of that game’s Rf6 (a **cross-type** correspondence); cross-type roles: our Be5 as their Rg7; transferred move b7d6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1804` (ply 166) [structural distance  $d=12$ , QAP obj=3]: the master played h7; under  $\mu$  our Nb7 plays the role of that game’s Ph6 (a **cross-type** correspondence); cross-type roles: our Be5 as their Rf5; transferred move b7d6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#780` (ply 166) [structural distance  $d=9$ , QAP obj=3]: the master played Be8; under  $\mu$  our Nb7 plays the role of that game’s Ba4 (a **cross-type** correspondence); cross-type roles: our Be5 as their Nh4; transferred move b7d6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nd6 (summed QAP objective 68, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



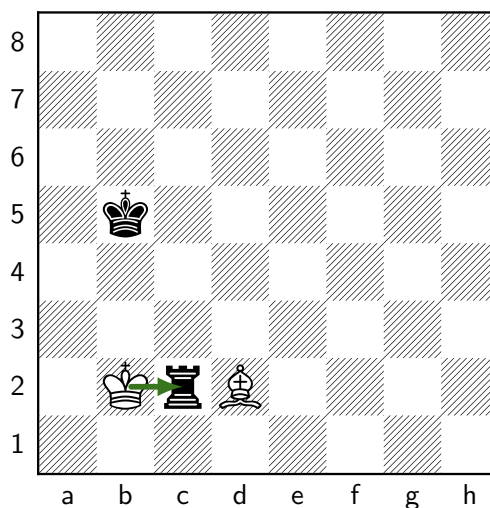
(a)



(b)

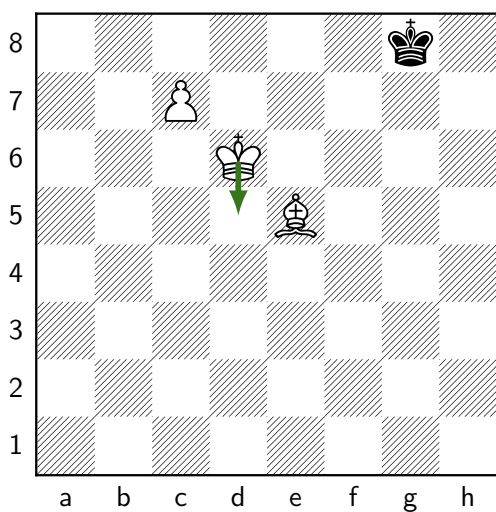


(c)

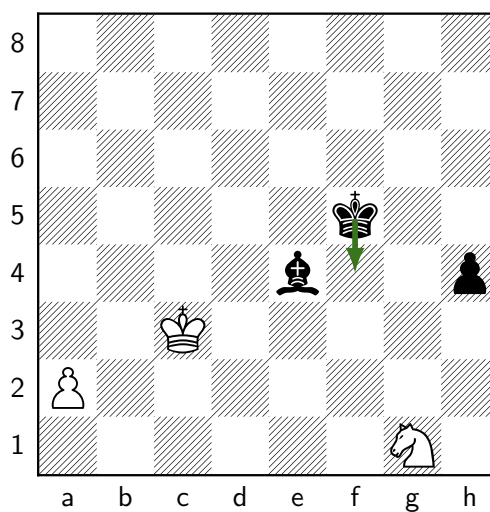


(d)

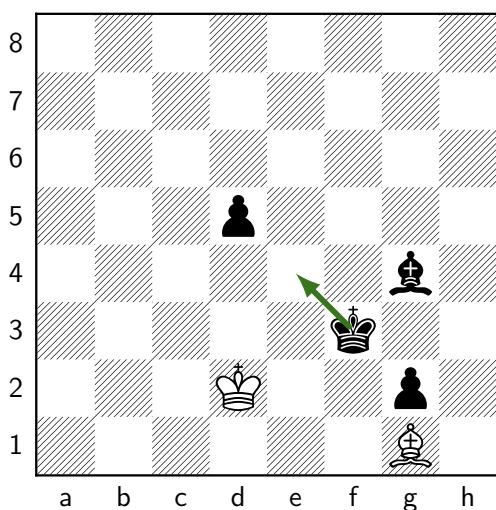
Figure 81: **Move 87, White: Kxd6.** (a) Our position; Melanie plays Kxd6 (d7d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1972` (ply 158) [structural distance  $d=16$ , QAP obj=71]: the master played Kc3; under  $\mu$  our  $Kd7$  plays the role of that game’s  $Kd4$ ; cross-type roles: our  $Be5$  as their  $Rc1$ ; transferred move d7d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#840` (ply 161) [structural distance  $d=17$ , QAP obj=71]: the master played Kf6; under  $\mu$  our  $Kd7$  plays the role of that game’s  $Kg3$ ; cross-type roles: our  $Be5$  as their  $Rb5$ , our  $rd6$  as their  $pf5$ ; transferred move d7d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#929` (ply 156) [structural distance  $d=11$ , QAP obj=68]: the master played Kxc2; under  $\mu$  our  $Kd7$  plays the role of that game’s  $Kb2$ ; transferred move d7d6 (**endorses this move**). Synthesis: 15 retrieved analogs transferred to Kxd6 (summed QAP objective 459, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



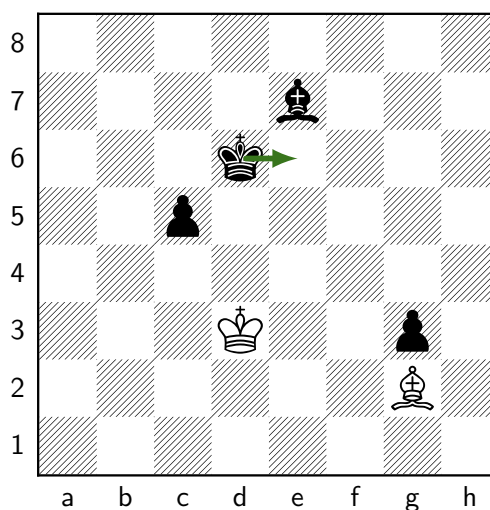
(a)



(b)

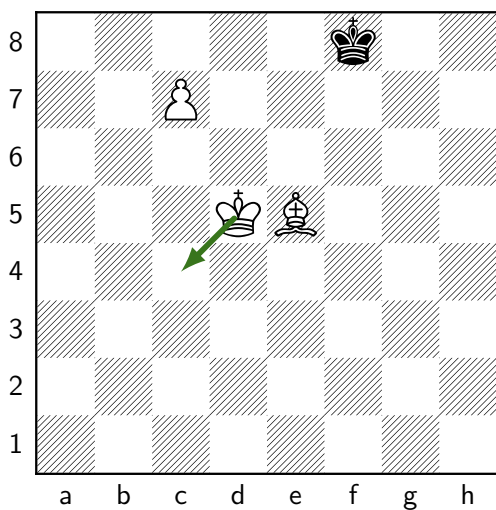


(c)

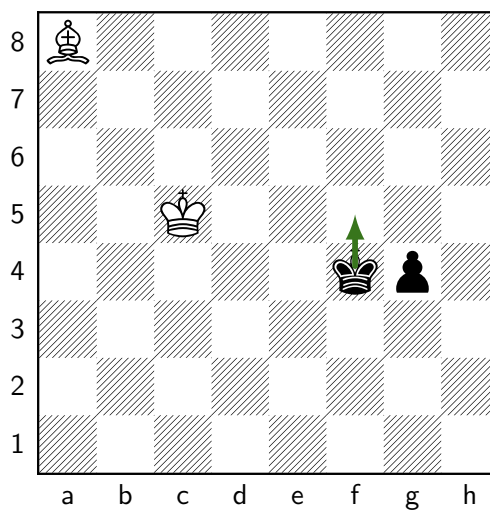


(d)

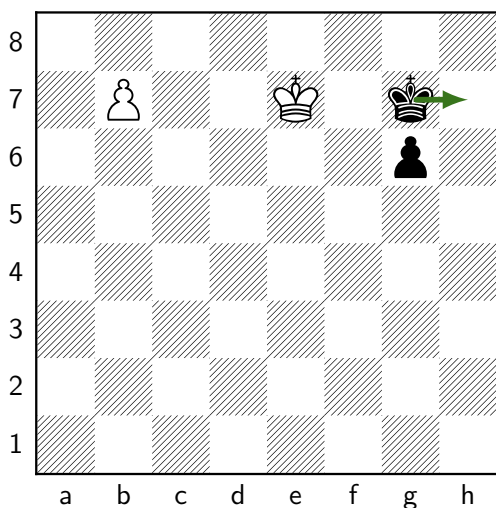
Figure 82: **Move 88, White: Kd5.** (a) Our position; Melanie plays Kd5 (d6d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#581` (ply 173) [structural distance  $d=11$ , QAP obj=2]: the master played Kf4; under  $\mu$  our Kd6 plays the role of that game's Kf4; cross-type roles: our Pc7 as their ng1; transferred move d6d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2484` (ply 187) [structural distance  $d=8$ , QAP obj=2]: the master played Ke4; under  $\mu$  our Kd6 plays the role of that game's Kf6; cross-type roles: our Pc7 as their bg1; transferred move d6d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#261` (ply 183) [structural distance  $d=8$ , QAP obj=2]: the master played Ke6; under  $\mu$  our Kd6 plays the role of that game's Kd3; transferred move d6d5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kd5 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



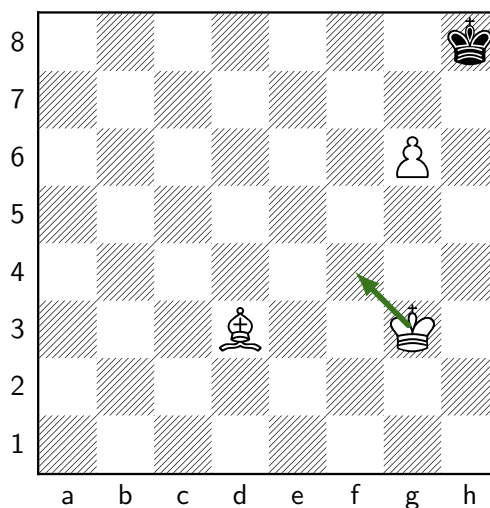
(a)



(b)

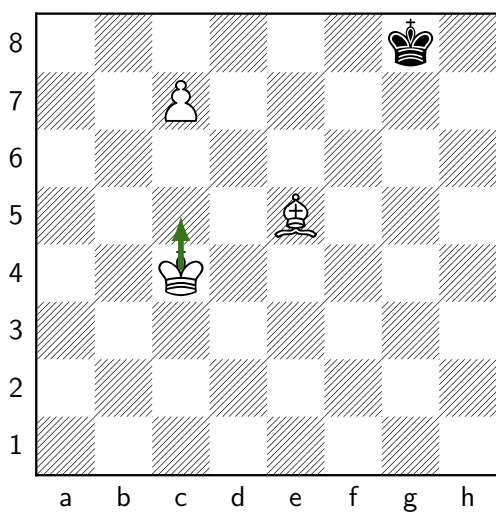


(c)

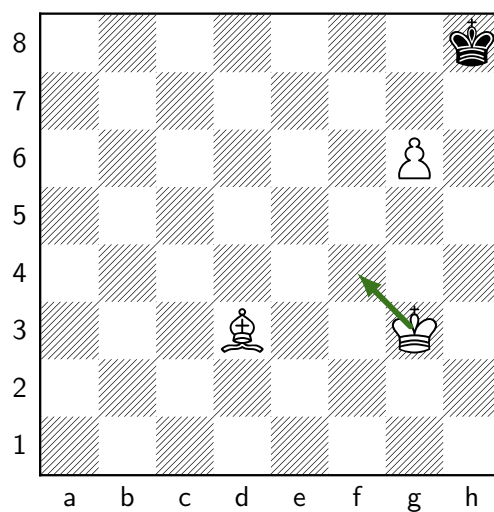


(d)

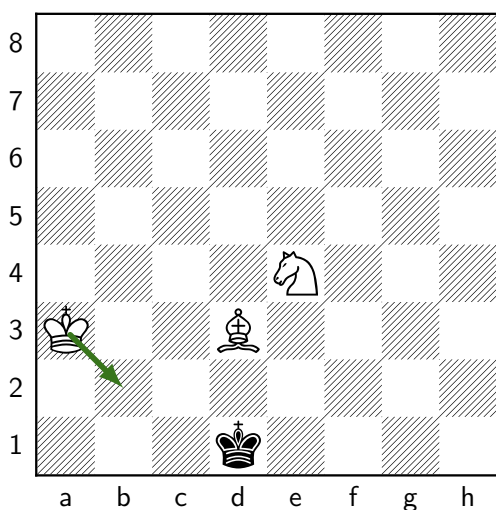
Figure 83: **Move 89, White: Kc4.** (a) Our position; Melanie plays Kc4 (d5c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3573` (ply 165) [structural distance  $d=7$ , QAP obj=1]: the master played Kf5; under  $\mu$  our Kd5 plays the role of that game’s Kf5; cross-type roles: our Pc7 as their ba8, our Be5 as their Pg4; transferred move d5c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2475` (ply 191) [structural distance  $d=7$ , QAP obj=1]: the master played Kh7; under  $\mu$  our Kd5 plays the role of that game’s Kg2; cross-type roles: our Be5 as their Pg6; transferred move d5c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#376` (ply 190) [structural distance  $d=5$ , QAP obj=0]: the master played Kf4; under  $\mu$  our Kd5 plays the role of that game’s Kg3; cross-type roles: our Pc7 as their Bd3, our Be5 as their Pg6; transferred move d5c4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kc4 (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



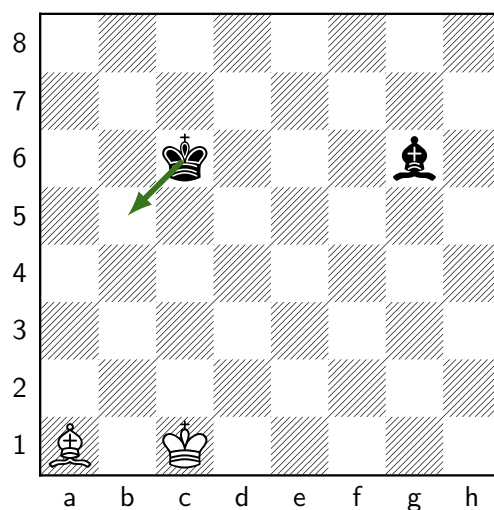
(a)



(b)

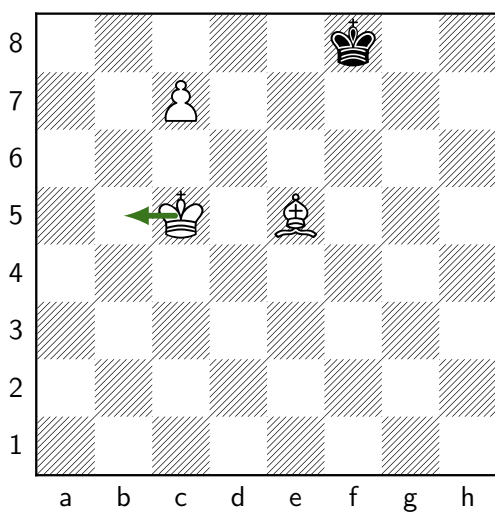


(c)

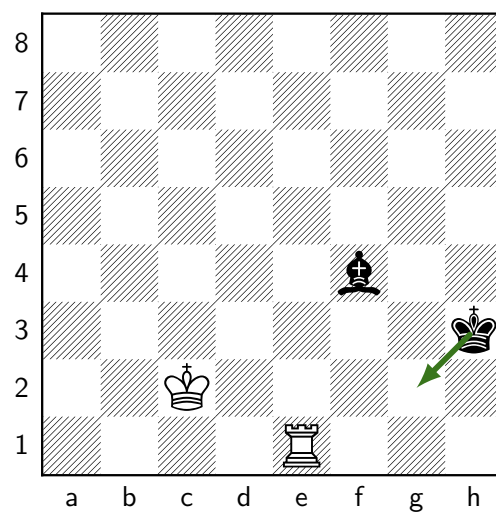


(d)

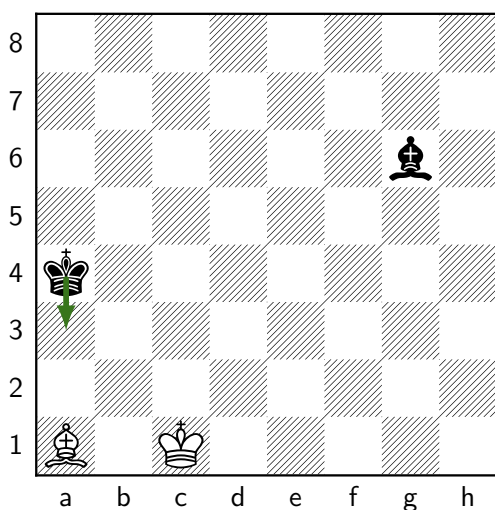
Figure 84: **Move 90, White: Kc5.** (a) Our position; Melanie plays Kc5 (c4c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#376` (ply 190) [structural distance  $d=1$ , QAP obj=1]: the master played Kf4; under  $\mu$  our *Kc4* plays the role of that game’s *Kg3*; transferred move c4c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1344` (ply 190) [structural distance  $d=6$ , QAP obj=1]: the master played Kb2; under  $\mu$  our *Kc4* plays the role of that game’s *Ka3*; cross-type roles: our *Pc7* as their *Ne4*; transferred move c4b3. (d) Retrieved from `ccr1_top_p0000#1609` (ply 181) [structural distance  $d=6$ , QAP obj=1]: the master played Kb5; under  $\mu$  our *Kc4* plays the role of that game’s *Kc3*; cross-type roles: our *Pc7* as their *ba1*; transferred move c4b4. Synthesis: 1 retrieved analog transferred to Kc5 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



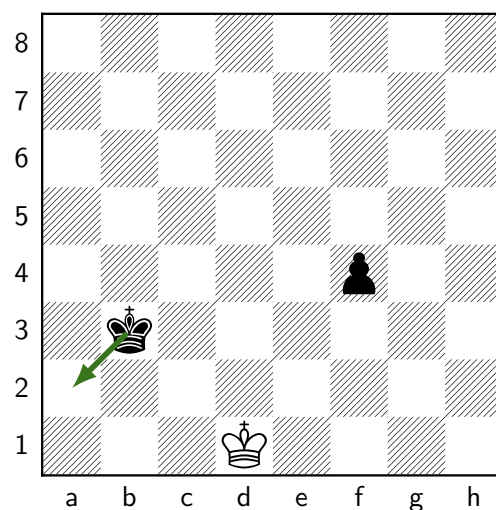
(a)



(b)



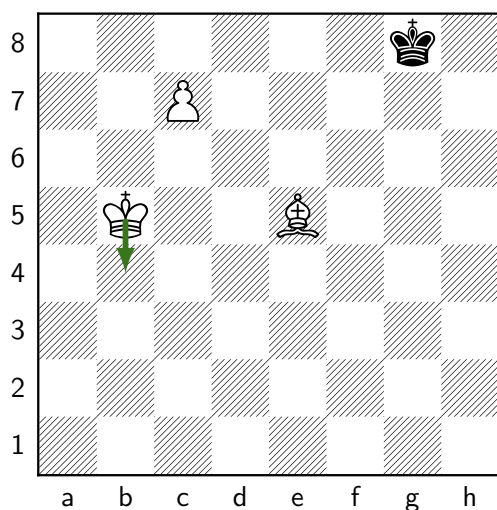
(c)



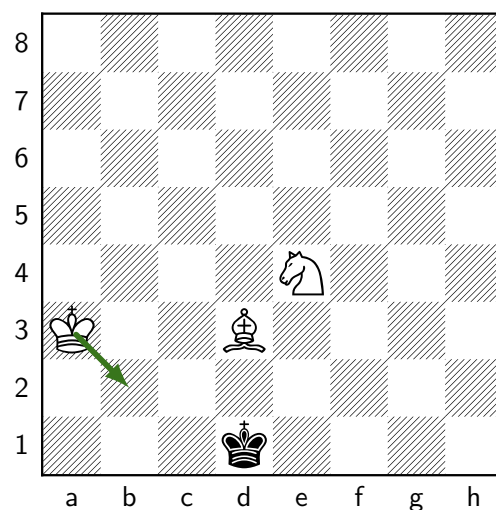
(d)

Figure 85: **Move 91, White: Kb5.** (a) Our position; Melanie plays Kb5 (c5b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1587` (ply 167) [structural distance  $d=6$ , QAP obj=1]: the master played Kg2; under  $\mu$  our *Kc5* plays the role of that game's *Kh6*; cross-type roles: our *Pc7* as their *re1*; transferred move c5b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1609` (ply 185) [structural distance  $d=6$ , QAP obj=0]: the master played Ka3; under  $\mu$  our *Kc5* plays the role of that game's *Ka5*; cross-type roles: our *Pc7* as their *ba1*; transferred move c5b5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1488` (ply 165) [structural distance  $d=6$ , QAP obj=0]: the master played Ka2; under  $\mu$  our *Kc5* plays the role of that game's *Kb6*; cross-type roles: our *Be5* as their *Pf4*; transferred move c5b5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kb5 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

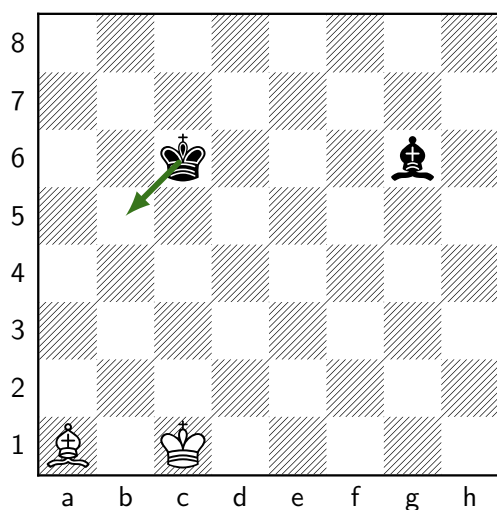




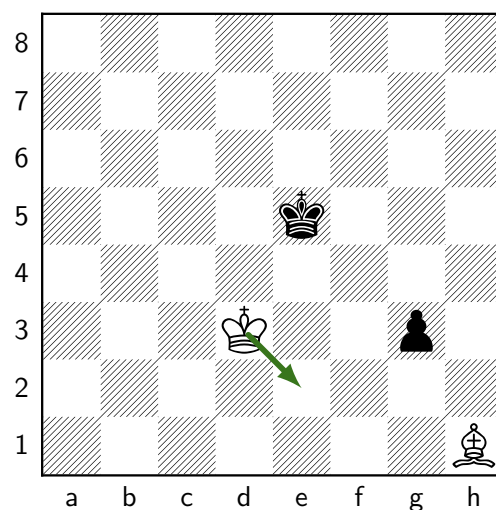
(a)



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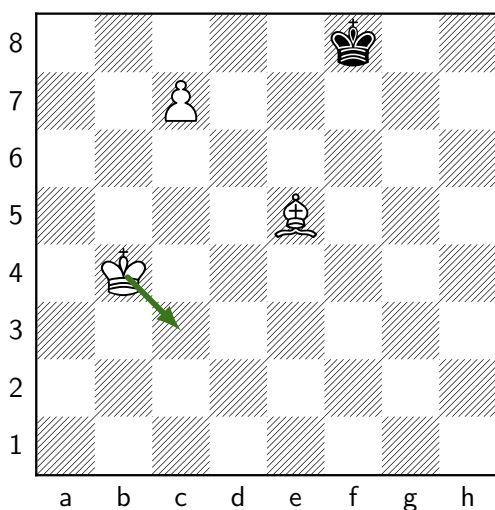


(c)

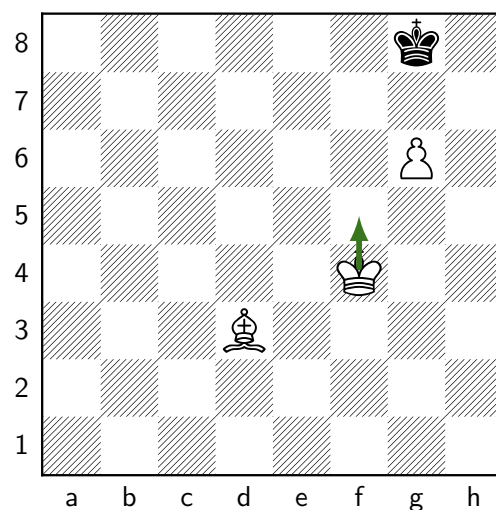


(d)

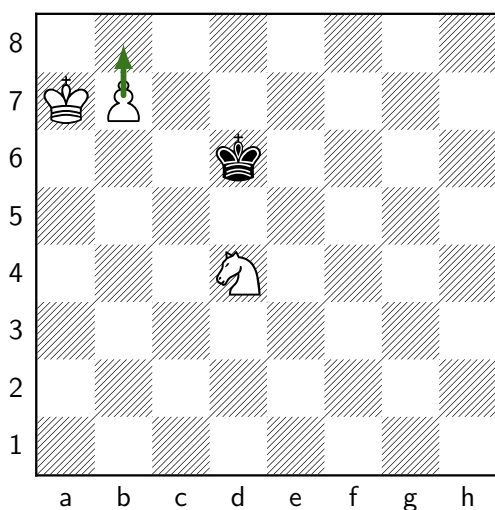
Figure 86: **Move 92, White: Kb4**. (a) Our position; Melanie plays Kb4 (b5b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1344` (ply 190) [structural distance  $d=6$ , QAP obj=1]: the master played Kb2; under  $\mu$  our *Kb5* plays the role of that game's *Ka3*; cross-type roles: our *Pc7* as their *Ne4*; transferred move b5b4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1609` (ply 181) [structural distance  $d=6$ , QAP obj=1]: the master played Kb5; under  $\mu$  our *Kb5* plays the role of that game's *Kc3*; cross-type roles: our *Pc7* as their *ba1*; transferred move b5b4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3573` (ply 172) [structural distance  $d=6$ , QAP obj=0]: the master played Ke2; under  $\mu$  our *Kb5* plays the role of that game's *Kd3*; transferred move b5b4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kb4 (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



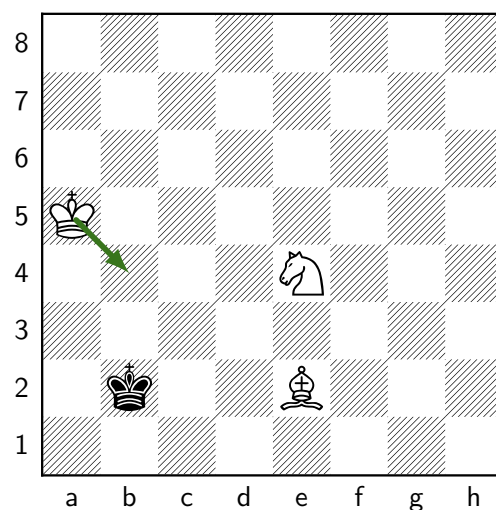
(a)



(b)

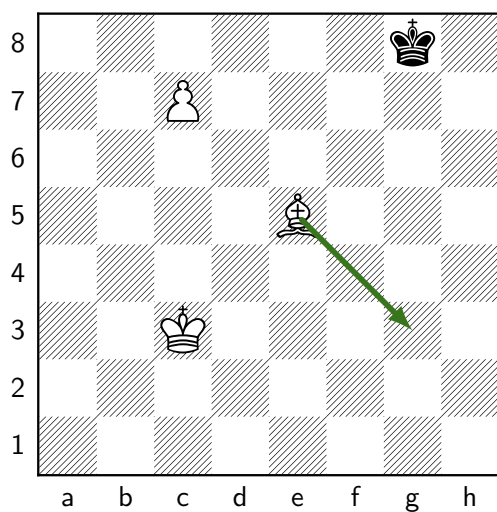


(c)

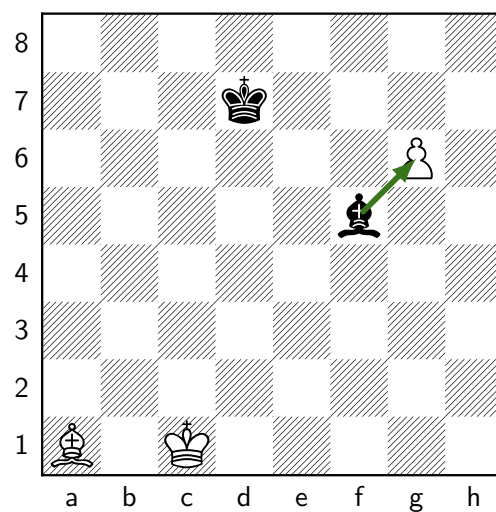


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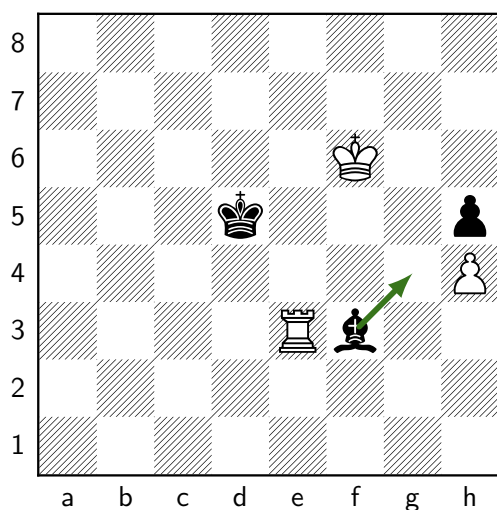
Figure 87: **Move 93, White: Kc3**. (a) Our position; Melanie plays Kc3 (b4c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#376` (ply 192) [structural distance  $d=2$ , QAP obj=1]: the master played Kf5; under  $\mu$  our  $Kb4$  plays the role of that game's  $Kf4$ ; transferred move b4c3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#576` (ply 184) [structural distance  $d=6$ , QAP obj=1]: the master played b8=Q+; under  $\mu$  our  $Pc7$  plays the role of that game's  $Pb7$ ; cross-type roles: our  $Be5$  as their  $Nd4$ ; transferred move c7c8q. (d) Retrieved from `ccr1_top_p0001#1344` (ply 178) [structural distance  $d=6$ , QAP obj=0]: the master played Kb4; under  $\mu$  our  $Kb4$  plays the role of that game's  $Ka5$ ; cross-type roles: our  $Pc7$  as their  $Be2$ , our  $Be5$  as their  $Ne4$ ; transferred move b4a4. Synthesis: 1 retrieved analog transferred to Kc3 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



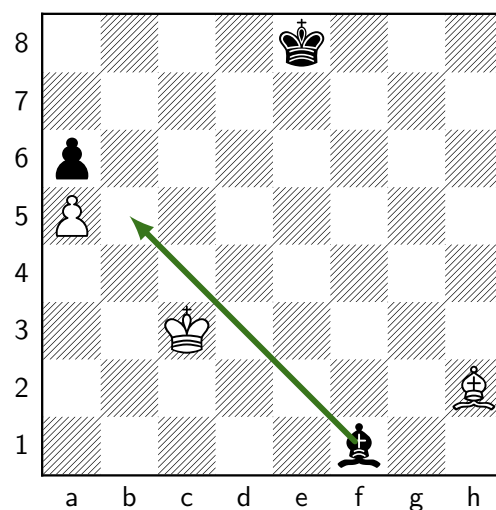
(a)



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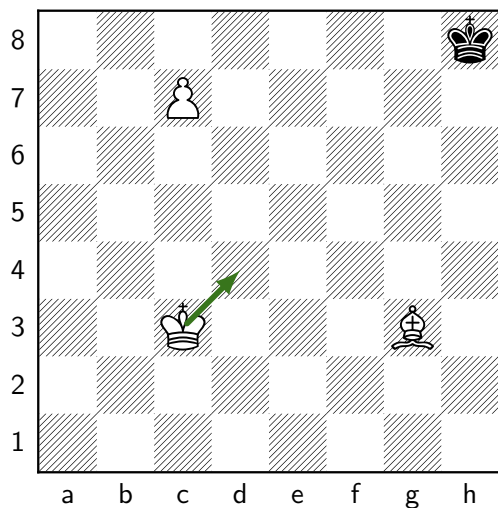


(c)

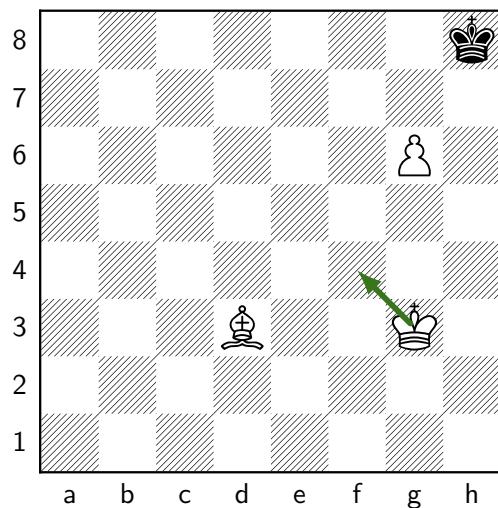


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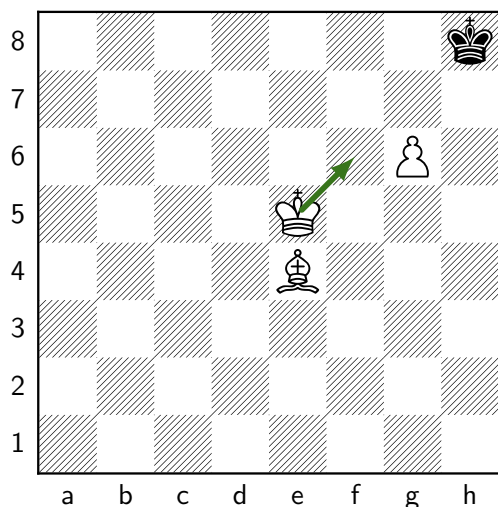
Figure 88: **Move 94, White: Bg3.** (a) Our position; Melanie plays Bg3 (e5g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1609` (ply 177) [structural distance  $d=11$ , QAP obj=3]: the master played Bxg6; under  $\mu$  our *Be5* plays the role of that game’s *Bf4*; transferred move e5g3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#729` (ply 193) [structural distance  $d=12$ , QAP obj=2]: the master played Bg4; under  $\mu$  our *Be5* plays the role of that game’s *Bf6*; transferred move e5g3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2457` (ply 199) [structural distance  $d=12$ , QAP obj=0]: the master played Bb5; under  $\mu$  our *Be5* plays the role of that game’s *Bf8*; transferred move e5g3 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Bg3 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



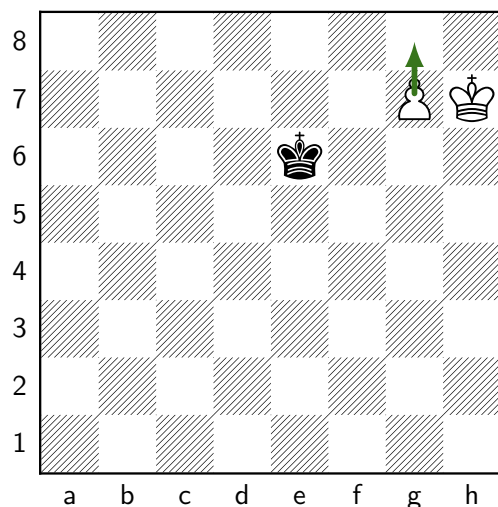
(a)



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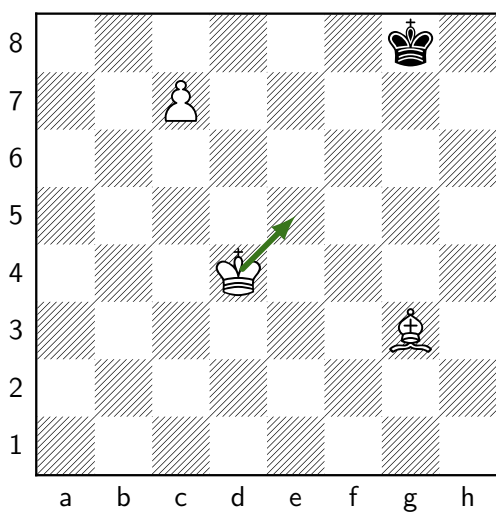


(c)

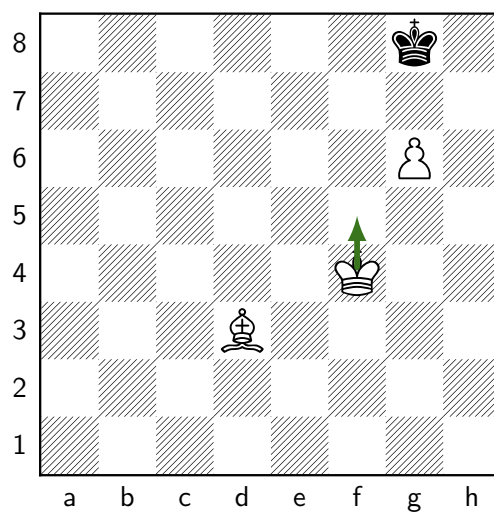


(d)

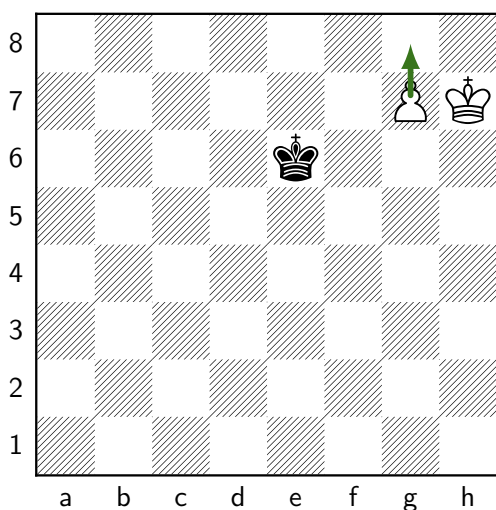
Figure 89: **Move 95, White: Kd4.** (a) Our position; Melanie plays Kd4 (c3d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#376` (ply 190) [structural distance  $d=1$ , QAP obj=1]: the master played Kf4; under  $\mu$  our  $Kc3$  plays the role of that game’s  $Kg3$ ; transferred move c3d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1483` (ply 202) [structural distance  $d=5$ , QAP obj=0]: the master played Kf6; under  $\mu$  our  $Kc3$  plays the role of that game’s  $Ke5$ ; transferred move c3d4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1379` (ply 190) [structural distance  $d=6$ , QAP obj=0]: the master played g8=Q+; under  $\mu$  our  $Bg3$  plays the role of that game’s  $Pg7$  (a **cross-type** correspondence); transferred move g3h4. Synthesis: 3 retrieved analogs transferred to Kd4 (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



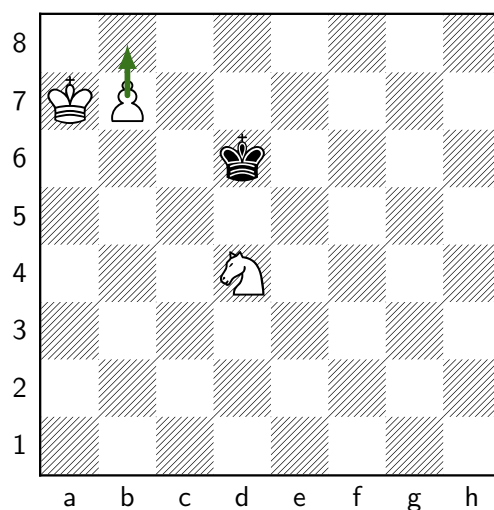
(a)



(b)

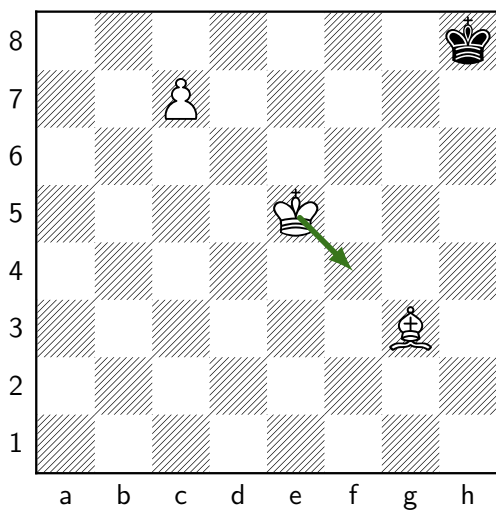


(c)

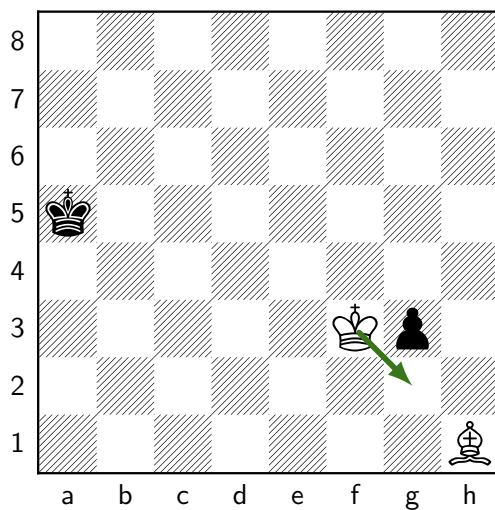


(d)

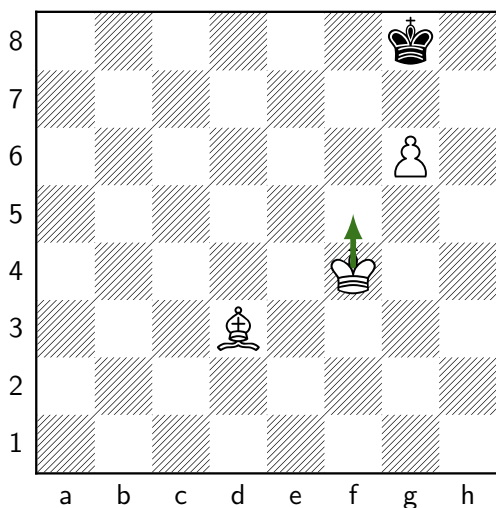
Figure 90: **Move 96, White: Ke5.** (a) Our position; Melanie plays Ke5 (d4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#376` (ply 192) [structural distance  $d=2$ , QAP obj=1]: the master played Kf5; under  $\mu$  our *Kd4* plays the role of that game's *Kf4*; transferred move d4e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1379` (ply 190) [structural distance  $d=6$ , QAP obj=0]: the master played g8=Q+; under  $\mu$  our *Bg3* plays the role of that game's *Pg7* (a **cross-type** correspondence); transferred move g3e5. (d) Retrieved from `ccr1_top_p0001#576` (ply 184) [structural distance  $d=6$ , QAP obj=0]: the master played b8=Q+; under  $\mu$  our *Pc7* plays the role of that game's *Pb7*; cross-type roles: our *Bg3* as their *Nd4*; transferred move c7c8q. Synthesis: 1 retrieved analog transferred to Ke5 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



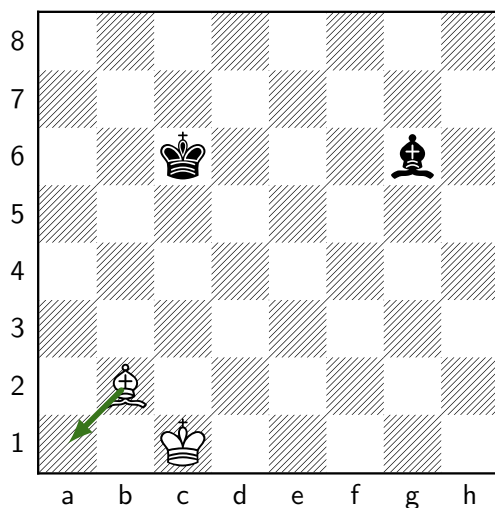
(a)



(b)

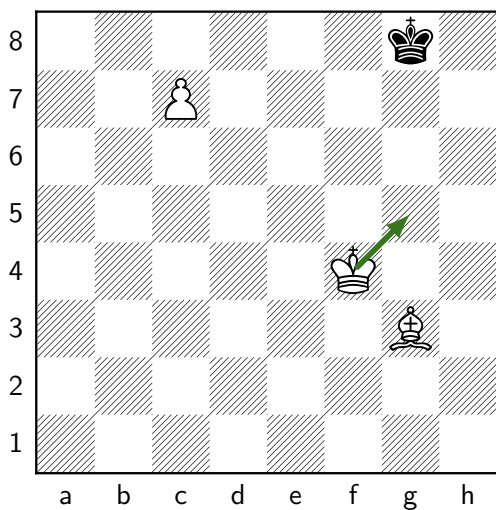


(c)

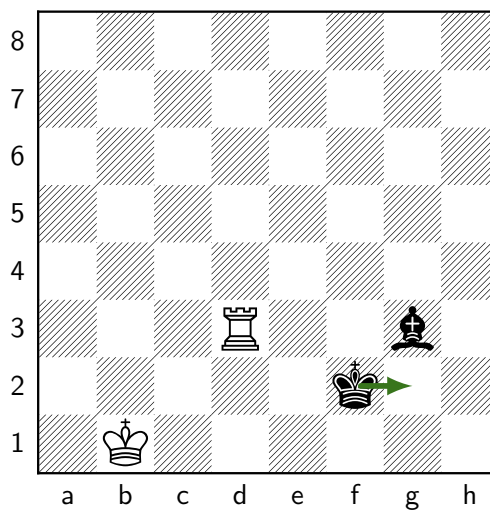


(d)

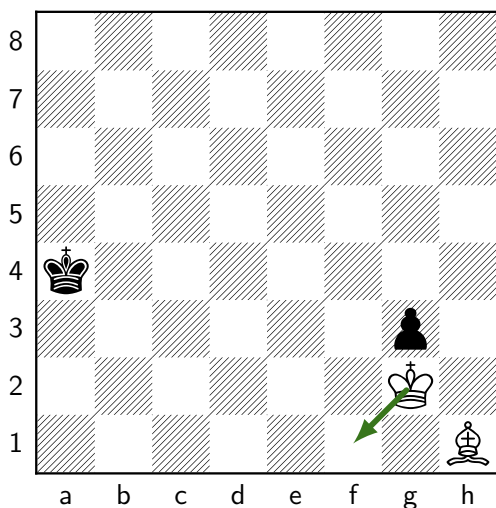
Figure 91: **Move 97, White: Kf4**. (a) Our position; Melanie plays Kf4 (e5f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3573` (ply 180) [structural distance  $d=7$ , QAP obj=2]: the master played Kg2; under  $\mu$  our Ke5 plays the role of that game's Kf3; transferred move e5f4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#376` (ply 192) [structural distance  $d=7$ , QAP obj=0]: the master played Kf5; under  $\mu$  our Ke5 plays the role of that game's Kf4; cross-type roles: our Pc7 as their Bd3, our Bg3 as their Pg6; transferred move e5f4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1609` (ply 180) [structural distance  $d=7$ , QAP obj=1]: the master played Ba1; under  $\mu$  our Bg3 plays the role of that game's Bb2; cross-type roles: our Pc7 as their bg6; transferred move g3e1. Synthesis: 2 retrieved analogs transferred to Kf4 (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



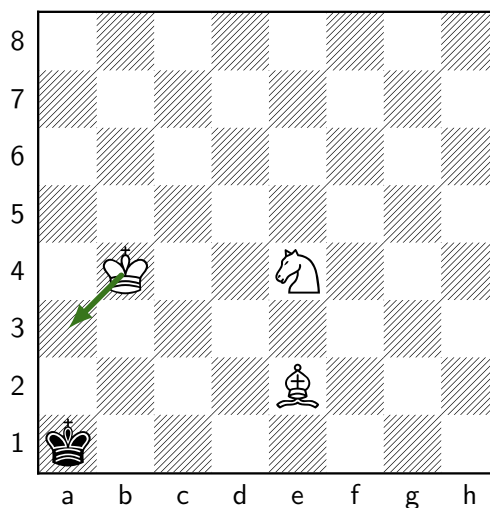
(a)



(b)

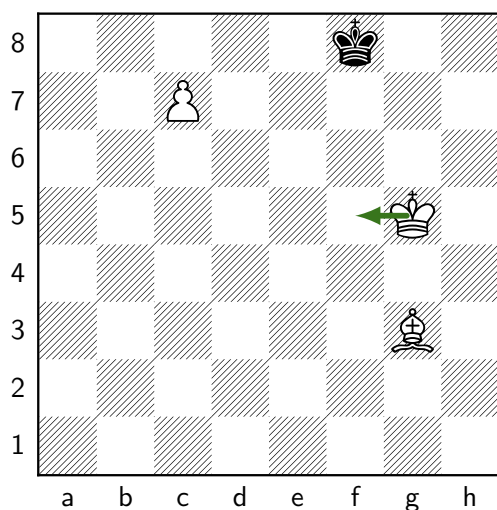


(c)

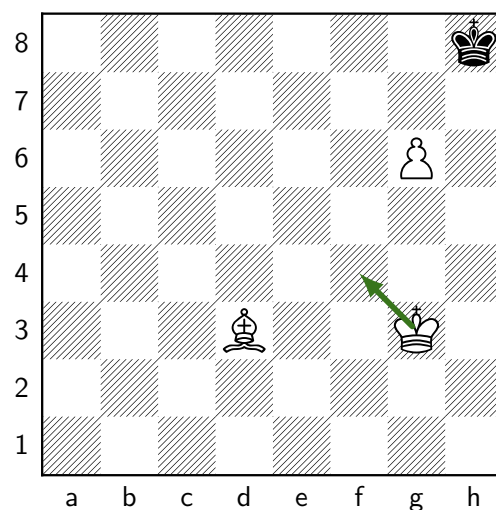


(d)

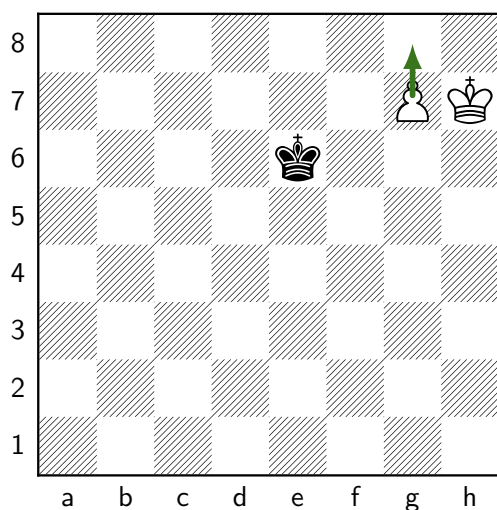
Figure 92: **Move 98, White: Kg5.** (a) Our position; Melanie plays Kg5 (f4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1587` (ply 179) [structural distance  $d=8$ , QAP obj=2]: the master played Kg2; under  $\mu$  our  $Kf_4$  plays the role of that game's  $Kf_7$ ; cross-type roles: our  $Pc_7$  as their  $rd_3$ ; transferred move f4g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3573` (ply 182) [structural distance  $d=8$ , QAP obj=2]: the master played Kf1; under  $\mu$  our  $Kf_4$  plays the role of that game's  $Kg_2$ ; transferred move f4g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1344` (ply 180) [structural distance  $d=10$ , QAP obj=0]: the master played Ka3; under  $\mu$  our  $Kf_4$  plays the role of that game's  $Kb_4$ ; cross-type roles: our  $Pc_7$  as their  $Ne_4$ ; transferred move f4g5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg5 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



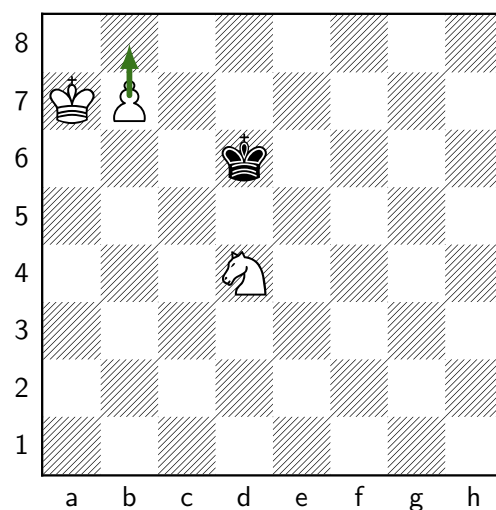
(a)



(b)



(c)



(d)

Figure 93: **Move 99, White: Kf5.** (a) Our position; Melanie plays Kf5 (g5f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#376` (ply 190) [structural distance  $d=1$ , QAP obj=1]: the master played Kf4; under  $\mu$  our Kg5 plays the role of that game's Kg3; transferred move g5f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1379` (ply 190) [structural distance  $d=6$ , QAP obj=0]: the master played g8=Q+; under  $\mu$  our Bg3 plays the role of that game's Pg7 (a **cross-type** correspondence); transferred move g3h4. (d) Retrieved from `ccr1_top_p0001#576` (ply 184) [structural distance  $d=6$ , QAP obj=0]: the master played b8=Q+; under  $\mu$  our Pc7 plays the role of that game's Pb7; cross-type roles: our Bg3 as their Nd4; transferred move c7c8q. Synthesis: 1 retrieved analog transferred to Kf5 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



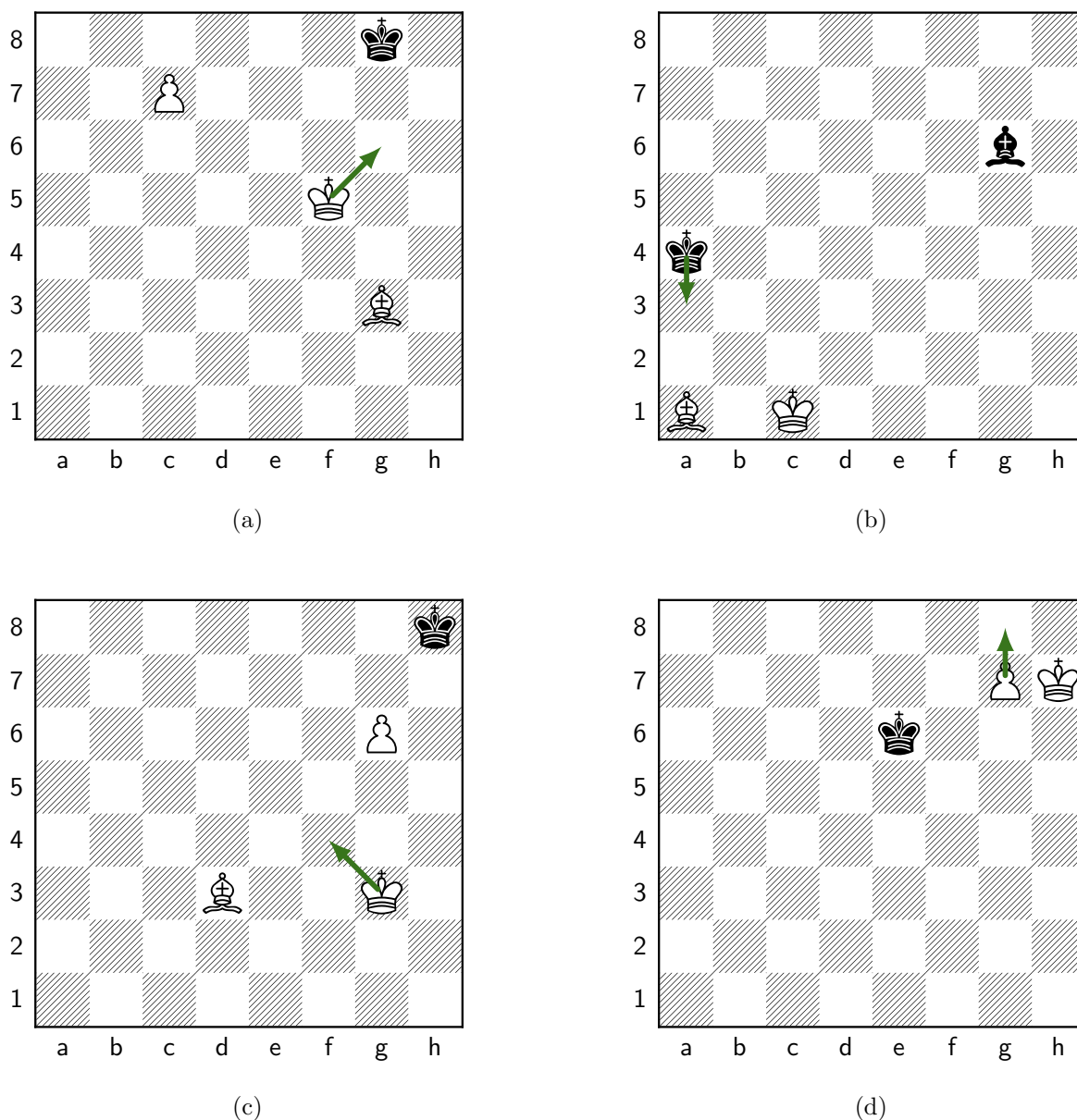
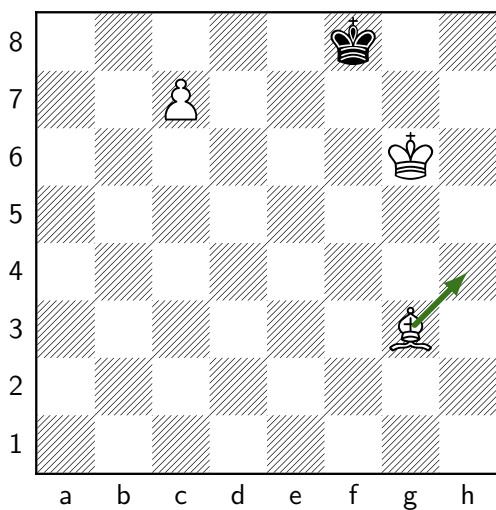
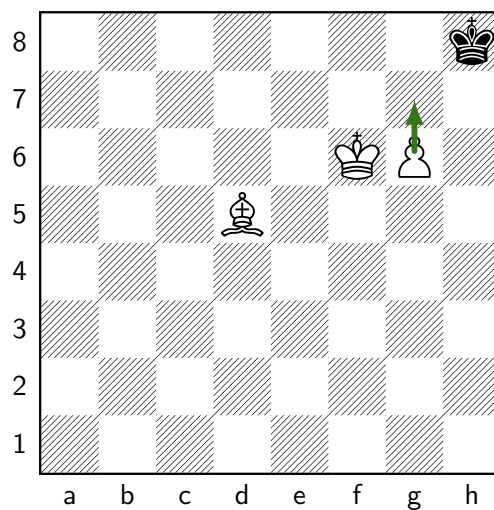


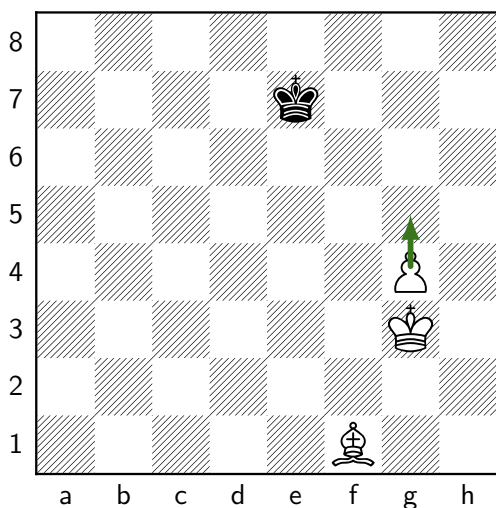
Figure 94: **Move 100, White: Kg6.** (a) Our position; Melanie plays Kg6 (f5g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1609` (ply 185) [structural distance  $d=6$ , QAP obj=0]: the master played Ka3; under  $\mu$  our Kf5 plays the role of that game’s Ka5; cross-type roles: our Pc7 as their Bg6; transferred move f5g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#376` (ply 190) [structural distance  $d=1$ , QAP obj=1]: the master played Kf4; under  $\mu$  our Kf5 plays the role of that game’s Kg3; transferred move f5e4. (d) Retrieved from `ccr1_top_p0003#1379` (ply 190) [structural distance  $d=6$ , QAP obj=0]: the master played g8=Q+; under  $\mu$  our Bg3 plays the role of that game’s Pg7 (a **cross-type** correspondence); transferred move g3e5. Synthesis: 1 retrieved analog transferred to Kg6 (summed QAP objective 0, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



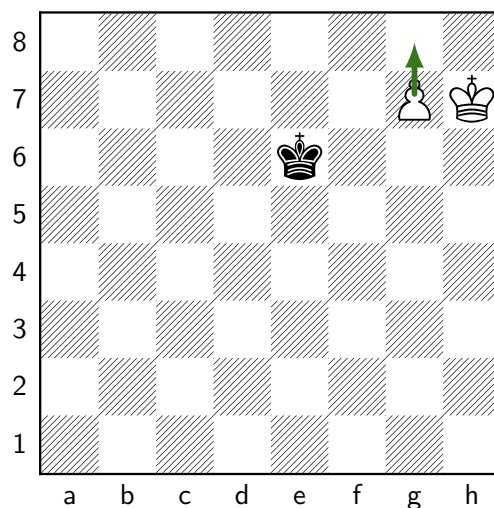
(a)



(b)

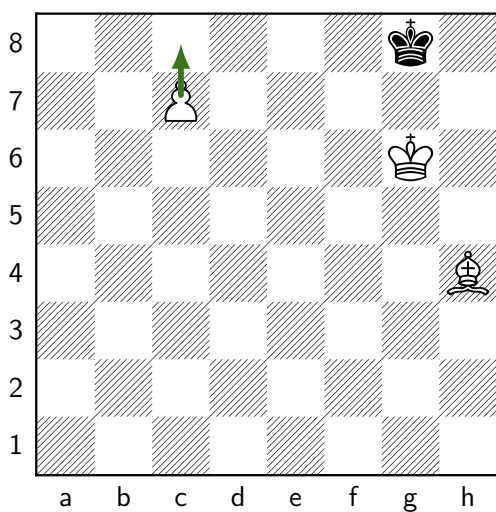


(c)

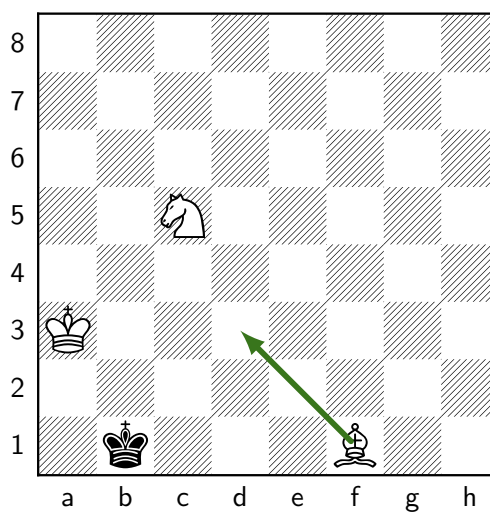


(d)

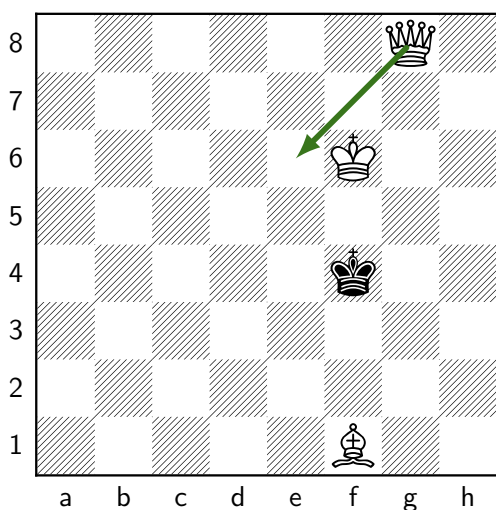
Figure 95: **Move 101, White: Bh4**. (a) Our position; Melanie plays Bh4 (g3h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0003#1483 (ply 206) [structural distance  $d=5$ , QAP obj=0]: the master played g7+; under  $\mu$  our Bg3 plays the role of that game’s Pg6 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Bd5; transferred move g3h4 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0003#376 (ply 184) [structural distance  $d=4$ , QAP obj=0]: the master played g5; under  $\mu$  our Bg3 plays the role of that game’s Pg4 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Bf1; transferred move g3h4 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0003#1379 (ply 190) [structural distance  $d=6$ , QAP obj=0]: the master played g8=Q+; under  $\mu$  our Bg3 plays the role of that game’s Pg7 (a **cross-type** correspondence); transferred move g3h4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bh4 (summed QAP objective 0, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



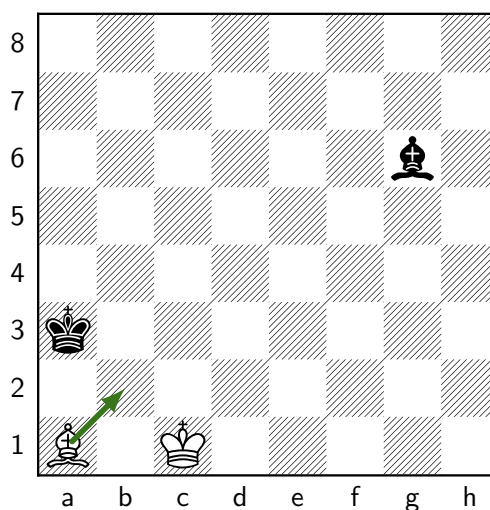
(a)



(b)



(c)



(d)

Figure 96: **Move 102, White: c8=Q#.** (a) Our position; Melanie plays c8=Q# (c7c8q). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1344` (ply 186) [structural distance  $d=2$ , QAP obj=1]: the master played Bd3+; under  $\mu$  our *Bh4* plays the role of that game’s *Bf1*; cross-type roles: our *Pc7* as their *Nc5*; transferred move h4f2. (c) Retrieved from `ccr1_top_p0003#376` (ply 202) [structural distance  $d=2$ , QAP obj=1]: the master played Qe6; under  $\mu$  our *Pc7* plays the role of that game’s *Qg8* (a **cross-type** correspondence); transferred move c7c8b. (d) Retrieved from `ccr1_top_p0000#1609` (ply 186) [structural distance  $d=2$ , QAP obj=0]: the master played Bb2+; under  $\mu$  our *Bh4* plays the role of that game’s *Ba1*; cross-type roles: our *Pc7* as their *bg6*; transferred move h4g5. Synthesis: no retrieved analog transferred to c8=Q# – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.